UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Software Engineering

NED University of Engineering and Technology





Speech Spectrum

Group Number: SE-22018 **Batch: 2022-2026**

Group Member Names:

Vaiz Hasan SE-22047

Syed Noor-ul-Talha SE-22072

Syed Ahmed Hussain Bukhari SE-22086

Syed Muhammad Reyan Imam SE-22091

Approved by

. . . . . .

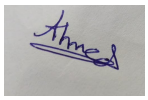
Engr. Sidra Masood

Project Advisor

© NED University of Engineering & Technology. All Rights Reserved – Aug 2025

Author's Declaration

We declare that we are the sole authors of this project. It is the actual copy of the project that was accepted by our advisor(s) including any necessary revisions. We also grant NED University of Engineering and Technology permission to reproduce and distribute electronic or paper copies of this project.

Signature and Date	Signature and Date	Signature and Date	Signature and Date
			
.....	.....	.....	.....
Vaiz Hasan SE-22047 hasan4505906@ cloud.neduet.edu.pk	Syed Noor-ul-Talha SE-22072 talha4503657@ cloud.neduet.edu.pk	S. Ahmed Hussain SE-22086 bukhari4500447@ cloud.neduet.edu.pk	S.M. Reyan Imam SE-22091 imam4503341@ cloud.neduet.edu.pk

Statement of Contributions

The project was a collaborative effort among the team members,

Vaiz Hasan (SE-22047)

- Developed the complete web application, including the admin panel, using Next.js with a responsive and scalable architecture.
- Designed and implemented role-based dashboards for parents, therapists, and administrator.
- Implemented core application modules, including questionnaire and speech analysis workflows with interactive UI screens.
- Integrated RESTful APIs with the frontend to enable secure and efficient communication with backend services.
- Ensured application usability and responsiveness through UI optimization and cross-device compatibility testing.

Syed Noor-ul-Talha (SE-22072)

- Developed the complete backend of SpeechSpectrum using Node.js and Express.js, designing and implementing RESTful APIs to support all core platform features.
- Designed and managed the entire relational database schema on Supabase.
- Integrated multiple third-party services into the backend including Safepay, Cloudinary, Zoom, and Firebase Cloud Messaging.
- Implemented the AI screening pipeline by connecting the backend to two external machine learning microservices responsible for screenings.
- Deployed and maintained the backend on Vercel with continuous deployment configured through GitHub.

Syed Ahmed Hussain Bukhari (SE-22086)

- Developed the mobile application using Flutter with responsive and user-friendly UI.
- Integrated RESTful APIs to connect frontend with backend services.
- Implemented AI screening flow and speech (voice) features in the app.
- Managed state and navigation efficiently using GetX.
- Integrated Firebase notifications, built payment gateway UI, added Google Meet, and performed testing and debugging.

Syed Muhammad Reyan Imam (SE-22091)

- Collected and prepared datasets required for machine learning tasks for model training.
- Performed comprehensive data preprocessing, including data cleaning, transformation, and feature engineering to enhance model performance.
- Developed and trained a Random Forest model for the questionnaire-based assessment module to support classification and decision-making.
- Designed and trained a Convolutional Neural Network (CNN) model for speech analysis to extract meaningful patterns from audio data.
- Deployed the trained models for integration into the final system for real-world application use.

Executive Summary

The report presents SpeechSpectrum, a system designed to address the challenge of early detection of speech delays and Autism Spectrum Disorder (ASD) in children. The problem arises due to delayed diagnosis, limited access to specialists, and reliance on traditional screening methods that are often time-consuming, subjective, and inaccessible for many families. Early identification is critical, as timely intervention significantly improves developmental outcomes, yet many cases go undetected during crucial early stages.

The background of the study highlights the importance of early screening and the limitations of existing approaches, including questionnaire-based assessments and conventional speech evaluation methods. While tools such as standardized checklists exist, they lack scalability and often do not incorporate advanced computational techniques. Recent advancements in machine learning and speech analysis provide an opportunity to enhance diagnostic accuracy and accessibility, motivating the development of a more efficient, technology-driven solution.

To solve this problem, the project adopts a hybrid methodology combining machine learning with a web-based application. The system integrates speech analysis models with questionnaire-based screening to provide a more comprehensive evaluation. Audio data is processed using feature extraction techniques and deep learning models to analyze speech patterns. The application is deployed on a cloud-based infrastructure to handle computational tasks efficiently. A modular architecture is implemented, consisting of frontend interfaces for users, a backend for processing and API communication, a machine learning component for prediction, and a database system for secure data storage.

The major findings indicate that combining speech-based analysis with traditional screening methods improves the reliability and accessibility of early ASD detection. The system demonstrates the ability to process user input efficiently and generate meaningful insights through automated analysis, reducing dependency on manual assessments.

In conclusion, the report establishes that SpeechSpectrum provides a scalable and effective solution for early screening of speech-related developmental issues. By leveraging machine learning and cloud technologies, the system enhances accessibility, supports timely intervention, and contributes to improving diagnostic practices in the domain of child development.

Acknowledgment

We would like to thank Allah for providing us with the ability and the competence to think and work. In addition to that, we would like to thank our supervisor Engr. Sidra Masood and our Industrial Advisor Dr. Danish Mehmood Khan for helping us throughout our project.In addition, we would like to appreciate the contribution of our teachers who have guided, taught, and helped us throughout our educational journey. Our sincere thanks also go to the staffs of our department and university for their assistance during our stay at the university.

Table of Contents

Author's Declaration ii
Statement Of Contribution iii
Executive Summary v
Acknowledgment vi
List of Tables xi
List of Figures xii
List of Abbreviations xv
United Nations Sustainable Development Goals xvi
Similarity Index Report xvii
Chapter1: Introduction 1
1.1 Background Information 1
1.2 Problem Statement 1
1.3 Proposed Solution 1
1.4 Aims and Objectives 2
1.5 Methodology 2
1.6 Significance and Motivation 3
Chapter2: Literature Review 3
2.1 Introduction 4
2.2 Speech Delays and ASD 4
2.3 Early Screening and Intervention 5
2.3.1 Impact of Early Intervention on Development 5
2.3.2 Barriers to Timely Diagnosis and Intervention Access 5
2.4 Digitized Autism Screening 6
2.5 Questionnaire-Based Screening (No ML) 6
2.5.1 Q-CHAT (Quantitative Checklist for Autism in Toddlers) 6
2.5.2 ASSQ (Autism Spectrum Screening Questionnaire) 7
2.5.3 M-CHAT (Modified Checklist for Autism in Toddlers) 7
2.6 ML-Enhanced Questionnaire-Based ASD Screening 7
2.6.1 Q-CHAT Related ML Work 7
2.6.2 M-CHAT Related ML Work 8
2.6.3 ASSQ Related ML Work 8
2.7 Speech-Based Screening (Conventional / Non-ML Approaches) 9
2.7.1 Pre–Machine Learning Era 10
2.7.2 Standardized Language Assessments 10
2.7.3 Audiological Screening 10

2.7.4 Developmental Milestone Tracking . . . 10
2.7.5 Practical Limitations . . . 10
2.8 Speech-Based Screening (ML/DL Methods) . . . 11
2.8.1 Natural / Conversational Speech Screening . . . 11
2.8.2 Nonword Repetition / Controlled Speech Tasks . . . 11
2.8.3 Diagnostic / ADOS-Style Conversations . . . 12
2.8.4 Infant Cry Analysis . . . 12
2.8.5 Prosodic / Natural Speech Meta-Analysis . . . 12
2.8.6 Multimodal / Fusion Approaches . . . 12
2.9 Post Analysis Interventions . . . 12
2.9.1 Traditional Behavioral and Speech Therapy . . . 13
2.9.2 Applied Behavior Analysis (ABA) . . . 14
2.9.3 Telehealth and Teletherapy . . . 14
2.9.4 Naturalistic Developmental Behavioral Interventions (NDBIs) . . . 14
2.9.5 Gamified Learning . . . 14
2.9.6 AI Based Applications . . . 14
Chapter3: System Features And Specifications . . . 15
3.1 Introduction . . . 16
3.2 Product Perspective and Users . . . 16
3.2.1 Product Scope . . . 16
3.2.2 User Classes and Characteristics . . . 16
3.3 Functional Requirements . . . 17
3.3.1 Parent/Guardian Functions . . . 17
3.3.2 Expert User Functions . . . 18
3.3.3 System Administrator . . . 19
3.4 Quality Factors . . . 20
3.4.1 Performance Attributes . . . 20
3.4.2 Security Attributes . . . 20
3.4.3 Usability (UI/UX) Attributes . . . 21
3.5 Operating Environment . . . 21
3.6 Data Flow Diagram . . . 22
3.6.1 Level 0 Data Flow Diagram . . . 22
3.6.2 Level 1 Data Flow Diagram . . . 22
3.7 Summary . . . 23
Chapter4: Architecture And Design . . . 23
4.1 Introduction . . . 24
4.2 Software Architecture . . . 24
4.2.1 Frontend System . . . 24
4.2.2 Backend System . . . 25

4.2.3 Machine Learning System . . . 25
4.2.4 Database System . . . 25
4.3 Database Design . . . 25
4.3.1 Entity Relationship Model . . . 26
4.4 Backend Design . . . 40
4.5 Frontend Design . . . 44
4.5.1 Web Application Design Components . . . 45
4.5.2 Mobile Application Design Components . . . 45
4.6 ML Model Design . . . 47
4.6.1 Random Forest for Questionnaire . . . 47
4.6.2 CNN Model for Speech Analysis . . . 48
4.7 Summary . . . 49
Chapter5: Construction and Development . . . 49
5.1 Introduction . . . 50
5.2 Database Setup . . . 50
5.2.1 Tools Used . . . 50
5.2.2 Methodology . . . 51
5.3 Backend Development . . . 52
5.3.1 Tools Used . . . 52
5.3.2 Methodology . . . 53
5.4 Web App Development . . . 54
5.4.1 Tools Used . . . 54
5.4.2 Methodology . . . 55
5.5 Mobile App Development . . . 56
5.5.1 Tools Used . . . 56
5.5.2 Methodology . . . 56
5.6 Machine Learning Model Training . . . 57
5.6.1 Questionnaire Based Screening . . . 57
5.6.2 Speech Analysis . . . 58
5.7 Summary . . . 62
Chapter6: Deployment . . . 63
6.1 Introduction . . . 64
6.2 Database Deployment . . . 64
6.2.1 Introduction . . . 64
6.2.2 Provisioning the Database . . . 64
6.2.3 Connection Configuration . . . 64
6.3 Backend Deployment . . . 65
6.3.1 Introduction . . . 65
6.3.2 Preparing the Environment . . . 65

6.3.3 Deployment Process 65
6.4 Mobile Application Build 65
6.4.1 Introduction 65
6.4.2 Preparing the Build 66
6.4.3 Building the App 66
6.5 Web Application Deployment 66
6.5.1 Introduction 66
6.5.2 Preparing the Web Frontend 66
6.5.3 Hosting the Web Application 67
6.5.4 Final Checks 67
6.6 Machine Learning Deployment 67
6.6.1 Questionnaire Based Analysis Model 67
6.6.2 Speech Analysis Model 68
6.7 Summary 68
Chapter7: Results 68
7.1 Introduction 69
7.2 Application Results 69
7.2.1 Login and Signup 69
7.2.2 Parent View 77
Future Enhancements 110
References 111

References **111**

List of Table

2.1 Validation of commonly used ASD screening questionnaires. 7
2.2 ML-Enhanced Questionnaire-Based ASD Screening 9
2.3 Limitations of Conventional Speech-Based Screening Methods 11
2.4 Speech-Based ML/DL Screening Methods 13
3.1 User Classes and Characteristics 17

List of Figures

1 Similarity Report 1 xviii
2 Similarity Report 2 xix
3 Similarity Report 3 xx
4 Similarity Report 4 xxi
5 Similarity Report 5 xxii
6 Similarity Report 6 xxiii
7 Similarity Report 7 xxiv
8 Similarity Report 8 xxv
3.1 Parent Use Case Diagram 18
3.2 Expert User/Specialist Use Case Diagram 19
3.3 Administrator Use Case Diagram 20
3.4 Level 0 DFD Diagram 22
3.5 Level 1 DFD Diagram 22
4.1 Schema Design of SpeechSpectrum System 38
4.2 Backend Architecture Design of SpeechSpectrum System 41
4.3 Frontend Architecture Design of SpeechSpectrum System 44
4.4 Machine Learning Model Architecture for SpeechSpectrum System 47
5.1 Database Tables 51
5.2 Trigger Functions 52
7.1 Web Login 69
7.2 Mobile Login 70
7.3 Web Signup 71
7.4 Sign Up As Parent Step 1 71
7.5 Sign Up As Parent Step 2 72
7.6 Sign Up As Parent Step 3 72
7.7 Sign Up As Parent Step 4 73
7.8 Sign Up As Parent Step 5 74
7.9 Signup as Expert step 1 74
7.10 Signup as Expert step 2 75
7.11 Signup as Expert step 3 75
7.12 Signup as Expert step 4 76
7.13 Signup as Expert step 5 76
7.14 Signup as Expert step 6 77
7.15 Web Dashboard 77
7.16 Mobile dashboard 78

7.17 Web Parent's Sidebar 79
7.18 Mobile Parent's Sidebar 80
7.19 My children page 80
7.20 My children Tab 81
7.21 Web progress and Results view 81
7.22 Mobile progress and Results view 82
7.23 Web Find Therapist view 82
7.24 Mobile Find Therapist View 83
7.25 Web My Appointments 83
7.26 Requested 84
7.27 Completed 84
7.28 confirmed 85
7.29 Setting 85
7.30 Web Questionnaire Screening 86
7.31 Mobile Questionnaire Screening step 1 86
7.32 Web Questionnaire Screening step 2 87
7.33 Mobile Questionnaire Screening step 2 87
7.34 Web Questionnaire screening Result 88
7.35 Mobile Questionnaire screening Result 88
7.36 Questionnaire Screening Report 89
7.37 Web Speech Analysis view 90
7.38 Mobile Speech Analysis view 90
7.39 Web Speech Analysis Result view 91
7.40 Mobile Speech Analysis Result view 92
7.41 Speech Analysis Report 93
7.42 Web Questionnaire Assessment History view 93
7.43 Mobile Questionnaire Assessment History view 94
7.44 Web Speech Assessment History view 95
7.45 Mobile Speech Assessment History view 96
7.46 Web Parent Profile Management view 97
7.47 Mobile Parent Profile Management view 97
7.48 Web Expert Dashboard view 98
7.49 Mobile Expert Dashboard 98
7.50 Web Expert Sidebar view 99
7.51 Mobile Expert Sidebar 100
7.52 Web locations view 101
7.53 Mobile locations Tab 1 view 101
7.54 Mobile locations Tab 2 view 102
7.55 Mobile locations Tab 3 view 102

7.56 Web Slots Page 103
7.57 Slots Page 1 103
7.58 Slots Page 2 104
7.59 Web Appointments Page 104
7.60 Mobile Appointments Page 1 105
7.61 Mobile Appointments Page 2 105
7.62 Mobile Appointments Page 3 106
7.63 Payment Gateway 106
7.64 Web Appointment Detail Page 107
7.65 Appointment Detail Page 1 107
7.66 Appointment Detail Page 2 108
7.67 Web Expert Profile Management 108
7.68 Expert Profile Management 109

List of Abbreviations

ASD	Autism Spectrum Disorder
CNN	Convolutional Neural Network
ML	Machine Learning
AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
ERD	Entity Relationship Diagram
VPS	Virtual Private Server
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token

United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace, and justice. There are a total of 17 SDGs as mentioned below. Check the appropriate SDGs related to the project.

- ☐ No Poverty
- ☐ Zero Hunger
- ☑ Good Health and Well-being
- ☑ Quality Education
- ☐ Gender Equality
- ☐ Clean Water and Sanitation
- ☐ Affordable and Clean Energy
- ☐ Decent Work and Economic Growth
- ☐ Industry, Innovation, and Infrastructure
- ☑ Reduced Inequalities
- ☐ Sustainable Cities and Communities
- ☐ Responsible Consumption and Production
- ☐ Climate Action
- ☐ Life Below Water
- ☐ Life on Land
- ☐ Peace, Justice, and Strong Institutions
- ☐ Partnerships to Achieve the Goals

Similarity Index Report

Following students have compiled the final year report on the topic given below for partial fulfillment of the requirement for Bachelor's degree in Software Engineering.

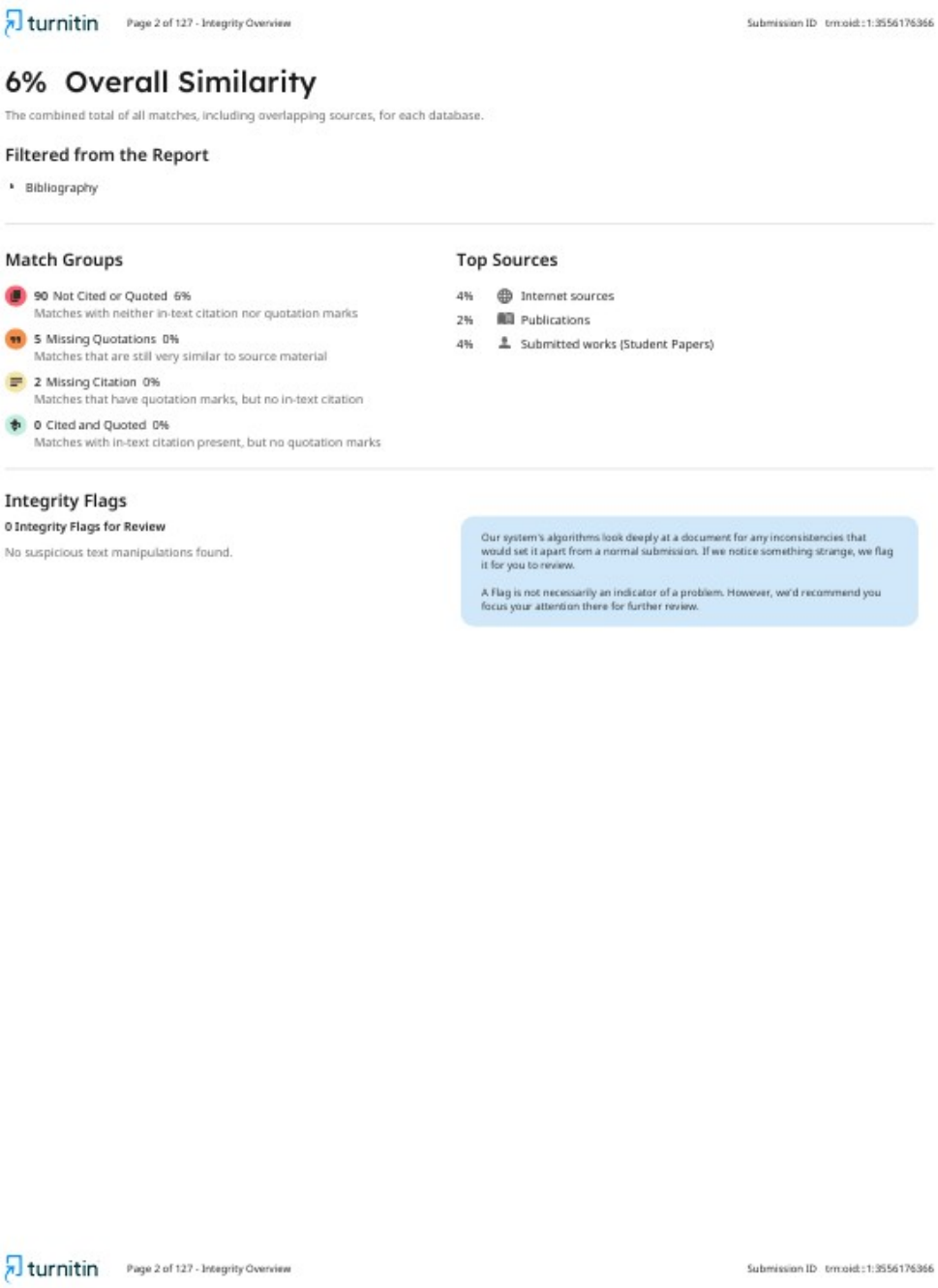
Project Title SpeechSpectrum

S. No.	Student Name	Seat Number
1.	Vaiz Hasan	SE-22047
2.	Syed Noor-ul-Talha	SE-22072
3.	Syed Ahmed Hussain Bukhari	SE-22086
4.	Syed Muhammad Reyan Imam	SE-22091

This is to certify that Plagiarism test was conducted on the complete report, and overall similarity index was found to be less than 20%, with maximum 5% from a single source, as required.

Signature and Date

Engr. Sidra Masood
Project Advisor



turnitin Page 2 of 127 - Integrity Overview Submission ID trn:oid:::1:3556176366

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

- 90 Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- 5 Missing Quotations 0%
  Matches that are still very similar to source material
- 2 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
4% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

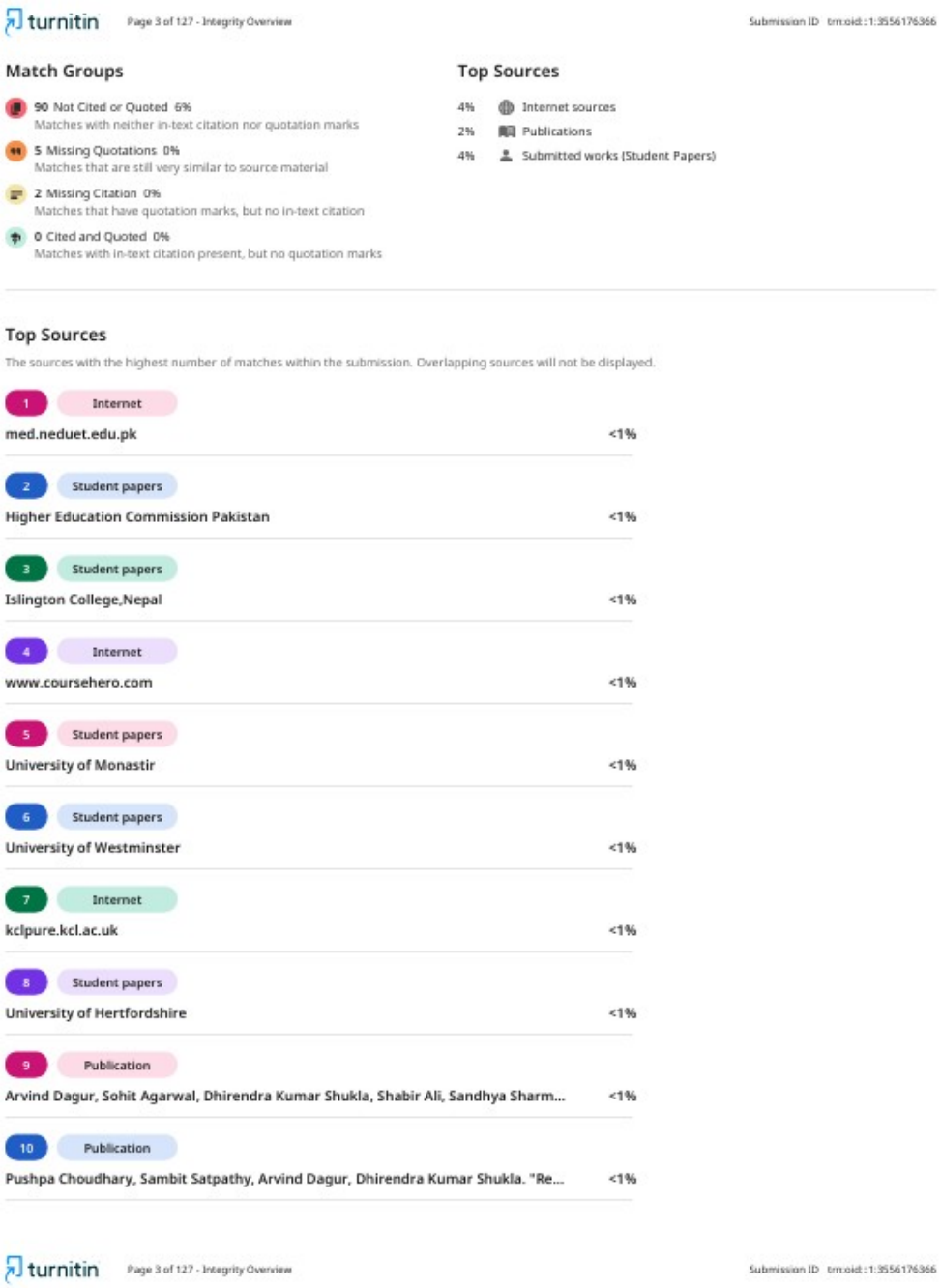
No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

turnitin Page 2 of 127 - Integrity Overview Submission ID trn:oid:::1:3556176366

Figure 1: Similarity Report 1



turnitin Page 3 of 127 - Integrity Overview Submission ID trn:oid::1:3556176366

Match Groups

- 90 Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- 5 Missing Quotations 0%
  Matches that are still very similar to source material
- 2 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
4% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	
med.neduet.edu.pk		<1%
2	Student papers	
Higher Education Commission Pakistan		<1%
3	Student papers	
Islington College,Nepal		<1%
4	Internet	
www.coursehero.com		<1%
5	Student papers	
University of Monastir		<1%
6	Student papers	
University of Westminster		<1%
7	Internet	
kclpure.kcl.ac.uk		<1%
8	Student papers	
University of Hertfordshire		<1%
9	Publication	
Arvind Dagur, Sohit Agarwal, Dhirendra Kumar Shukla, Shabir Ali, Sandhya Sharm...		<1%
10	Publication	
Pushpa Choudhary, Sambit Satpathy, Arvind Dagur, Dhirendra Kumar Shukla. "Re...		<1%

turnitin Page 3 of 127 - Integrity Overview Submission ID trn:oid::1:3556176366

Figure 2: Similarity Report 2

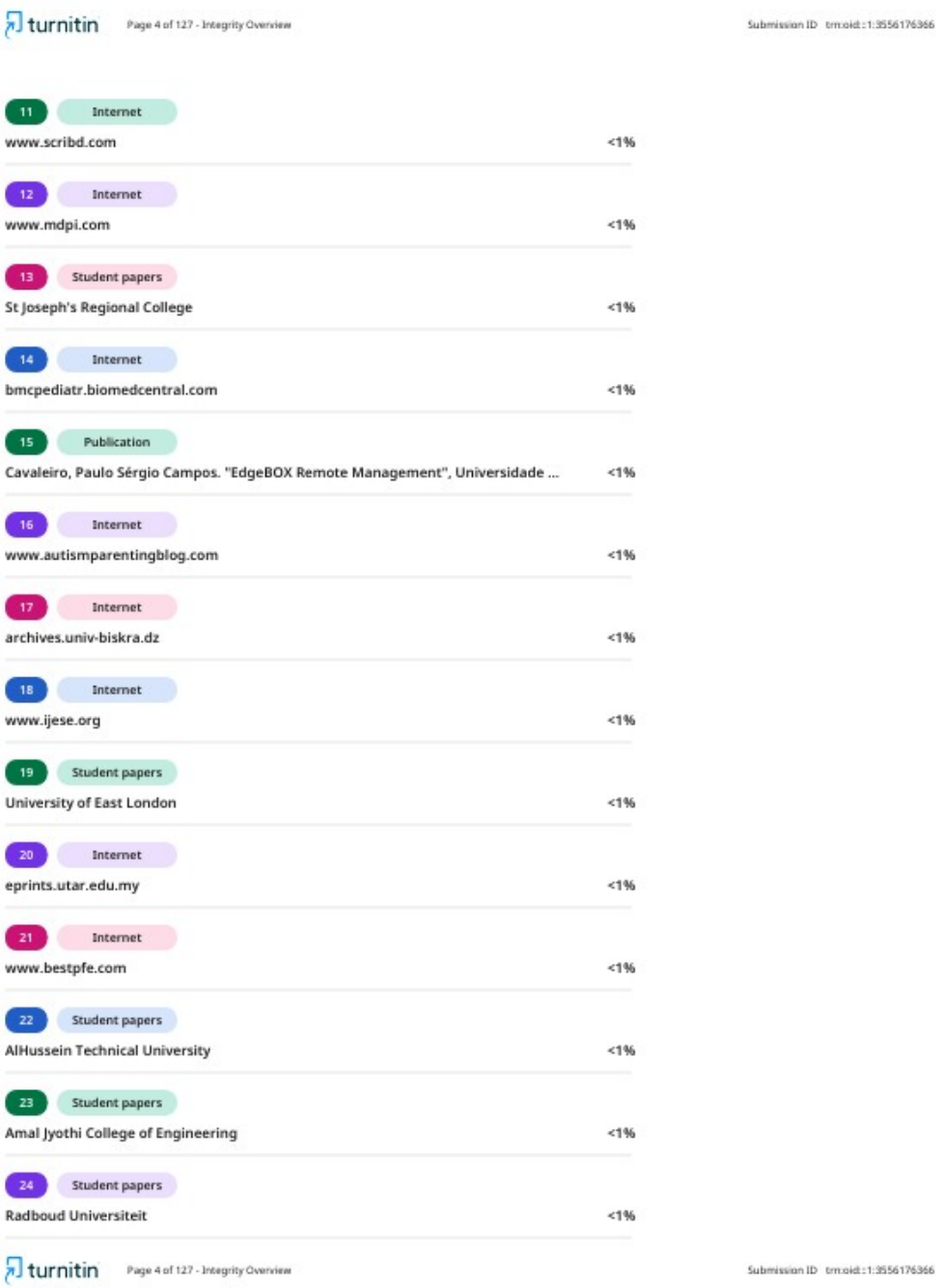


Figure 3: Similarity Report 3

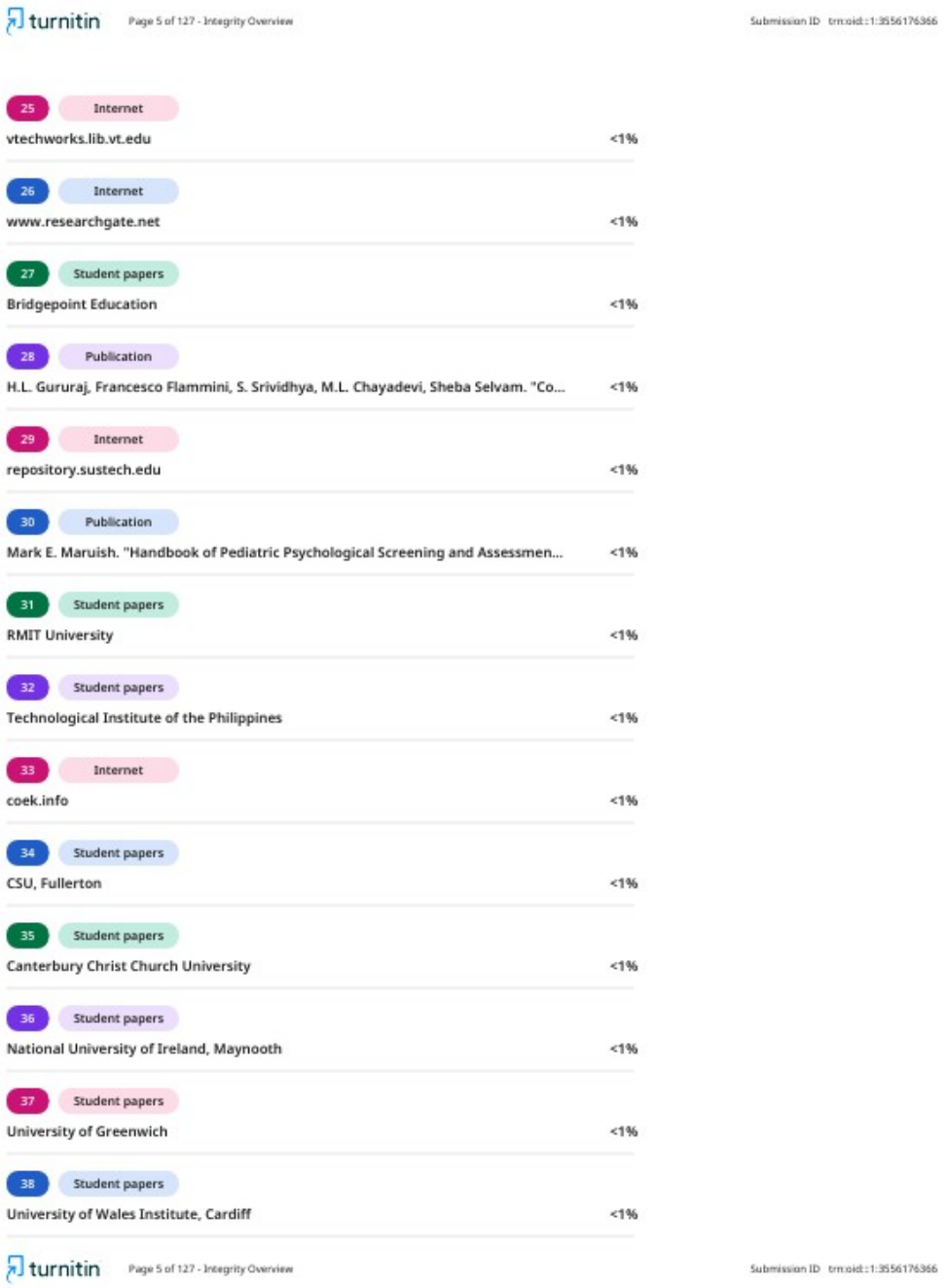


Figure 4: Similarity Report 4

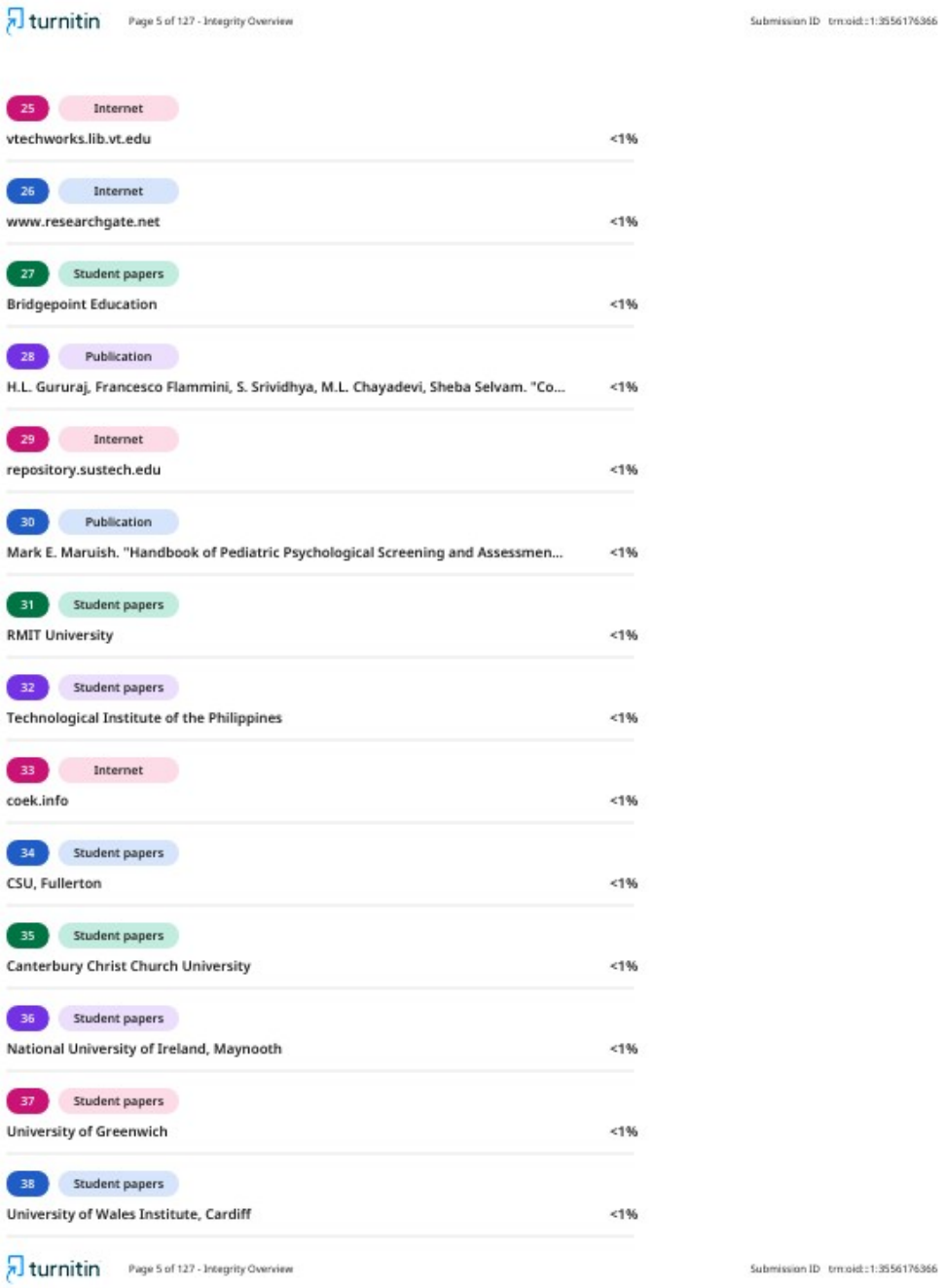


Figure 5: Similarity Report 5

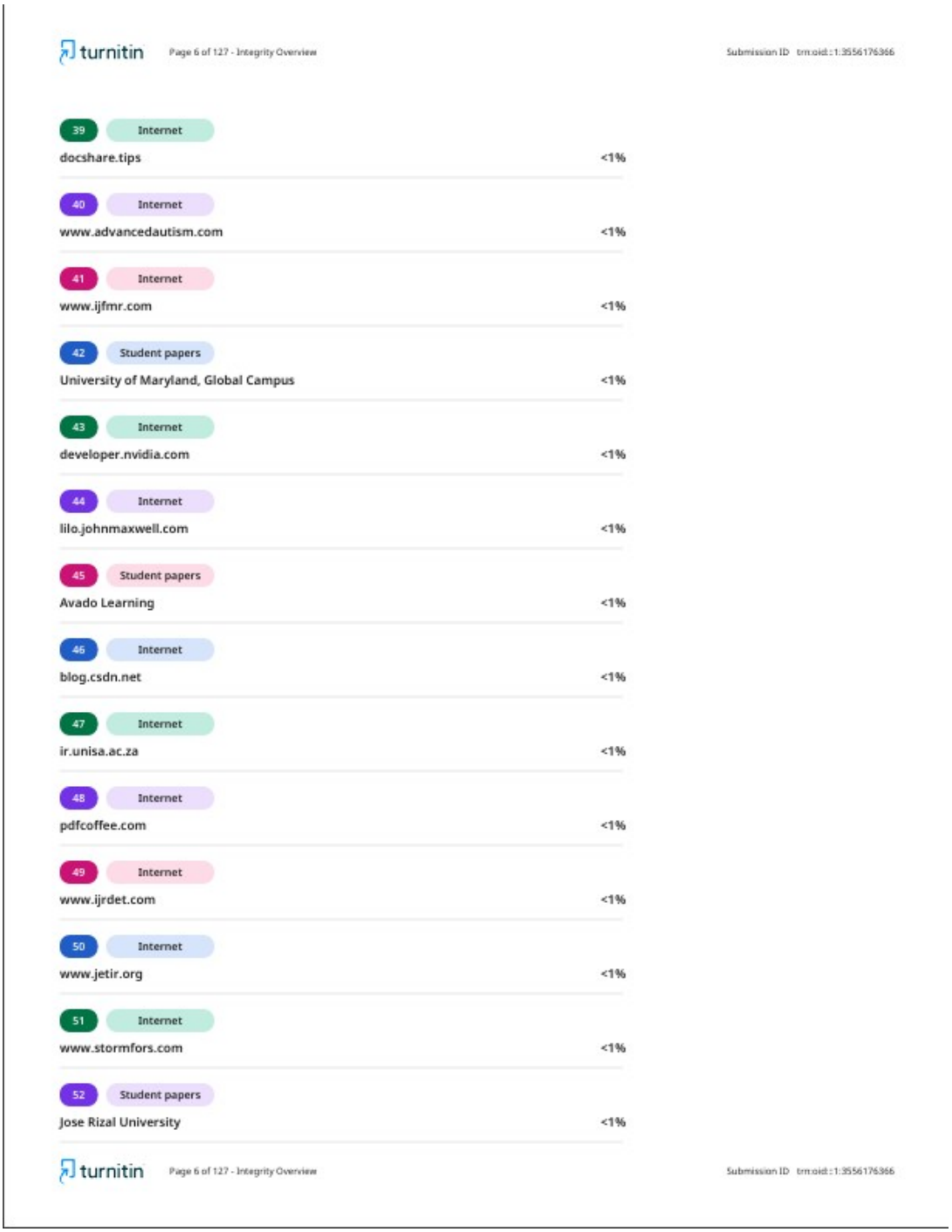


Figure 6: Similarity Report 6

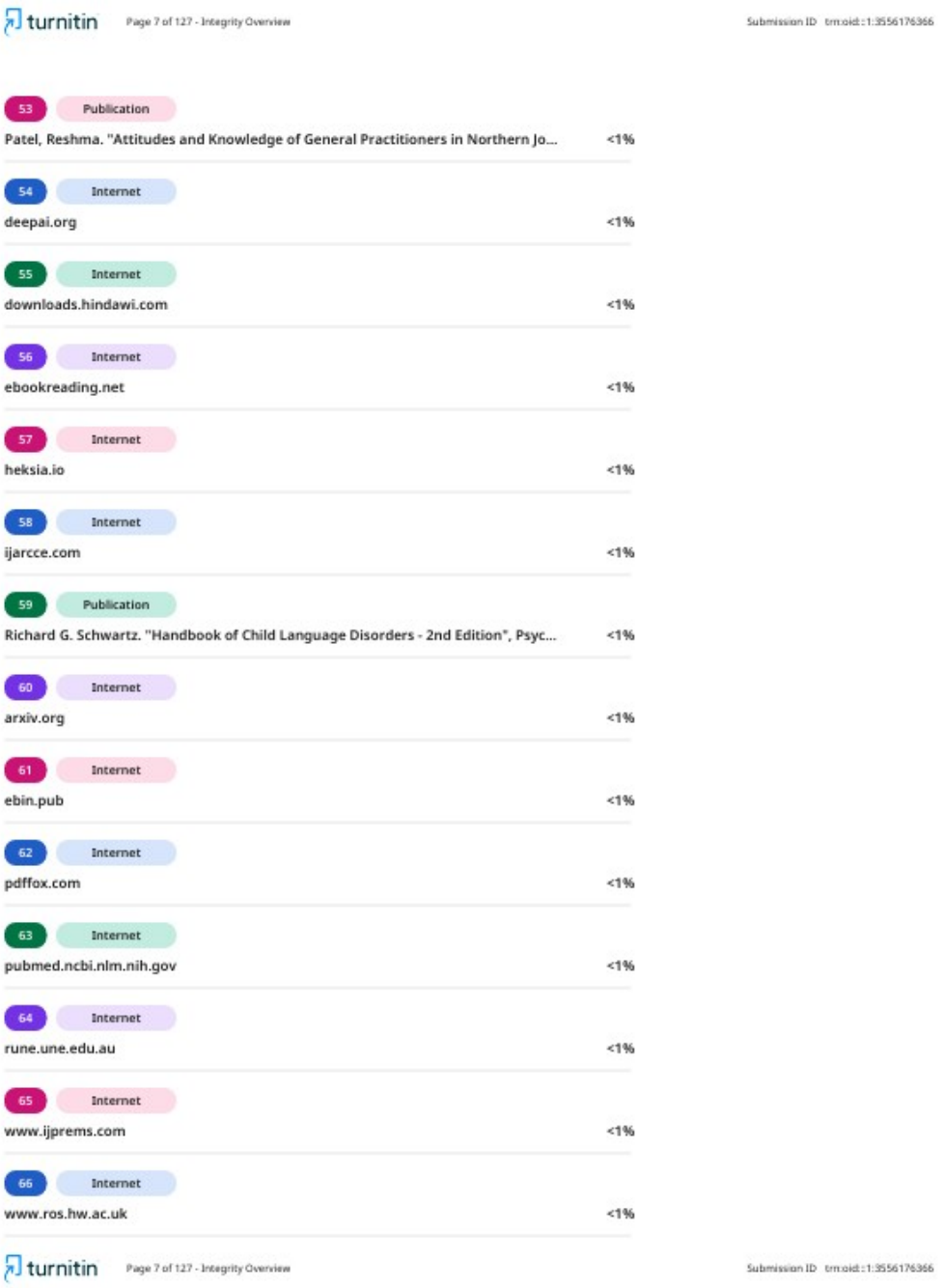


Figure 7: Similarity Report 7

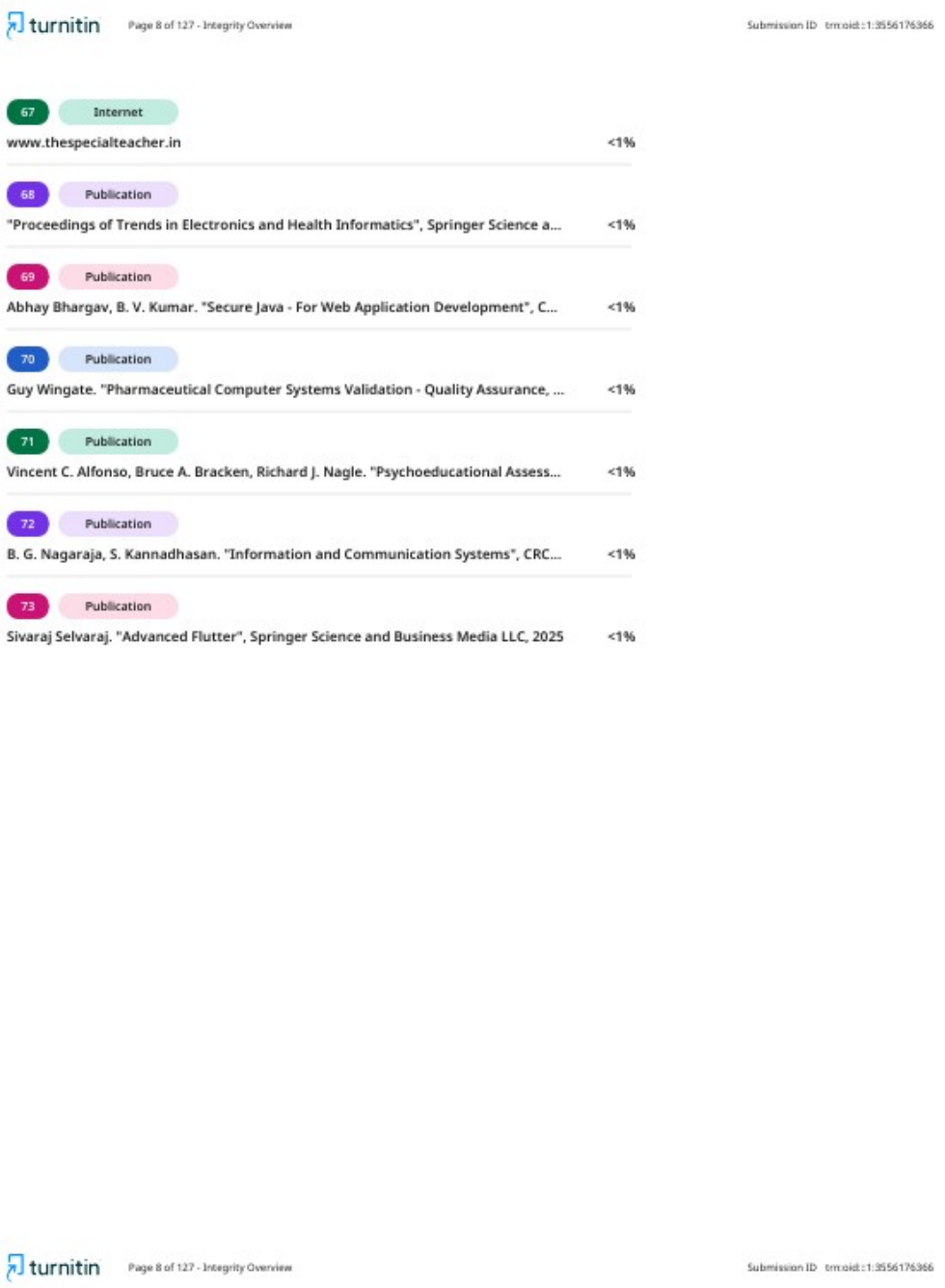


Figure 8: Similarity Report 8

Chapter 1
Introduction

1.1 Background Information

Speech and language delays are among the earliest indicators of developmental issues in children aged 1–10 years. Autism Spectrum Disorder (ASD) is one of the major causes, yet early identification is often missed because parents overlook subtle signs such as limited vocabulary, poor social interaction, or delayed gestures. Early screening using digital tools can greatly improve intervention opportunities and long-term communication outcomes in children. By utilizing modern technology, these tools can make early detection more accessible and practical for families.

1.2 Problem Statement

Early autism-related speech problems are often missed by parents, and professional tests are costly and demand specific knowledge. This disparity leads to missed chances for early treatment and delayed intervention. A sophisticated, user-friendly, and accessible technology that can help parents easily identify possible autism-related speech problems at home is desperately needed. Early intervention during the most critical developmental stage can be made possible by this kind of system, which can facilitate immediate consultation with medical specialists.

1.3 Proposed Solution

SpeechSpectrum, the suggested solution, is a smartphone app (also a web application) that combines a structured questionnaire with speech recordings that are subjected to machine learning algorithms in order to calculate the likelihood of autism. The program will handle questionnaire data using supervised models like Random Forest while Convolutional Neural Networks (CNNs) will be used to analyze voice aspects. Beyond risk prediction, SpeechSpectrum supports a post-analysis workflow by providing trackable intervention activities and facilitating professional consultation through expert user involvement. While preserving ease of use, accessibility, and data protection, this integrated strategy guarantees precise prediction and offers parents insightful information.

1.4 Aims and Objectives

The primary goal is to create a mobile application driven by AI that uses behavioral and voice data to check for autism early.
Among the goals are:

- By Using an intuitive UI to gather behavioral and vocal data from children.
- Utilizing conventional ML algorithms and CNN to accurately predict autism risk.
- Delivering educational guidance, knowledge, and unambiguous feedback to parents.
- Facilitating post analysis intervention and professional consultations.
- Ensuring privacy and accessibility for individuals with different backgrounds.

These goals center on enabling parents to take preventative measures by means of technology-assisted screening and awareness.

1.5 Methodology

The proposed methodology involves several key stages:
Data Collection:

- Collect behavioral data through structured questionnaires.
- Gather speech samples (audio recordings) from children for vocal analysis.

Feature Extraction:

- Extract audio features such as MFCCs, pitch, and prosody from recorded speech.
- Prepare structured questionnaire data for machine learning processing.

Model Development:

- Use Convolutional Neural Networks (CNNs) to analyze speech data.
- Apply supervised algorithms such as Random Forest, K-Nearest Neighbors (KNN), and XGBoost for behavioral data.

Model Integration:

- Combine both models to generate an autism risk prediction score.
- Implement backend integration using TensorFlow or PyTorch for inference.

App & Web Development:

- Design a user-friendly mobile interface using Flutter for parents and caregivers.
- Develop a web-based platform using Next.js for wider accessibility.
- Implement backend services using Node.js and Flask for secure API communication and model inference.
- Ensure seamless integration between web, mobile, and backend components.

Testing and Deployment:

- Conduct unit, integration, and user acceptance testing to validate performance.
- Deploy the final application on web and mobile platforms, monitoring for accuracy, usability, and reliability.

1.6 Significance and Motivation

SpeechSpectrum promotes early detection, awareness, and intervention for autism-linked speech delays through AI-driven analysis. It bridges the gap between parental observation and expert diagnosis, offering a fast, private, and affordable screening solution. The project supports the broader goals of Good Health and Well-being, Quality Education, and Reduced Inequalities, by making early autism screening accessible to families from all backgrounds. This initiative reflects the growing motivation to use technology for inclusive healthcare and early developmental support.

Chapter 2
Literature Review

2.1 Introduction

This chapter presents a critical review of the foundational research underpinning the Speech-Spectrum project. The primary goal is to establish the clinical justification and technical feasibility of an integrated machine learning (ML) system for the early screening of Autism Spectrum Disorder (ASD) linked speech delays in children. The review proceeds in a structured manner, first detailing the clinical necessity for early diagnosis, followed by an in-depth analysis of ASD-related speech pathology and the quantitative acoustic features that serve as objective biomarkers. Subsequently, the literature validating the two core components of the proposed system behavioral screening via the Quantitative Checklist for Autism in Toddlers (Q-CHAT) and deep learning models for acoustic feature analysis is examined. By synthesizing existing research, this review articulates the methodological advantages of integrating classical supervised ML for structured data with Convolutional Neural Networks (CNNs) for complex signal processing, thereby defining the unique contribution and positioning of the SpeechSpectrum solution within the field of digital healthcare.

2.2 Speech Delays and ASD

Autism Spectrum Disorder (ASD) often affects early speech and language, but the problems can vary and are not unique to ASD [1] [2]. Language difficulties are often one of the first signs of ASD – ranging from not speaking at all to unusual tone or rhythm of speech [2] – and parents often notice these issues early. Research shows that by 12 months, babies who later develop ASD already have unusual patterns in understanding and using language compared to other children [3]. Other studies also report that children with ASD show differences in how they use pitch, rhythm, and phrasing in speech [1].

- **Early language differences:** Babies who later develop ASD often score lower in understanding and using language by 12 months [3]. Studies suggest that both the scores and the way babies use language at 1 year can help identify children who may later be diagnosed with ASD[3].
- **Speech development traits:** Even with language delays, some speech skills may develop normally. Research using sound analysis found that toddlers at high risk for ASD had normal average voice-onset times and error rates but produced fewer sounds and showed more variation in these sounds[4]. So, while overall speech may seem typical, subtle differences in how sounds are used can appear in ASD.

- **Behavioral cues in speech-delayed children:** Studies of children aged 18–48 months with speech delays show that certain behaviors can help predict ASD. A combined "behavioral response" score – including tantrums, lack of eye contact, and slow response to their name – showed high accuracy in identifying ASD in children with speech delays. This means that even among children who are slow to talk, specific social behaviors in speech can indicate higher ASD risk.

2.3 Early Screening and Intervention

Early screening and intervention for Autism Spectrum Disorder (ASD) are critical because developmental gains are much more likely when support is initiated during early childhood. A comprehensive meta-analysis of 29 early-intervention studies showed that children receiving early behavioral therapies made significantly greater improvements in social communication compared to those who received later or no intervention [6]. At the same time, computational approaches leveraging speech data have shown promise: for example, deep neural networks trained on acoustic and prosodic features of young children's speech can reliably estimate autism severity, suggesting a scalable, objective screening pathway [7]. Although early detection offers clear advantages, traditional ASD diagnostics rely heavily on specialists and are often delayed, underscoring the need for accessible automated tools [8]. Collectively, this body of work emphasizes the importance of combining early behavioral interventions with scalable screening technologies to maximize developmental outcomes.

2.3.1 Impact of Early Intervention on Development

Early intervention has a significant impact on the developmental outcomes of children with Autism Spectrum Disorder (ASD). Studies show that structured behavioral therapies delivered during infancy and early toddlerhood improve social communication, adaptive behaviors, and cognitive functioning [6], [8]. Additionally, early intervention can enhance speech and language development, which is critical for long-term social and academic success [6]. Timely therapy underscores the importance of detecting ASD symptoms early and applying targeted interventions to maximize developmental gains [8].

2.3.2 Barriers to Timely Diagnosis and Intervention Access

Many families face significant barriers to early ASD diagnosis and intervention, which are rooted in systemic, logistical, and sociocultural factors. Parents often report long wait times for assessments, limited availability of specialists, and healthcare providers who may dismiss or overlook early developmental concerns [9]. Socioeconomic disadvantage, cultural stigma, and limited awareness about ASD further delay the recognition of symptoms and access to ser-

vices [10]. In rural or underserved regions, geographic isolation, lack of transportation, and insufficiently trained professionals compound the problem, creating substantial gaps in timely intervention. These barriers not only delay the start of critical early therapy but also increase stress for families and reduce the potential developmental benefits for children. Addressing these obstacles is therefore essential to ensure equitable and timely access to intervention services [11].

2.4 Digitized Autism Screening

Using digital tools for ASD screening has many benefits: it can automatically score results, use new types of data like videos or audio, and reduce errors or biases that happen with paper questionnaires. Research has shown that digital screening tools can improve accuracy when used in real-world settings. For example, studies on tablet-based screening apps for children aged 18–36 months show that these apps can present videos and tasks to elicit behaviors related to ASD and use machine learning to analyze the child's responses[5]. The algorithms in these tools have achieved high accuracy, with around 88% sensitivity and 81% specificity, in distinguishing children with ASD from those without[5].
Importantly, research also indicates that digital methods perform consistently across different sexes, races, and ethnicities, which can help reduce disparities in screening[5]. Studies have also shown that traditional questionnaires often underperform in typical clinical settings (for example, some widely-used tools detected less than 40% of cases in large samples of children). Overall, digital screening tools provide a more objective, automated, and scalable way to identify children at risk for ASD.

2.5 Questionnaire-Based Screening (No ML)

Questionnaire-based screening tools are commonly used to detect early signs of ASD. These tools are completed by parents or teachers and do not involve machine learning. They provide a structured way to identify children who may need further evaluation. Below, we describe the main screening questionnaires:

2.5.1 Q-CHAT (Quantitative Checklist for Autism in Toddlers)

The Q-CHAT is a parent-reported checklist for toddlers to identify early autistic traits. Research shows that it can flag children at risk, though its accuracy is moderate. Studies report good reliability and validity for early ASD screening[12]. Q-CHAT is often used alongside other assessment methods to improve early detection[12][13].

2.5.2 ASSQ (Autism Spectrum Screening Questionnaire)

The ASSQ is a 27-item checklist completed by parents or teachers, typically for children aged 7–16. Studies on translated versions, including Urdu adaptations, report moderate internal consistency but strong test–retest reliability. Research concludes that ASSQ is a valid and reliable tool for screening ASD in school-aged children[14].

2.5.3 M-CHAT (Modified Checklist for Autism in Toddlers)

The M-CHAT is a 20–23 item parent-reported questionnaire for children aged 16–30 months. Validation studies of local adaptations, including Urdu versions, report high sensitivity (87%) and specificity (83%). Factor analysis of responses identifies coherent domains of language and social behavior, confirming M-CHAT as a reliable early screening tool[15][16].

Questionnaire	Age/Use	Key Finding (Validation Study)	Citation
Q-CHAT	Toddlers (16–30 months)	Used in research; original Q-CHAT-10 shows good reliability and 83% accuracy in screening ASD	[12][13]
ASSQ	School-age (7–16 years)	Moderate internal consistency (Cronbach $\alpha \approx 0.58$); validated as a reliable screening tool for Urdu-speaking children	[14]
M-CHAT/F	Toddlers (16–30 months)	High diagnostic accuracy: sensitivity 87%, specificity 83%; factor analysis shows coherent domains of language and social behavior	[15][16]

Table 2.1: Validation of commonly used ASD screening questionnaires.

2.6 ML-Enhanced Questionnaire-Based ASD Screening

Machine learning (ML) has been applied to traditional ASD questionnaires to improve accuracy, reduce the number of items, and provide a quantitative risk score for each child. ML models use item responses sometimes combined with demographics to identify the most predictive items, shorten the questionnaire, and compute a probability of ASD.

2.6.1 Q-CHAT Related ML Work

The Quantitative Checklist for Autism in Toddlers (Q-CHAT) is a parent-reported questionnaire designed for toddlers. ML techniques such as Support Vector Machines (SVM), Random Forest (RF), and Recursive Feature Elimination (RFE) have been applied to the full 25-item Q-CHAT to:

- Identify the most predictive questions for ASD screening.
- Reduce the questionnaire length without significant loss of accuracy.
- Compute a risk score indicating the probability of ASD.

Research shows that:

- A 14-item subset of Q-CHAT achieved 91% accuracy in predicting ASD.
- A 3-item mini-Q-CHAT retained 83% accuracy, demonstrating that even a very short set of questions can screen effectively when optimized with ML [17].

2.6.2 M-CHAT Related ML Work

The Modified Checklist for Autism in Toddlers (M-CHAT-R/F) is a widely used screening tool for children aged 16–30 months. Machine learning has been applied to M-CHAT data to:

- Identify the most informative items through feature-selection methods such as entropy ranking, ROC-based selection, and statistical tests.
- Develop shorter versions of the questionnaire while maintaining strong screening performance.
- Improve predictive scoring by generating probabilistic ASD-risk outputs rather than simple pass/fail results.

A publicly available ML study by Achenie et al. used a feed-forward artificial neural network (fANN) to optimize M-CHAT-R by selecting a smaller subset of highly predictive items. Their best reduced model (18 items) achieved 99.72% overall correct classification, with specificity reaching 99.9% and a significantly improved positive predictive value (PPV) compared to the original M-CHAT scoring method. This demonstrates that ML can refine M-CHAT screening by improving diagnostic efficiency and reducing false referrals [18].

2.6.3 ASSQ Related ML Work

The Autism Spectrum Screening Questionnaire (ASSQ) is primarily used for school-age children (7–16 years). Currently, no research studies applying ML to ASSQ responses have been published. Traditional ASSQ scoring is based on fixed cutoffs, but future work could explore ML for item selection and risk scoring.

Table 2.2: ML-Enhanced Questionnaire-Based ASD Screening

Questionnaire	**Target Age / Population**	**ML Model**	**Input Features**	**Outcome / Accuracy**	**Citation**
Q-CHAT	Toddlers / preschool (25 items)	SVM, RF + RFE	Full Q-CHAT responses	14-item model: 91% accuracy; 3-item mini-QCHAT: ∼83%	[17]
M-CHAT-R/F	Toddlers (16–30 months)	Neural network + feature selection	Full M-CHAT-R (20 items) reduced to 18 important items	18-item ML model: ∼99.7% accuracy; very high specificity (99.9%); better PPV (0.789) meaning fewer false positives	[18]
ASSQ	School-age children	N/A	ASSQ responses	No ML studies available	–

2.7 Speech-Based Screening (Conventional / Non-ML Approaches)

Before machine learning approaches were widely used, speech-based screening for Autism Spectrum Disorder (ASD) relied primarily on manual assessment, observation, and structured questionnaires. Clinicians and researchers focused on identifying differences in speech patterns, prosody, and vocal behaviors without computational analysis. Despite technological advances, these conventional methods remain foundational in developmental evaluation, providing clinically validated benchmarks and structured frameworks for early detection [19].

These traditional screening methods not only helped in identifying overt speech and language delays but also provided insights into subtler markers of atypical development, such as unusual intonation patterns, speech rhythm, and social use of language. By combining standardized tests, parent-reported questionnaires, and direct clinician observation, practitioners could form a holistic understanding of a child's communicative abilities. Furthermore, these approaches enabled early referral to intervention services, which is critical for improving developmental outcomes. While effective, their reliance on human judgment and manual scoring introduced variability, highlighting the potential for more objective and scalable methods in the future [19, 20].

2.7.1 Pre–Machine Learning Era

Before machine learning models were applied, speech-based screening relied on structured observations and manual analyses to identify early language and communication differences. Clinicians assessed articulation, prosody, and speech rhythm, often using language sampling and standardized checklists, while parents provided critical insights through questionnaires such as M-CHAT and ASQ [19]. These approaches enabled early detection of atypical speech patterns and social communication delays, forming the foundation upon which later automated and ML-based screening methods were developed [20].

2.7.2 Standardized Language Assessments

Structured, validated tools are widely used to assess expressive and receptive language skills. Instruments such as the Preschool Language Scale (PLS), Clinical Evaluation of Language Fundamentals (CELF), and MacArthur–Bates Communicative Development Inventories (CDI) evaluate vocabulary, sentence structure, comprehension, and pragmatic language use [19, 21]. These assessments provide clinicians with benchmarks relative to developmental norms, making them effective for identifying significant speech delays. However, these assessments often require trained professionals and controlled testing environments, which can limit accessibility in low-resource or remote settings [19].

2.7.3 Audiological Screening

Hearing impairment can mimic or exacerbate speech delays; therefore, audiological testing is routinely conducted to rule out auditory deficits. Ensuring that language difficulties are not attributable solely to sensory limitations is critical for differentiating between primary speech-language disorders and secondary delays caused by hearing loss [19].

2.7.4 Developmental Milestone Tracking

Tracking developmental milestones remains a widely used conventional approach. Typical expectations include the production of first words by approximately 12 months and combining words into simple phrases by 24 months [20]. While milestone tracking is simple and accessible, it lacks sensitivity to subtle variations in early speech and social communication development.

2.7.5 Practical Limitations

Despite their clinical value, traditional speech-based screening methods have well-documented limitations:

Table 2.3: Limitations of Conventional Speech-Based Screening Methods

Limitation	**Description**
Subjectivity	Heavy reliance on parental and clinician judgment introduces variability [23].
Accessibility	Requires trained professionals; limited availability in underserved areas [22].
Time and Cost	Comprehensive assessments are lengthy and expensive, limiting repeated use [22].
Delayed Diagnosis	Subtle early signs may go unnoticed until later childhood, missing critical intervention windows [20].

2.8 Speech-Based Screening (ML/DL Methods)

Machine learning (ML) and deep learning (DL) are increasingly applied to children's speech recordings to screen for autism spectrum disorder (ASD) and speech delays. Unlike questionnaires, these methods can analyze both what children say (linguistic content) and how they say it (voice quality and prosody). Researchers use different types of speech tasks as "tests" to extract meaningful features for ML models. The models then classify children as likely ASD or typically developing, often achieving high accuracy.

2.8.1 Natural / Conversational Speech Screening

One of the most well-known publicly available studies in this space is by Cho, Liberman, Ryant, Cola, Schultz & Parish-Morris (2019). They recorded brief, natural conversations (5 min) between children and a conversational partner on everyday topics (e.g., family, hobbies). From these conversations, they extracted a large set of acoustic features (352 features) and text / lexical features (272 features) from transcribed speech. They used gradient-boosting machines as their classifier. To avoid overfitting, they applied feature reduction (PCA) after removing irrelevant features [24].

Their final model performed "substantially above chance," showing that combining what children say plus how they say it can be informative for ASD screening in unstructured, real-world-style conversations [24].

2.8.2 Nonword Repetition / Controlled Speech Tasks

Nonword repetition tasks standardize the linguistic input, allowing models to focus on phonetic and articulatory patterns. Briend et al. (2023) had children repeat made-up words and extracted low-level acoustic features such as MFCCs, formants, jitter/shimmer, and harmonics-to-noise ratio (HNR). The data were analyzed using k-means clustering combined with an ROC-supervised classifier [25].

Performance: Accuracy reached 91% when distinguishing ASD from TD children, and approximately 85% when including a mixed-developmental control group [25].

2.8.3 Diagnostic / ADOS-Style Conversations

Semi-structured conversations during ADOS assessments provide both social and linguistic cues. Features such as prosody, pitch, speech rate, pauses, and volume are extracted. Using gradient boosting models, researchers classified ASD versus TD children [24].
Performance: The models consistently performed above chance, demonstrating that clinician-guided conversation recordings contain predictive features for ML-based screening [24].

2.8.4 Infant Cry Analysis

Early, nonverbal vocalizations like infant cries have also been explored. Chen et al. (2025) applied a Recursive Convolutional Neural Network (R-CNN) to infant cries, analyzing acoustic features including jitter, shimmer, HNR, pitch, and energy [26].
Performance: The R-CNN model achieved 90.28% accuracy with F1 = 91.75%, showing strong predictive potential for ASD in very young children [26].

2.8.5 Prosodic / Natural Speech Meta-Analysis

Meta-analyses show that prosodic features (pitch, intonation, rhythm) consistently differ in children with ASD. Features such as mean pitch, pitch range, and speech rhythm were examined across multiple studies [27].
Performance: Effect sizes for pitch-related measures were moderate-to-large, confirming that prosodic differences are robust and can inform feature selection for ML-based classifiers [27].

2.8.6 Multimodal / Fusion Approaches

Emerging approaches combine multiple audio features (MFCCs, HNR, GFCCs) with behavioral cues. Deep learning or ensemble ML methods (e.g., CNN + RF/SVM fusion) have been applied. These methods aim to leverage complementary information from multiple modalities to increase predictive accuracy [28].
Performance: Using multimodal fusion, studies have reported up to 97.5% accuracy in developmental disorder classification, which can be adapted for ASD screening [28].

2.9 Post Analysis Interventions

Following the identification of autism-related speech delays, a range of post-analysis interventions can support developmental progress in affected children. These interventions span

Table 2.4: Speech-Based ML/DL Screening Methods

Speech Task / Test	**ML / DL Model**	**Input Features**	**Population / Age**	**Accuracy / Performance**	**Reference**
Natural / Conversational Speech	Gradient Boosting	Acoustic + lexical/text features (ASR transcripts)	Preschoolers	Substantially above chance	Cho et al., 2019
Nonword Repetition / Controlled Speech	K-means + ROC-supervised classifier	MFCCs, formants, jitter/shimmer, HNR	Children (ASD vs TD)	91% (ASD vs TD), 85% mixed group	Briend et al., 2023
ADOS-Style / Diagnostic Conversations	Gradient Boosting	Prosody, pitch, rate, pauses, volume	ADOS Module 3/4	Significantly above chance	Cho et al., 2019
Infant Cry Analysis	Recursive CNN	Jitter, shimmer, HNR, pitch, energy	Infants 18–54 mo	90.28% accuracy, F1=91.75%	Chen et al., 2025
Prosodic Meta-Analysis	Feature Classifier	Pitch, intonation, rhythm	Children	Moderate-to-large effect sizes	Behav. Sci., 2024
Multimodal / Fusion	Deep learning / RF/SVM ensemble	MFCC, HNR, GFCC + behavioral cues	Children	97.5% accuracy	MDPI, 2025

traditional therapeutic approaches, behavioral frameworks, remote or technology-assisted delivery models, and modern AI-driven solutions. Each method targets communication, social engagement, and adaptive functioning through different mechanisms—whether therapist-led, parent-mediated, or digitally facilitated. The following subsections outline key evidence-based intervention categories shown to improve speech, language, and overall developmental outcomes in children with ASD.

2.9.1 Traditional Behavioral and Speech Therapy

A Karachi study reported that integration of speech therapy with occupational therapy led to better social, learning, and behavioral skills in autistic children [29]. These findings suggest that treating speech delay alongside other domains (e.g., motor skills, daily living activities) can yield broader developmental gains.

2.9.2 Applied Behavior Analysis (ABA)

Applied Behavior Analysis (ABA) remains the most extensively studied therapy. A scoping review found that ABA-based programs yielded improvements in language, cognition, social and communication skills in most participants. Gitimoghaddam et al. (2022) report that in ABA studies, improvements were observed across seven of the eight outcome measures (cognitive, language, social/communication, etc.) [30]. This shows ABA can significantly enhance speech and related development when implemented intensively by trained therapists.

2.9.3 Telehealth and Teletherapy

Systematic reviews indicate that telehealth-delivered interventions can be as effective as in-person therapy for improving communication skills. A recent meta-analysis found that telehealth social-communication interventions produced significant pre-post improvements in language skills, especially social language, in children with ASD [31]. In practice, speech-language pathologists report that telepractice offers substantial benefits; an Indian qualitative study found clinicians viewing teletherapy as advantageous for "the majority of children with ASD", provided training and technical support are in place [32].

2.9.4 Naturalistic Developmental Behavioral Interventions (NDBIs)

Naturalistic Developmental Behavioral Interventions (NDBIs) represent a well-supported, ecologically grounded class of early interventions for children with ASD. A study synthesized findings from multiple meta-analyses on NDBIs for autistic children under eight [33]. Across reviews, they found consistent positive effects on communication, language, cognition, and adaptive behavior, confirming NDBIs as a strong evidence-based practice for early intervention.

2.9.5 Gamified Learning

Interactive game-based interventions target motivation. Systematic reviews report that serious games improve social skills in ASD (e.g., emotion identification, turn-taking) [34]. These games often incorporate speech tasks (e.g., answering questions to progress, repeating phrases) and can enhance engagement. In practice, therapists may prescribe selected educational apps or game software as a supplement to therapy.

2.9.6 AI Based Applications

Artificial intelligence (AI) and educational apps are emerging tools in ASD therapy. One recent study evaluated CognitiveBotics, an AI-driven platform for structured learning plus standard therapy. Children who used the platform showed substantial gains versus controls: expressive

language increased by almost 60% (vs 40% in controls) and receptive language by 56% [35]. AI-powered and app-based interventions are promising supplements to conventional speech therapy, though they typically require oversight and integration with clinician-led work.

Chapter 3
System Features And Specifications

3.1 Introduction

The purpose of this Software Requirements Specification (SRS) is to provide a detailed, comprehensive, and unambiguous description of the requirements for the SpeechSpectrum system. This document serves as a blueprint for development, defining the functional and non-functional requirements necessary to create a mobile application designed for the early, proactive screening of Autism Spectrum Disorder (ASD)-linked speech delays in children.

3.2 Product Perspective and Users

3.2.1 Product Scope

The SpeechSpectrum application provides a two-phase screening process designed for parents/guardians:

1. **Phase 1: Questionnaire-Based Risk Assessment**: Utilizes a pre-trained classical machine learning model on standardized questionnaire items and demographic data to instantly predict the likelihood of ASD traits.
2. **Phase 2: Speech Spectrum Analysis**: Acquires a child's natural speech and processes it using Digital Signal Processing (DSP) and a Deep Learning model (CNN) to extract acoustic features and generate an objective ASD-linked speech pattern score.

The system's final output is a combined risk report, actionable intervention sessions, and a secure sharing mechanism for healthcare professionals.

3.2.2 User Classes and Characteristics

The system supports three distinct user classes:

User Class	Age/Use	Description	Key Characteristics
Parent/Guardian	Primary User	Individuals using the app to screen their child. This class requires a highly usable, intuitive mobile interface.	Sign up, Log in, Manage Child Profiles, Complete Assessment, Record Speech, View Results, Access Educational Content, Consult expert users.
Expert User (Therapist)	Healthcare Professional	Healthcare professionals who review the generated results. This user class requires authenticated access.	Sign up/Log in, Access shared results, View patient history, Manage therapy plans, Evaluate progress
Admin (System Administrator)	System Manager	Manages users, content, system settings, and ensures compliance.	Login, Manage User Accounts, Approve/Reject Expert Onboard

Table 3.1: User Classes and Characteristics

3.3 Functional Requirements

The functional requirements detail the required behavior of the system, organized by the user role performing the action.

3.3.1 Parent/Guardian Functions

The Parent/Guardian is the primary actor responsible for data input and viewing results.

1. **User Signup**: The system shall allow new users to register securely using email/password.
2. **User Login**: The system shall authenticate returning users based on stored credentials and use token-based validation for session management.
3. **Manage Child Profile**: The system shall allow a parent to create, edit, and securely store records for multiple children.
4. **Complete Questionnaire**: The system shall present the A1-A10 questionnaire (Q-CHAT items) and demographic fields in a guided, intuitive flow.
5. **Record Child Speech**: The system shall provide a clear, one-button interface to record and securely upload child vocalizations to cloud storage.

6. **Get Risk Assessment**: The system shall submit validated questionnaire data to the ML Service API and display the immediate, calculated risk level (Low/Moderate/High Chances).

7. **View Detailed Report**: The system shall display a comprehensive risk report presenting questionnaire-based and speech-based screening results.

8. **View Test History**: The system shall display a history of all previous tests performed.

9. **Access Educational Content**: The system shall provide a dedicated, easily navigable section featuring educational content (videos, articles) on early speech milestones and ASD awareness.

10. **Follow Post-Analysis Care and Re-Assessment**: For low risk cases, the system shall generate intervention activities and support re-assessment for progress comparison. For high-risk cases, the system shall recommend professional consultation by providing a list of Expert Users, enabling parents to select a doctor and schedule appointments for consultations.

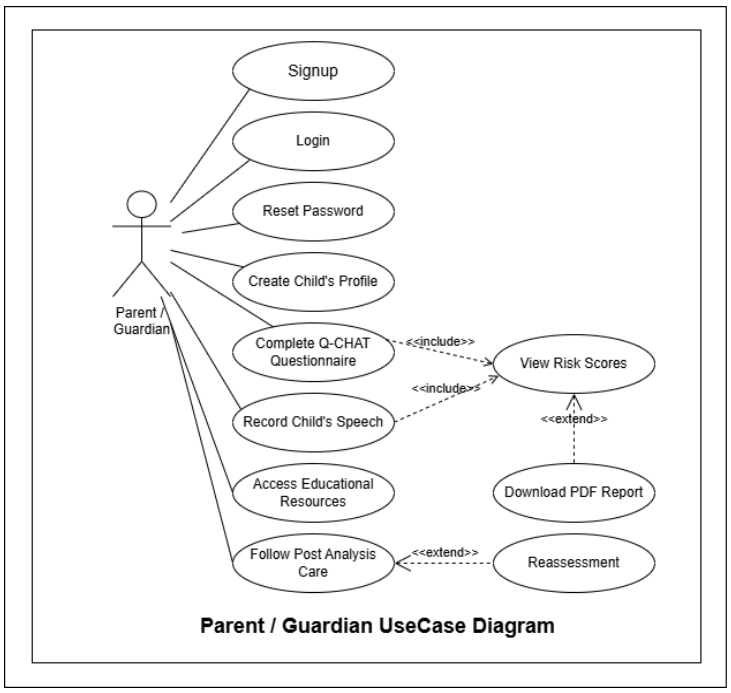


Figure 3.1: Parent Use Case Diagram

3.3.2 Expert User Functions

The Expert User is an authenticated specialist responsible for consulting with patients, reviewing shared screening results, planning intervention activities, and maintain appointment records.

1. **Expert Signup/Login**: The system shall allow Expert Users to register and log in via a dedicated, authenticated portal/interface.
2. **Access Child Profile/Results**: The Expert User shall be able to access and view the detailed child's profile and screening results.
3. **View Patient History**: The system shall allow the Expert User to view the complete history of tests and results for a patient whose data has been shared with them.
4. **Manage Appointments**: The system shall allow the Expert User to schedule, modify, and document appointment sessions for high risk patients.

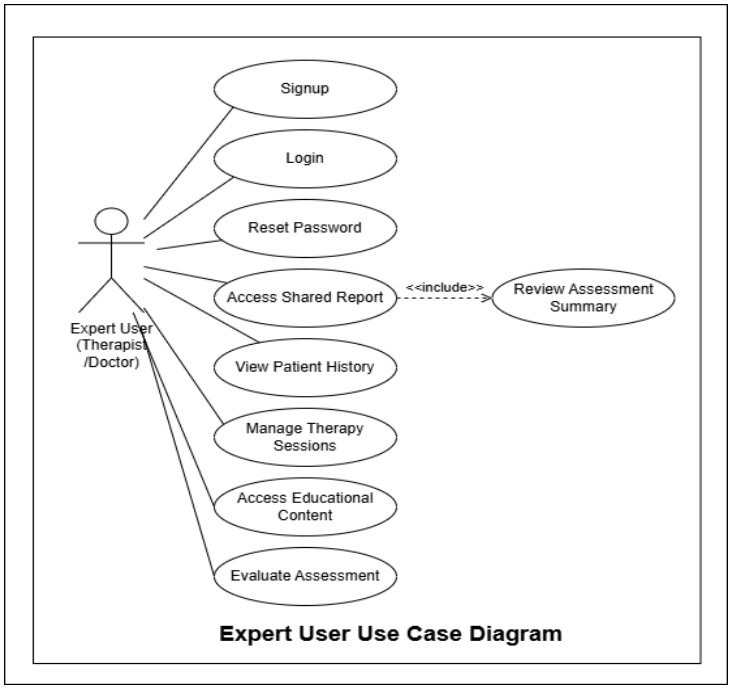


Figure 3.2: Expert User/Specialist Use Case Diagram

3.3.3 System Administrator

The System Administrator (Admin) is responsible for maintaining system integrity, managing user data, overseeing quality control, and configuring core services.

1. **Login as Admin**: The system shall allow the Admin User to securely authenticate into the dedicated administration portal using validated credentials.
2. **Manage User Accounts**: The system shall allow the Admin to view, suspend, or delete both Parent/Guardian and Expert User accounts.
3. **Handle Expert Registration**: The system shall provide a tool for the Admin to review expert user's registration application and accept or reject their onboarding.

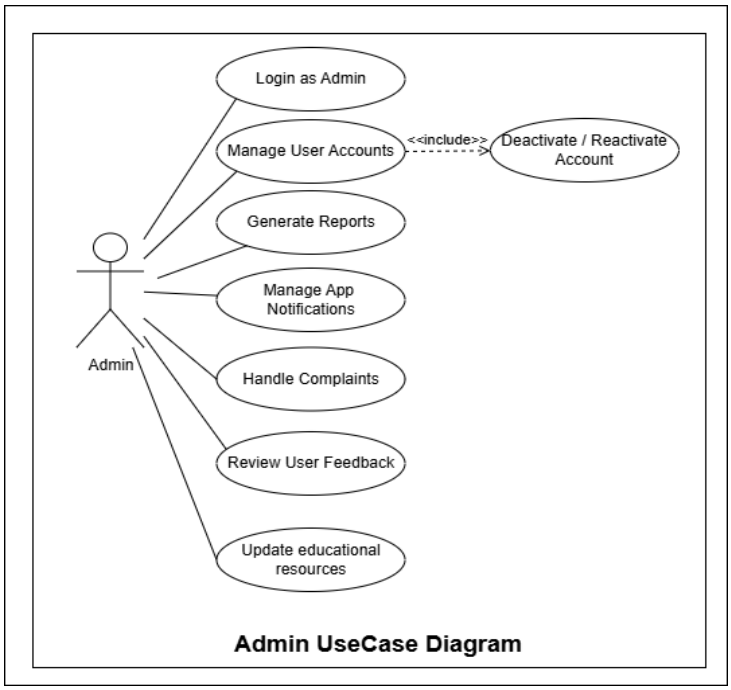


Figure 3.3: Administrator Use Case Diagram

3.4 Quality Factors

Quality Factors define the essential constraints and attributes that ensure the quality and effective operation of the system.

3.4.1 Performance Attributes

1. **ML Latency**: The system shall provide the risk assessment results within a reasonable time frame to ensure a smooth and responsive user experience.

2. **Scalability**: The system architecture shall be designed to scale efficiently to support increasing numbers of users and concurrent analysis requests without noticeable degradation in performance.

3. **App Response**: The application shall ensure prompt loading and interaction across key user interfaces, providing a responsive experience under typical network conditions.

3.4.2 Security Attributes

1. **Data Encryption**: All PII (Personal Identifiable Information) and stored speech recordings must be encrypted both in transit and at rest.

2. **Authentication**: The system must utilize secure token-based authentication (OAuth 2.0 / JWT) for all user sessions and API calls.

3.4.3 Usability (UI/UX) Attributes

1. **Interface**: The application must be visually simple, intuitive, and accessible to parents with varying technical expertise.
2. **Clarity**: Assessment results must be presented using clear, non-technical risk bands (Low, Moderate, High) alongside the numerical probability score.
3. **Feedback**: The system shall provide immediate visual feedback (e.g., loading indicators) during lengthy operations like speech file upload.

3.5 Operating Environment

- **Platform**: The system shall support both mobile applications (Android and iOS) and a web-based application.
- **Frontend**: The mobile application shall be developed using a cross-platform framework (Flutter), while the web application shall be developed using Next.js.
- **Backend & Data Storage**: The system shall utilize a cloud-based backend platform (e.g., Supabase) for user authentication, role-based access control, secure storage of user profiles, child data, screening results, and generated reports.
- **Machine Learning Service**: A dedicated machine learning backend service implemented using Python and Flask shall be deployed on a cloud server to perform signal processing, feature extraction, and inference using trained models (Random Forest and CNN).

3.6 Data Flow Diagram

3.6.1 Level 0 Data Flow Diagram

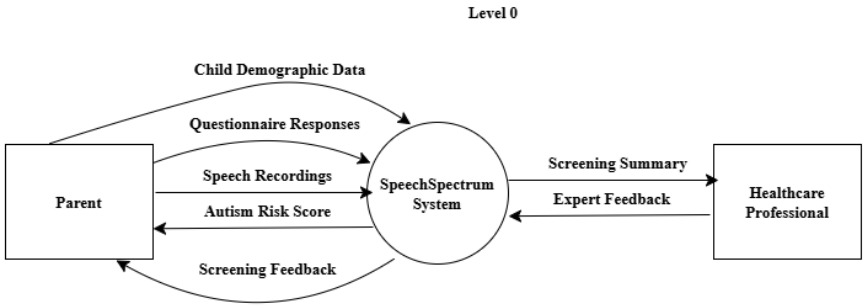


Figure 3.4: Level 0 DFD Diagram

3.6.2 Level 1 Data Flow Diagram

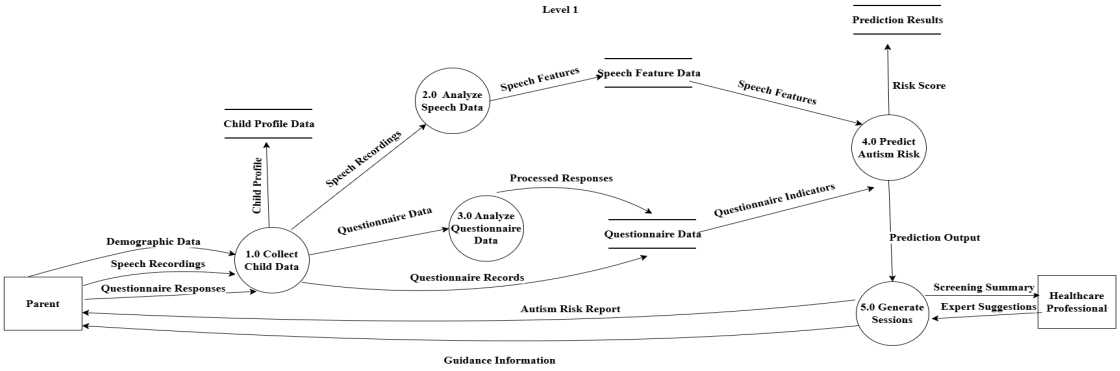


Figure 3.5: Level 1 DFD Diagram

3.7 Summary

The system features and specifications of SpeechSpectrum have been designed through careful architectural planning, appropriate technology selection, and a strong focus on usability and clinical relevance. By integrating a two-phase autism screening approach combining questionnaire-based machine learning analysis with speech-based deep learning techniques, the system delivers early, accessible risk assessment for children. The use of modern cross-platform and web technologies, including Flutter and Next.js, alongside a cloud-based backend powered by Supabase and a dedicated Python-based machine learning service, ensures scalability, security, and maintainability. Functional requirements were structured to clearly define the responsibilities of Parent/Guardian, Expert User, and Administrator roles, enabling seamless post-analysis intervention, progress tracking, and professional oversight. Emphasis on performance, security, and usability further ensures a responsive, secure, and intuitive experience. Collectively, these specifications demonstrate a structured and robust system design that supports early intervention and provides a strong foundation for future enhancements in autism screening and care.

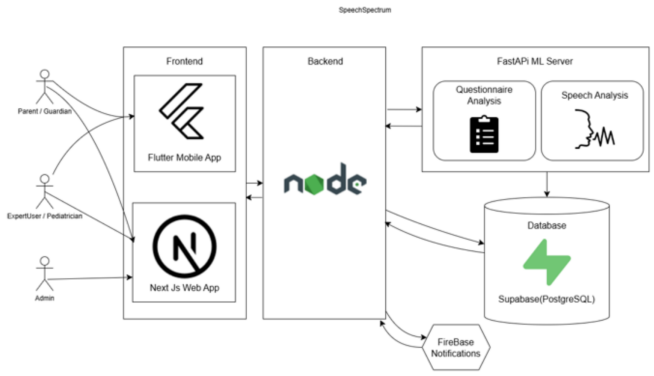
Chapter 4
Architecture And Design

4.1 Introduction

This chapter outlines the key components, modules, and functionalities of the system, along with the logical and physical structure.

4.2 Software Architecture

The SpeechSpectrum system is meant to be intelligent, easy to use, and useful for screening kids for Autism Spectrum Disorder (ASD) early on. The architecture diagram shows that the system is made up of many integrated modules and services for three main types of users: Parent / Guardian, Expert User / Pediatrician, and Admin / System Manager.



In overall, the proposed system architecture uses Flutter for the mobile front end, Next.js for the web front end, a Node.js back end for core application services and API management, and a FastAPI-based Machine Learning server for predicting ASD risk using questionnaire and speech analysis models. Supabase Database is used to keep data safe, and Firebase Notifications is used to send and receive real-time alerts. The architecture is designed to provide a reliable, scalable, and user-friendly platform for successful implementation of SpeechSpectrum.

4.2.1 Frontend System

The client-side system of SpeechSpectrum where interaction with the system takes place. It involves the Flutter Mobile Application and Next.js Web Application. Parents and experienced users can be able to interact using both mobile and web-based interfaces, whereas the admins primarily use the web-based interface. The purpose of the frontend system is to make sure

that there is an easy way of performing tasks such as login, taking questionnaires, recording speeches, and checking results.

4.2.2 Backend System

Backend System serves as the main processing unit for SpeechSpectrum. This system is created utilizing Node.js technology and takes care of all API calls, authentications, business logic, and communications within the frontend, database, and Fast API ML servers. This system is designed to ensure that there is a safe flow of data from different users, processes the reports, and sends notifications.

4.2.3 Machine Learning System

The Machine Learning System is developed for doing predictions of the risks of ASD. This system consists of two models: Random Forest that will analyze the questionnaire and CNN (Convolutional Neural Network) that will analyze speech/audio. The functionality of the module includes processing input data and outputting screening results like low risk, medium risk, and high risk.

4.2.4 Database System

The database and services system will serve as a place where application data is stored and managed in a secure way. Supabase Database is used for user account management, children's profiles, questionnaires, report creation, and predictions. Firebase Notifications is used for notifications. Its main function is to provide secure storage and quick access to services.

4.3 Database Design

The database design forms an essential part of the SpeechSpectrum application, ensuring that all data pertaining to users, children, ASD testing, scheduling, payments, and alerts can be efficiently stored and managed. The database design involves aspects such as organizing data in logical structures, the connections between different entities, and securing data. Proper database design guarantees efficient data storage and retrieval for entities within the system. One of the most important elements that should be considered when designing the database is data security. Supabase will provide access control, row-level security, and user authentication capabilities.

4.3.1 Entity Relationship Model

The entity-relationship model serves as a conceptual data model that can be used to illustrate the structure and relationship between data elements contained within the SpeechSpectrum database. This model provides a graphical representation of the entities, attributes, and relationships. The ER model is based on the database structure that runs on Supabase (PostgreSQL).

Entities

An entity is an object or a concept whose information is recorded in the database. An entity relates to an actual item that pertains to the SpeechSpectrum database. The entities shown in the ER diagram are shown below.

- profiles
- parents
- expert_users
- admins
- children
- child_health_profiles
- questionnaire_submissions
- questionnaire_results
- speech_submissions
- speech_results
- expert_locations
- appointment_slots
- appointments
- appointment_records
- payments
- notifications
- notification_schedule
- notification_preferences

- notification_deliveries
- User_push_tokens

Attributes

These attributes refer to the unique characteristics of entities. These attributes specify the type of data that is stored in the database about each individual entity.

profiles

- user_id: Unique identifier for each user, serves as the primary key and links to Supabase Auth.
- role: Enumerated role of the user — parent, expert, or admin.
- is_approved: Boolean flag indicating whether an expert account has been approved by an admin.
- created_at: Timestamp of when the profile was created.

parents

- user_id: Primary key, references the corresponding profiles record.
- full_name: Full name of the parent.
- phone: Contact phone number of the parent.
- created_at: Timestamp of record creation.

expert_users

- expert_id: Primary key, references the corresponding profiles record.
- full_name: Full name of the expert.
- specialization: Area of clinical specialisation (e.g. speech therapy, developmental paediatrics).
- organisation: Name of the institution or clinic the expert is affiliated with.
- contact_email: Expert's professional email address.
- phone: Contact phone number.

- pmdc_number: Unique Pakistan Medical and Dental Council registration number for identity verification.
- profile_image_public_id: Cloudinary public ID of the expert's profile photograph.
- degree_doc_public_id: Cloudinary public ID of the uploaded degree document.
- certificate_doc_public_id: Cloudinary public ID of the uploaded certification document.
- approval_status: Current approval state — pending, approved, or rejected.
- approved_by: UUID of the admin who approved or rejected the expert.
- approved_at: Timestamp of the approval decision.
- created_at: Timestamp of record creation.

admins

- admin_id: Primary key for the admin record.
- full_name: Full name of the administrator.
- created_at: Timestamp of record creation.

children

- child_id: Unique identifier for each child profile.
- parent_user_id: Foreign key referencing the parent who owns this child profile.
- child_name: Full name of the child.
- date_of_birth: Date of birth used for age calculation and WHO percentile computation.
- gender: Gender of the child — male, female, or other.
- created_at: Timestamp of record creation.

child_health_profiles

- profile_id: Unique identifier for the health profile.
- child_id: Foreign key referencing the associated child — unique, enforcing one profile per child.
- blood_group: Blood group of the child, constrained to standard ABO/Rh values.

- known_allergies: Free-text description of known allergies.
- family_history_asd: Boolean flag for family history of ASD.
- family_history_speech_disorders: Boolean flag for family history of speech disorders.
- family_history_hearing_loss: Boolean flag for family history of hearing loss.
- genetic_disorders: Free-text description of any known genetic conditions.
- chronic_conditions: Free-text description of chronic health conditions.
- weight_kg: Child's weight in kilograms.
- height_cm: Child's height in centimetres.
- bmi: Body mass index — a generated column computed automatically by PostgreSQL from weight and height.
- weight_percentile: WHO growth chart weight percentile recorded at time of measurement.
- height_percentile: WHO growth chart height percentile recorded at time of measurement.
- medical_records: JSONB array of uploaded medical document references.
- current_prescriptions: JSONB array of current prescriptions.
- created_at: Timestamp of profile creation.
- updated_at: Timestamp of last update.

questionnaire_submissions

- submission_id: Unique identifier for the questionnaire submission.
- parent_user_id: Foreign key referencing the parent who submitted the questionnaire.
- child_id: Foreign key referencing the child being screened.
- responses: JSONB object containing the parent's answers.
- submitted_at: Timestamp of submission.

questionnaire_results

- q_result_id: Unique identifier for the result record.
- submission_id: Foreign key referencing the associated questionnaire submission.
- parent_user_id: Foreign key referencing the parent.
- child_id: Foreign key referencing the child.
- result: JSONB object containing the Random Forest model's prediction output.
- generated_at: Timestamp of when the result was produced.

speech_submissions

- speech_submission_id: Unique identifier for the speech submission.
- parent_user_id: Foreign key referencing the submitting parent.
- child_id: Foreign key referencing the child being screened.
- recording_public_id: Cloudinary public ID of the uploaded audio recording.
- recording_duration_seconds: Duration of the audio recording in seconds.
- recording_format: File format of the recording.
- submitted_at: Timestamp of submission.

speech_results

- speech_result_id: Unique identifier for the speech result.
- speech_submission_id: Foreign key referencing the associated speech submission.
- parent_user_id: Foreign key referencing the parent.
- child_id: Foreign key referencing the child.
- result: JSONB object containing the CNN model's prediction output.
- generated_at: Timestamp of when the result was produced.

expert_locations

- location_id: Unique identifier for the location.
- expert_id: Foreign key referencing the expert.
- label: Human-readable label for the location.
- address: Full street address.
- city: City of the location.
- maps_url: Optional Google Maps URL.
- is_active: Boolean flag.
- created_at: Timestamp of record creation.

appointment_slots

- slot_id: Unique identifier for the slot.
- expert_id: Foreign key referencing the expert.
- slot_date: Date of the appointment slot.
- start_time: Start time of the slot.
- end_time: End time of the slot.
- mode: Consultation mode.
- fee_online: Fee for online consultation.
- fee_physical: Fee for physical consultation.
- currency: Currency denomination.
- location_id: Optional foreign key.
- status: Availability status.
- is_recurring: Boolean flag.
- recurrence_rule: Recurrence pattern.
- created_at: Timestamp.
- updated_at: Timestamp.

appointments

- appointment_id: Unique identifier.
- slot_id: Foreign key.
- expert_id: Foreign key.
- parent_id: Foreign key.
- child_id: Foreign key.
- booked_mode: Mode selected.
- fee_charged: Fee amount.
- currency: Currency.
- scheduled_at: Datetime.
- duration_minutes: Duration.
- status: Lifecycle status.
- meet_link: Zoom link.
- cancelled_by: Cancellation initiator.
- cancellation_reason: Reason.
- cancelled_at: Timestamp.
- payment_status: Payment state.
- created_at: Timestamp.
- updated_at: Timestamp.

appointment_records

- record_id: Unique identifier.
- appointment_id: Foreign key.
- notes: Clinical observations.
- therapy_plan: Therapy plan.
- medication: JSONB object.
- progress_feedback: Assessment.
- created_at: Timestamp.

payments

- payment_id: Unique identifier.
- appointment_id: Foreign key.
- parent_user_id: Foreign key.
- safepay_tracker: Tracker token.
- safepay_reference: Reference code.
- amount: Amount in paisa.
- currency: Currency.
- status: Payment status.
- metadata: JSONB field.
- created_at: Timestamp.
- updated_at: Timestamp.

notifications

- notification_id: Unique identifier.
- recipient_user_id: Foreign key.
- event_type: Event classification.
- title: Notification title.
- body: Message body.
- entity_type: Related entity type.
- entity_id: UUID.
- deep_link: Mobile navigation link.
- web_path: Web route.
- is_read: Boolean.
- read_at: Timestamp.
- clicked_at: Timestamp.
- payload: JSONB object.
- created_at: Timestamp.

notification_schedule

- schedule_id: Unique identifier for the scheduled notification.
- recipient_user_id: Foreign key referencing the intended recipient.
- event_type: Type of the scheduled event.
- title: Notification title.
- body: Notification message body.
- entity_type: Type of the related entity.
- entity_id: UUID of the related entity.
- deep_link: Mobile deep link for navigation.
- web_path: Web route for navigation.
- payload: JSONB object with additional context for the notification.
- scheduled_for: Datetime at which the notification should be dispatched.
- status: Processing status — pending, sent, failed, or cancelled.
- processed_at: Timestamp of when the scheduler processed this record.
- dedupe_key: Unique key used to prevent duplicate scheduled notifications for the same event.
- created_at: Timestamp of record creation.

notification_preferences

- preference_id: Unique identifier for the preference record.
- user_id: Foreign key referencing the user — unique, enforcing one preferences record per user.
- push_enabled: Master toggle for all push notifications.
- appointments_enabled: Toggle for appointment-related notifications.
- payments_enabled: Toggle for payment-related notifications.
- reminders_enabled: Toggle for appointment reminder notifications.
- records_enabled: Toggle for clinical record update notifications.

- assessments_enabled: Toggle for screening result notifications.
- expert_approval_enabled: Toggle for expert account approval notifications.
- quiet_hours_enabled: Boolean flag enabling the quiet hours feature.
- quiet_hours_start: Time at which quiet hours begin — no notifications sent after this time.
- quiet_hours_end: Time at which quiet hours end.
- timezone: User's timezone used for quiet hours calculation.
- created_at: Timestamp of record creation.
- updated_at: Timestamp of last update.

notification_deliveries

- delivery_id: Unique identifier for the delivery attempt.
- notification_id: Foreign key referencing the parent notification.
- token_id: Foreign key referencing the push token the notification was sent to.
- provider: Delivery provider used — fcm.
- provider_message_id: Message ID returned by Firebase Cloud Messaging.
- status: Delivery status — sent, delivered, failed.
- error_code: Error code returned by the provider on failure.
- error_message: Descriptive error message on failure.
- sent_at: Timestamp of the delivery attempt.
- created_at: Timestamp of record creation.

user_push_tokens

- token_id: Unique identifier for the push token record.
- user_id: Foreign key referencing the registered user.
- fcm_token: Firebase Cloud Messaging device token — unique across the system.
- platform: Device platform — ios, android, or web.
- app_name: Name of the application that registered the token.

- device_id: Optional device identifier for tracking.
- device_name: Optional human-readable device name.
- app_version: Version of the app that registered the token.
- is_active: Boolean flag indicating whether the token is still valid and active.
- last_seen_at: Timestamp of the most recent successful delivery to this token.
- created_at: Timestamp of token registration.
- updated_at: Timestamp of last update.

Relationships

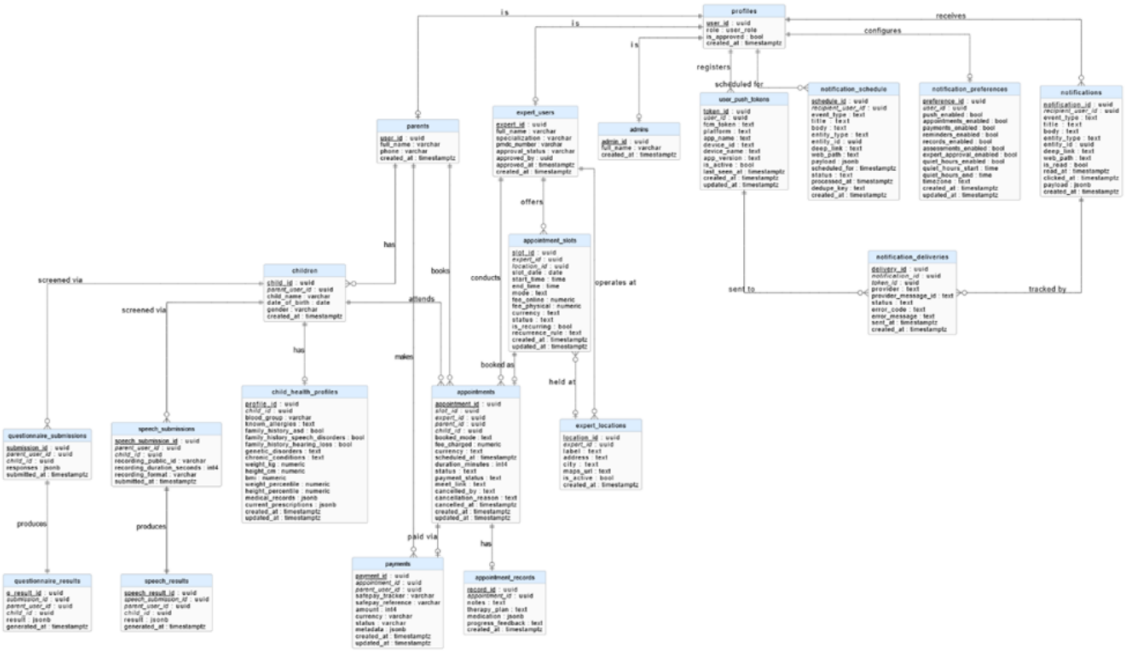
In the ER model, relationships are the associations between entities. They specify how entities relate to one another and set the logical dependencies among the components of the database.

- One profiles record maps to exactly one of parents, expert_users, or admins, depending on the user's role.
- One parent may have many children.
- One child has at most one child_health_profiles record.
- One child may have many questionnaire_submissions and many speech_submissions.
- One questionnaire_submission produces exactly one questionnaire_results record.
- One speech_submission produces exactly one speech_results record.
- One expert_user may have many expert_locations.
- One expert_user may offer many appointment_slots.
- One appointment_slot belongs to at most one expert_location.
- One appointment_slot is consumed by at most one appointment.
- One appointment is conducted by one expert_user and booked by one parent for one child.
- One appointment may have at most one appointment_records record.
- One appointment may have at most one payments record.
- One parent may make many payments.

- One profiles record may receive many notifications.
- One profiles record may have many notification_schedule entries.
- One profiles record has at most one notification_preferences record.
- One profiles record may register many user_push_tokens.
- One notification may have many notification_deliveries records tracking delivery to each device.
- One user_push_token may be associated with many notification_deliveries.

ER Diagram

The ER diagram is the graphic representation of all the entities involved, the main characteristics of those entities, and the relationships existing between them. The entities are presented as boxes carrying the name of the entity inside them.
Each entity box contains a list of the attributes associated with the entity, along with the data types. The relationship is indicated using an edge connecting the two entities.



Schema Design Components

The schema design defines the logical structure of the SpeechSpectrum database, including tables, attributes, relationships, constraints, and data types. It ensures that the database is organized, scalable, and optimized for efficient data storage and retrieval. The schema is imple-

mented using Supabase (PostgreSQL), which supports relational integrity and advanced security features.
The schema is divided into multiple interconnected components representing different functional modules of the system.

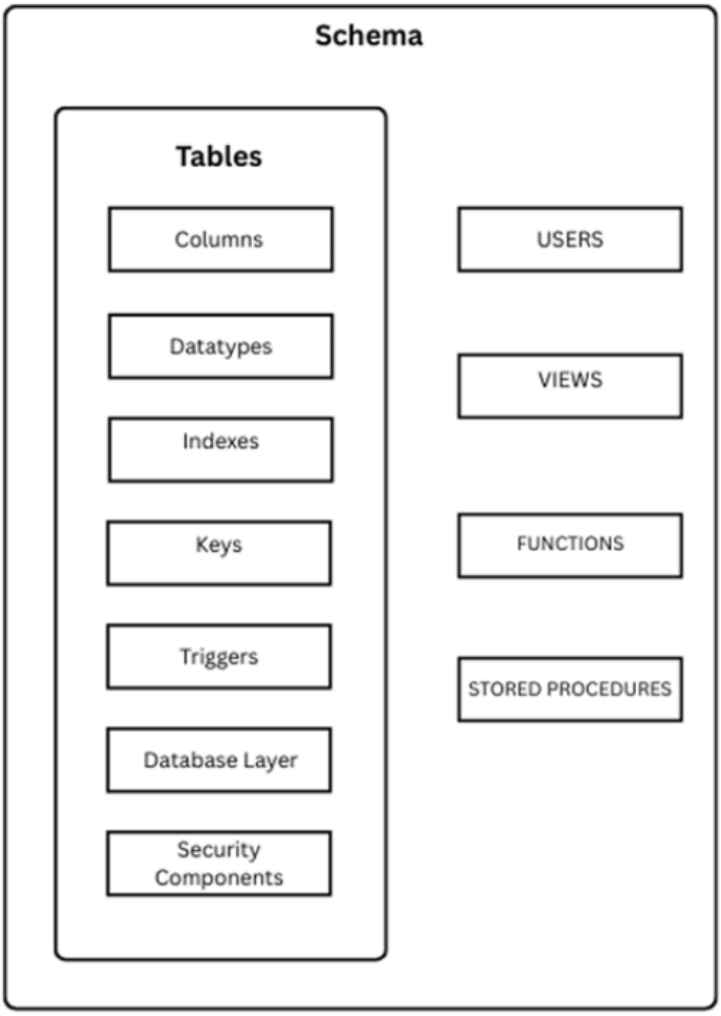


Figure 4.1: Schema Design of SpeechSpectrum System

Tables

The tables are the main data structures that help arrange information within the SpeechSpectrum database. Every table stands for some real-life object related to the application. Each column in the tables is of some specific type and has its constraints. The primary keys of the tables are always UUIDs obtained using the gen_random_uuid() command provided by PostgreSQL. Also, each table has foreign keys and, thus, refers to other tables in terms of relations between their records. In addition, cascade deletes can be performed in the case of deleting any parent record.

Data Types

At SpeechSpectrum, several PostgreSQL data types have been used, depending on the type of intrinsic properties each data possesses. Globally, universally unique identifiers are used for the creation of primary keys and foreign keys. VARCHAR will be used in situations where there will be some length requirements. TEXT data type is used for unstructured data. NUMERIC is used for money and measuring purposes in cases where accuracy of decimal is important. For integer counting, INT4 will be used. BOOLEAN will be used for binary values. DATE and TIME data types will be used for scheduling. TIMESTAMP WITH TIME ZONE data type will be used for storing audit log information, whereas JSONB will be used for storing nested data.

Indexes

In order to have the database execute faster, indexes are made on foreign key columns that are often used in queries. Some important indexes are `parent_user_id`, `child_id`, `expert_id`, `status`, `safepay_tracker`, `recipient_user_id`, and `user_id`. These indexes make searches faster, reduce the time it takes to run queries, and improve the way the app accesses data that is often used.

Keys

Each table uses a UUID primary key, while foreign keys maintain relationships between related tables. Unique constraints enforce one-to-one mappings on fields such as child_id, user_id, fcm_token, and pmdc_number. A composite unique constraint on dedupe_key in notification_schedule prevents duplicate scheduled notifications for the same event.

Triggers

Database triggers automatically update timestamps and calculated values. BEFORE UPDATE triggers on child_health_profiles and notification_preferences automatically update the updated_at timestamps every time a record changes. The bmi field in child_health_profiles is a generated column that is automatically calculated from weight_kg and height_cm. This makes sure that the BMI values are always correct and consistent.

Views

In Supabase, Row-Level Security (RLS) policies limit who can see data at the database level. Parents can only see their own records, experts can only see appointments and related child data that they are in charge of, and admins can see more. This makes sure that data is kept safe, no matter what the application logic is.

Stored Procedures and Functions

PostgreSQL functions do things at the database level, like gen_random_uuid() for primary keys and a reusable trigger function to keep track of updated_at timestamps. A PostgreSQL expression is used to calculate BMI in a generated column. The Express.js backend takes care of business logic like calculating WHO percentiles, Safepay payment sessions, and Firebase notifications. This keeps the database and application responsibilities separate.

4.4 Backend Design

The backend of SpeechSpectrum has been implemented on Node.js with the Express.js framework through ES Modules. The system follows a modular design with controllers. The backend runs as a serverless app on Vercel and talks to the Supabase hosted PostgreSQL database. The backend handles all business logic, data processing, AI models, payments, and notifications. Such an architecture makes it scalable and easy to maintain for handling API calls, implementing access control, and interacting with external services.

Backend Design Components

The backend architecture incorporates several key components to ensure a robust, secure, and extensible system. These components include routing, middleware, controllers, authentication, validation, database interaction, external service integrations, and a notification dispatch system. The building blocks of the SpeechSpectrum backend are described below.

Backend Design Description

The figure illustrates both the internal and external components of the backend system. The internal components are the structural elements of the Express.js application. External components include Supabase for database and authentication, Cloudinary for file storage, Safepay for payment processing, Zoom for meeting link generation, and Firebase Cloud Messaging for push notification delivery.

Routes

The application defines route modules for each domain area of the system. Each route module maps HTTP methods and URL paths to the appropriate controller method. Routes are registered in the central `app.js` file and are grouped by resource.

Middleware

Middleware functions intercept incoming requests before they reach the route handler. The primary middleware used in SpeechSpectrum is the `supabaseClientMiddleware`, which creates

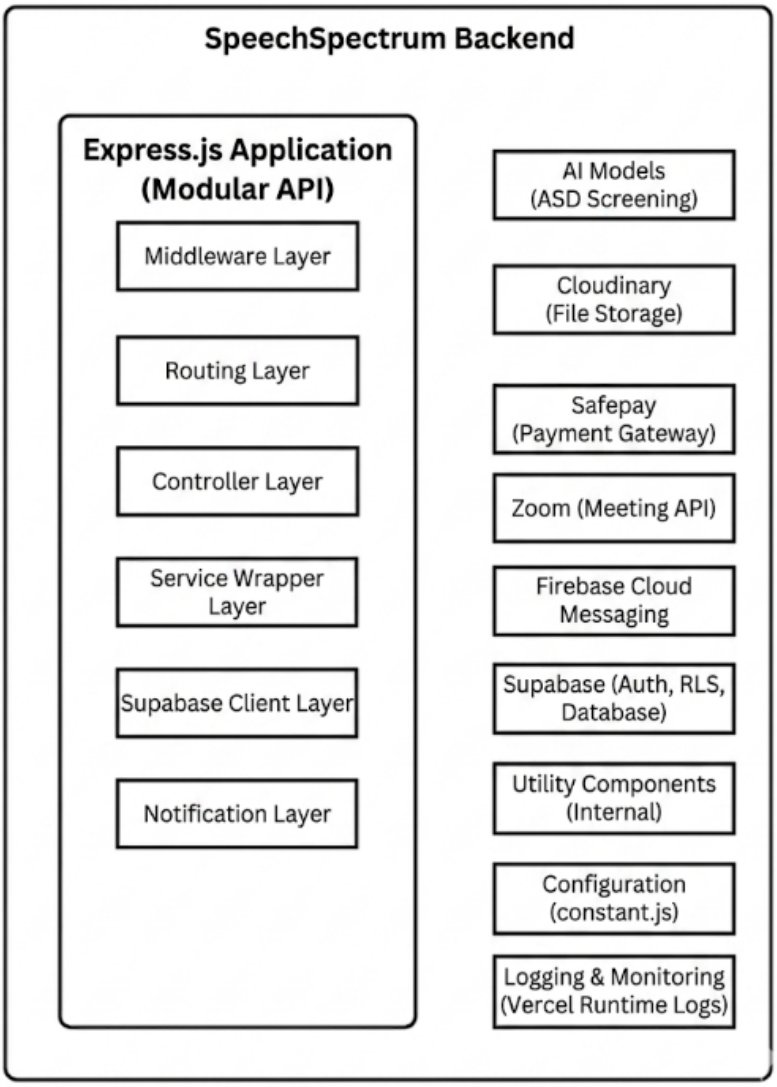


Figure 4.2: Backend Architecture Design of SpeechSpectrum System

a per-request Supabase client instance using the Authorization Bearer token from the request headers. Additional middleware includes Multer configured with Cloudinary storage for file upload interception on routes that accept audio recordings, images, and documents.

Controllers

Controllers are the core processing units of the backend. Each controller is implemented as an ES module class with static async methods, one per endpoint. Controllers receive the parsed request, execute the necessary business logic, interact with Supabase for data operations, and return a structured JSON response. The controller classes in SpeechSpectrum are organised by domain, and every controller method follows a consistent pattern: authenticate the user, check role permissions, validate inputs, perform database operations, and return a response with a status boolean field.

Authentication

Authentication is fully delegated to Supabase Auth. The backend does not implement or manage any custom JWT logic. Every protected route extracts the Bearer token from the Authorization header and passes it to `supabase.auth.getUser()` to resolve the authenticated user's identity. If the token is missing, expired, or invalid, the request is rejected with a 401 Unauthorized response.

Role-Based Access Control

Role-based access control is enforced within each controller method. After authenticating the user, the controller queries the profiles table to retrieve the user's role (parent, expert, or admin). The role is then checked against the requirements of the requested operation. This check is performed explicitly in every controller method that has role restrictions, ensuring that no role-sensitive operation can be accessed through an alternative code path.

Validation

Request validation is performed within controller methods before any database interaction occurs. Incoming request bodies and parameters are checked for the presence of required fields, correct data types, and valid enumerated values.

Database Interaction

All database operations are performed through the Supabase JavaScript client using its query builder API. The client provides methods for select, insert, update, upsert, and delete operations with support for filtering, ordering, pagination, and joins. Each query destructures a { data, error } response and checks for errors before proceeding.

AI Model Integration

The backend integrates with two external AI microservices for ASD screening. The questionnaire-based screening endpoint forwards the parent's Q-CHAT responses to a hosted Random Forest model and returns the risk score and classification. The speech-based screening endpoint uploads the audio recording to Cloudinary, then forwards the file to a hosted CNN model. Both integrations are wrapped in try-catch blocks, and the submission record is persisted to the database regardless of whether the prediction call succeeds, ensuring no data is lost due to a temporary model service outage.

File Storage

File storage is handled through Cloudinary, integrated into the Express.js request pipeline using multer-storage-cloudinary. Three Multer storage configurations are defined: `speechRecordingStorage` for audio files with a 30MB limit, `imageStorage` for profile photographs with a 10MB limit, and `documentStorage` for medical and credential documents with a 50MB limit. Upon a successful upload, `req.file.filename` contains the Cloudinary public ID, which is stored in the database for later retrieval and deletion.

Payment Integration

Payment processing is handled through the Safepay payment gateway. The payment flow consists of three steps executed by the backend: creating a payment session token, building a hosted checkout URL, and verifying payment completion after the user returns from the checkout page. Authoritative payment confirmation is handled through a Safepay webhook, which is verified using an HMAC-SHA512 signature computed over the raw request body using the webhook secret.

Zoom Integration

For online appointments, the backend integrates with the Zoom API to generate meeting links at the time of booking. A server-to-server OAuth token is obtained using the Zoom Account ID, Client ID, and Client Secret stored in environment variables. The token is used to call Zoom's meeting creation endpoint, and the resulting join URL is stored in the `meet_link` field of the appointment record.

Push Notification System

Push notifications are dispatched through Firebase Cloud Messaging. The backend maintains a `user_push_tokens` table where each device token registered by a Flutter or web client is stored against the owning user. When a notification event is triggered, the backend looks up all active tokens for the recipient user and sends a notification payload to each token. Notification

records are written to the `notifications` table for in-app inbox display, and delivery outcomes are recorded in the `notification_deliveries` table for each token.

4.5 Frontend Design

The frontend architecture of the system is developed using React with Next.js, enabling a scalable, responsive, and high-performance web-based platform. The design follows a role-based architecture to support multiple user types, including parents, therapists, and administrators. The system emphasizes user-centric design, modularity, and real-time interaction, ensuring smooth navigation and efficient access to AI-driven autism screening and analysis tools. The frontend communicates with backend services through RESTful APIs, enabling secure and structured data flow. The overall design focuses on delivering an intuitive user experience across all modules while maintaining consistency, accessibility, and performance optimization.



Figure 4.3: Frontend Architecture Design of SpeechSpectrum System

4.5.1 Web Application Design Components

The Web application frontend system is built using modular and reusable design principles. Key components include navigation structure, API integration, state handling, and UI components. These elements collectively ensure maintainability, scalability, and performance efficiency.

Navigation

Navigation is implemented using the Next.js routing system, enabling seamless transitions between different sections of the application. The system supports role-based navigation structures:

- Parent Interface: Screening, results, and child progress tracking
- Therapist Dashboard: Patient overview, assessment review, and case monitoring
- Admin Panel: System management, user control, and analytics

API Client

The frontend communicates with backend services using the Fetch API / Axios, enabling secure and efficient data exchange. It handles questionnaire submission, ML prediction requests, result retrieval, authentication, and session validation. This ensures smooth integration between frontend interfaces and backend AI/ML services.

State Management

Application state is managed using React Context API, ensuring lightweight and efficient global state handling without unnecessary complexity. It manages key states such as user authentication status, active child profile selection, questionnaire progress tracking, and result and session data. This ensures consistency across components and prevents data redundancy.

UI Components

The system is built using a component-based architecture, where each UI element is modular and reusable. Examples include header, footer, questionnaire cards, result visualization panels, dashboard widgets, navigation bars, and modals. This approach improves maintainability, scalability, and development efficiency.

4.5.2 Mobile Application Design Components

The mobile application of the FYP is built using Flutter, a cross-platform framework that enables a consistent and responsive mobile UI. It supports modular widget-based development,

ensuring smooth performance, reusable components, and a uniform design across Android and iOS devices.

Visual Styling System

The application enforces a consistent visual identity through a global theme and a small design system: a primary color palette, standardized typography scales, and spacing rules. Visual constants (colors, sizes) and shared styles are defined centrally so components inherit consistent colors, font sizes, and paddings, ensuring a cohesive look across screens and user roles.

Screen Layout Structure

Screens follow a predictable composition pattern—top app bar (or header), a primary content region (cards, lists or forms), and context-sensitive action areas—so users can quickly locate information and actions. Layout primitives are designed with mobile constraints in mind (single-column, scrollable content, clear visual grouping) and use consistent margins and card styling to improve readability and reduce visual clutter.

Navigation Structure

Navigation is implemented as a route-driven architecture that groups related screens into logical flows (onboarding/auth, parent, expert). Routes and navigation rules are centralized so transitions, deep links, and role-based landing screens are handled consistently. This separation keeps navigation logic out of UI code and makes it straightforward to add or reorganize flows without changing screen implementations.

Reusable Component Architecture

UI elements are built as composable, reusable widgets (buttons, input fields, cards, list items, and navigation shells) and consume shared style tokens. This component-first approach reduces duplication, enforces accessibility and interaction patterns uniformly, and accelerates UI changes.

Mobile Interaction and Responsiveness

Design decisions prioritize touch ergonomics and mobile constraints: controls use comfortable hit targets, primary actions appear within thumb reach, and layouts adapt to smaller screens using scrolling and vertical stacking. The app locks to portrait orientation and uses spacing, feedback states, and concise content to minimize cognitive load, producing a reliable, easy-to-use experience on handheld devices.

4.6 ML Model Design

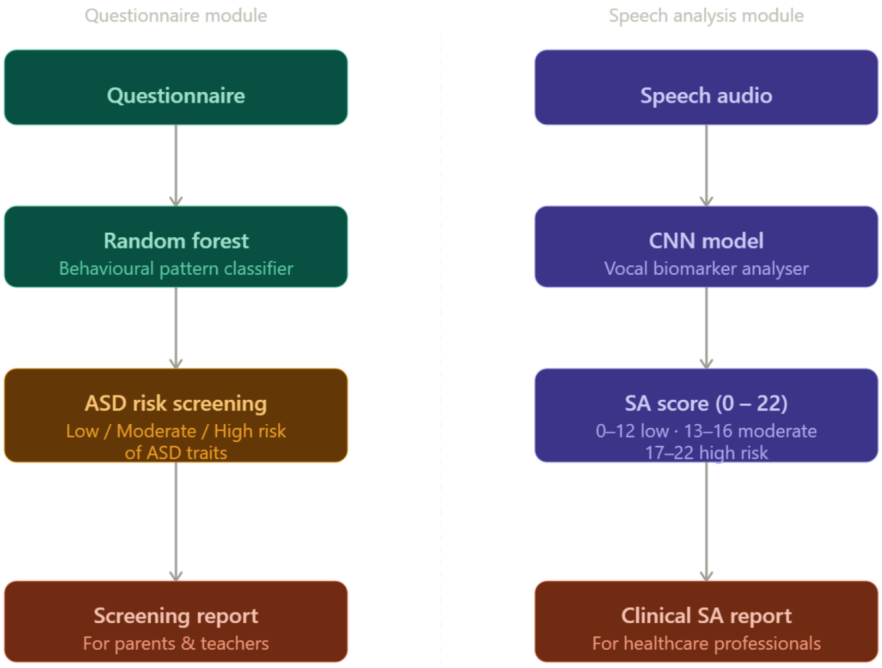


Figure 4.4: Machine Learning Model Architecture for SpeechSpectrum System

4.6.1 Random Forest for Questionnaire

The questionnaire screening module was developed using a Random Forest classifier to identify potential autism risk from caregiver responses. It analyzes structured behavioural and developmental answers and predicts whether the response pattern indicates typical development or ASD-related traits.

Questionnaire Features and Preprocessing

The questionnaire data is converted into numerical form so it can be processed by the model. Features include social interaction, communication, behavioural patterns, and developmental indicators. Missing values are handled and responses are encoded into a consistent format before training and prediction.

Random Forest Model

Random Forest was selected because it works well with tabular data and reduces overfitting. It builds multiple decision trees from different subsets of the data, and each tree contributes a vote. The final result is produced through majority voting, making the prediction more stable and reliable.

Screening Output

The model outputs a screening result indicating either low risk or high risk of autism. This helps provide an early decision-support tool for parents, teachers, and healthcare professionals. The prediction is not a diagnosis, but it can support further clinical evaluation.

4.6.2 CNN Model for Speech Analysis

To evaluate a child's autism severity through speech, a Convolutional Neural Network (CNN) was deployed. This section breaks down the complete architecture and design of the machine learning pipeline, which is specifically trained to predict a Social Affect (SA) score (ranging from 0 to 22).

Introduction to the Speech Analysis Model

Traditional speech AI focuses on Speech-to-Text (understanding what words are said). This model acts as a digital biomarker that analyzes how a child speaks. Children with Autism Spectrum Disorder (ASD) often exhibit unique vocal patterns such as monotone pitch, unusual rhythm, or lack of vocal emotion. The CNN is designed to analyze raw audio and assign a clinical Social Affect (SA) risk score based purely on voice quality.

The 49 Acoustic Features

The CNN does not process raw audio files directly. Instead, a Python-based audio processing pipeline extracts 49 specific vocal measurements from each audio clip. These include:

- Pitch & Intonation: Measures voice height, monotony, and variation.
- Vocal Energy: Captures changes in speech rhythm and loudness.
- Voice Quality (Jitter & Formants): Measures vocal cord instability and resonance characteristics.

These 49 features transform audio into a structured numerical representation for model processing.

Temporal Data Structuring

The CNN processes data in a fixed input shape of 100×49 (100 time frames and 49 features). An Adaptive Temporal Drift algorithm segments the audio into overlapping windows and maps them into sequential frames. This creates a temporal representation of speech dynamics for deep learning analysis.

Convolutional Neural Network (CNN) Architecture

Once structured, the data is fed into the CNN, which treats it as a 2D matrix.
The model:

- Uses convolution filters to detect hidden vocal patterns.
- Learns anomalies such as irregular pitch drops or energy shifts.
- Connects short-term vocal variations with overall emotional tone.

This allows the model to distinguish between typical and ASD-related speech patterns.

Predicting the Social Affect (SA) Score

Instead of binary classification, the model uses a regression-based output layer to predict a continuous Social Affect (SA) score between 0 and 22:

- 0 – 12: Low Risk (Typical development)
- 13 – 16: Moderate Risk (Some atypical vocal traits)
- 17 – 22: High Risk (Strong ASD speech markers)

To improve reliability, Ensemble Averaging is applied across 25 trained model configurations. The final SA score is the average of all predictions, reducing noise and improving stability.

4.7 Summary

This chapter details the architecture of the SpeechSpectrum system, an intelligent platform for early Autism Spectrum Disorder (ASD) screening. It employs a Flutter/Next.js frontend, Node.js backend, and Supabase database. The machine learning component uses Random Forest for questionnaire analysis and a Convolutional Neural Network (CNN) for speech analysis, predicting ASD risk via a Social Affect (SA) score ranging from low, moderate to high risk.

Chapter 5
Construction and Development

5.1 Introduction

This chapter details the robust construction and technical stack of the SpeechSpectrum system. It outlines the database, layered Node.js backend, cross-platform Next.js/Flutter frontends, and the training of machine learning models for questionnaire-based screening and acoustic speech analysis.

5.2 Database Setup

5.2.1 Tools Used

The choice of database tools was driven by the need for reliability, developer productivity, real-time capability, security, and seamless integration with the Node.js backend.

- **Supabase** – Selected as the primary database platform for SpeechSpectrum. Supabase is an open-source Backend-as-a-Service built on top of PostgreSQL, providing a fully managed relational database with built-in authentication, row-level security, real-time subscriptions, and a REST API generated automatically from the schema.
- **PostgreSQL** – The underlying relational database engine managed by Supabase. PostgreSQL was selected because of its robust support for advanced data types including JSONB for flexible structured storage.
- **Supabase Dashboard** – Its Table Editor is used as the primary visual interface for creating and inspecting tables during development. It provides a spreadsheet-like view of each table's data. The SQL Editor is used to execute raw SQL for operations including creation of tables with generated columns, definition of check constraints and unique constraints, creation of indexes, and registration of trigger functions.
- **Supabase Row-Level Security (RLS)** – Used to enforce data access policies directly at the database layer. RLS policies are defined per table and evaluated for every query.
- **Cloudinary** – Used as the file storage layer for all binary assets associated with database records. Expert profile images, degree and certificate documents, child speech recordings, and child medical report documents are stored on Cloudinary.

5.2.2 Methodology

The database configuration and management methodology for SpeechSpectrum focuses on security, referential integrity, maintainability, and consistency across development and production environments.

Schema Design and Creation

The database schema was designed upfront based on the system's entity model, with all tables, relationships, constraints, and indexes defined before application development began. Tables were created by executing SQL DDL statements directly in the Supabase SQL Editor. Each table uses a UUID primary key generated by PostgreSQL's `gen_random_uuid()` function.

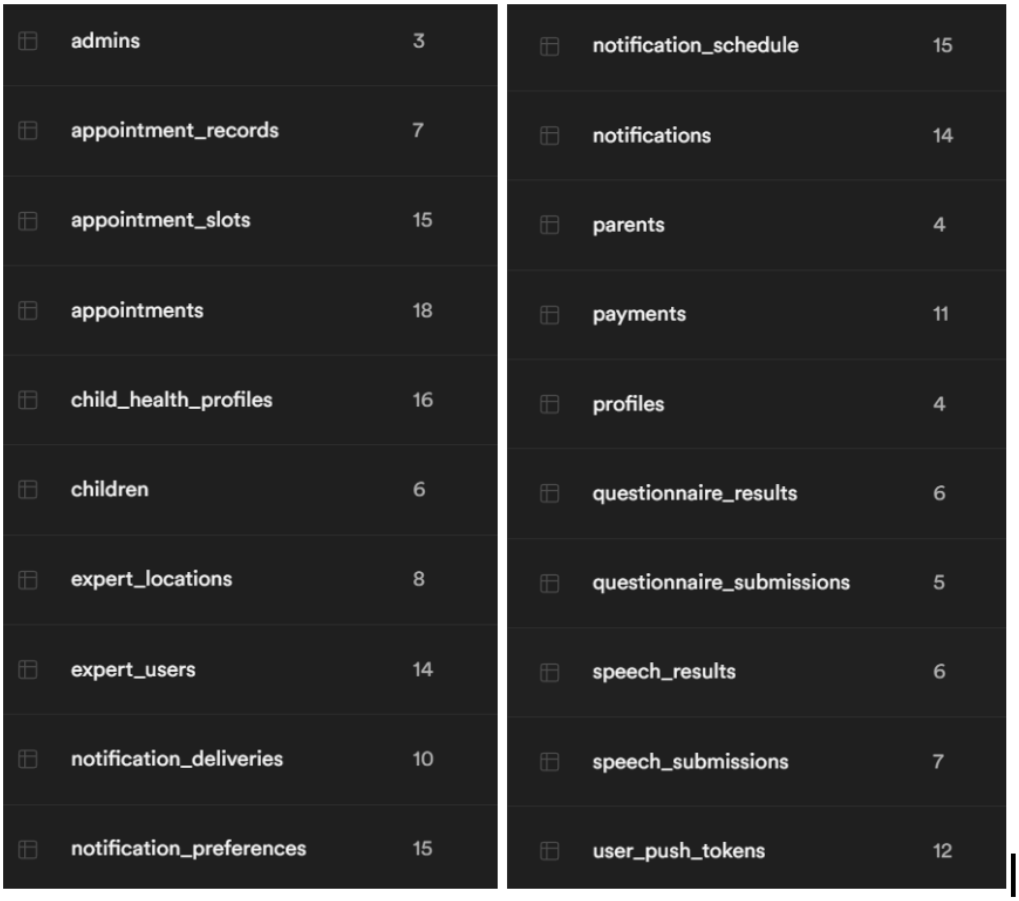


Figure 5.1: Database Tables

Row-Level Security

Row-Level Security (RLS) policies were enabled on all tables containing user-specific data immediately after table creation. Policies were defined using Supabase's policy editor and written in PostgreSQL policy syntax. Each policy specifies the operation it applies to (SELECT,

INSERT, UPDATE, DELETE), the role it targets (authenticated), and the condition that must be satisfied for the operation to be permitted.

Data Integrity Enforcement

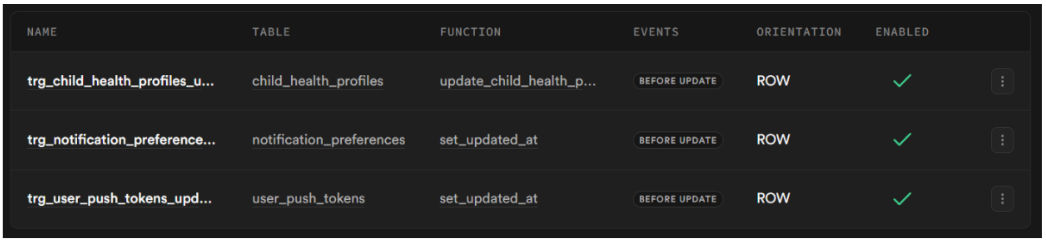
Data integrity is enforced through a combination of database-level constraints and application-level validation. Check constraints restrict enumerated fields to their valid values. These constraints ensure that invalid data cannot be persisted even if the application-layer validation is bypassed.

Indexes

Indexes were created on foreign key columns that are queried frequently to ensure efficient data retrieval as the dataset grows.

Triggers

Database triggers were created to automate the maintenance of `updated_at` timestamp columns. A `BEFORE UPDATE` trigger function was defined in SQL and attached to tables that require audit timestamps. The trigger function sets `NEW.updated_at = NOW()` on every row update, ensuring that the timestamp accurately reflects the most recent modification without requiring the application layer to manage this explicitly.



NAME	TABLE	FUNCTION	EVENTS	ORIENTATION	ENABLED
trg_child_health_profiles_u...	child_health_profiles	update_child_health_p...	BEFORE UPDATE	ROW	✓
trg_notification_preference...	notification_preferences	set_updated_at	BEFORE UPDATE	ROW	✓
trg_user_push_tokens_upd...	user_push_tokens	set_updated_at	BEFORE UPDATE	ROW	✓

Figure 5.2: Trigger Functions

5.3 Backend Development

5.3.1 Tools Used

Node.js (ES Modules) – Node.js was selected for its non-blocking I/O model, strong package ecosystem, and suitability for API-first systems that need to handle concurrent requests efficiently.

Express.js 5.x – Express 5.x is used as the core web framework to build HTTP APIs. It provides lightweight but powerful routing, middleware chaining, JSON request handling, and centralized

error processing. In this project, Express is responsible for registering all domain APIs while preserving a clear and maintainable request flow.

Supabase JavaScript SDK – Supabase acts as the primary backend data/auth platform. The project uses `@supabase/supabase-js` for user authentication, profile and role management, and CRUD operations on domain tables. This approach reduces custom infrastructure overhead while keeping database and auth interactions consistent from the backend.

Multer + Cloudinary Storage – For media and document handling, Multer is used for multipart upload parsing and Cloudinary is used as managed media storage. This combination supports secure and scalable upload workflows for speech files, profile images, and supporting documents without storing large binaries directly on the application server.

Firebase Admin SDK – Firebase Admin SDK is integrated for push notification delivery (FCM). The backend stores push tokens, manages notification preferences, logs delivery outcomes, and handles invalid tokens automatically. This allows real-time user-facing alerts for important events such as appointment reminders and payment status changes.

Safepay Node SDK – The project uses `@sfpy/node-sdk` to implement payment operations. The backend supports checkout initiation, tracker verification, and webhook-based payment confirmation. Signature validation and dual-format webhook handling are used to improve payment integrity and reliability across asynchronous transaction updates.

Nodemon – Nodemon improves development speed through automatic server restart.

Postman – Postman is used for API testing and endpoint verification.

5.3.2 Methodology

Layered API Structure

The backend follows a layered organization:

- Routes define API endpoints and attach middleware.
- Controllers handle HTTP request/response logic.
- Services encapsulate reusable business logic.

Middleware-Driven Authentication and Authorization

A middleware-based security pipeline is used to validate bearer tokens and inject per-request Supabase clients. Separate middleware paths are used for normal authenticated access and privileged admin/service operations. This ensures endpoint-level protection while maintaining flexibility for role-specific workflows.

Role-Based Access Control (RBAC)

Access control is implemented through role-aware logic centered on user profiles (e.g., parent, expert, admin). Sensitive operations verify both authentication and role permissions before executing. This is essential for healthcare-oriented workflows where each actor must be limited to appropriate actions.

RESTful API Design

The system APIs are designed around REST principles using clear resource-based routes and standard HTTP methods. Domain modules such as users, children, appointments, speech, payments, and notifications expose predictable endpoints, simplifying frontend integration and long-term maintenance.

Event-Oriented Notification Workflow

The backend applies event-style notification handling for important domain actions such as appointment booking, confirmation, cancellation, payment success or failure, and reminder scheduling.

Webhook-Centric Payment Confirmation

Payment processing follows a resilient asynchronous pattern: initial checkout creation is followed by webhook-driven final status updates. Signature validation and status reconciliation are used before mutating payment and appointment state. This reduces inconsistencies and improves transaction trustworthiness.

5.4 Web App Development

5.4.1 Tools Used

React (Next.js) — selected for building a scalable, server-side rendered web application with strong performance and SEO support. It enables the development of a unified system for parents, therapists, and administrators.
Tailwind CSS — used for utility-first styling, allowing rapid UI development with consistent design across the platform.
Axios / Fetch API — used for handling HTTP requests and communication with backend services including ML-based prediction APIs.
React Context API — used for lightweight global state management, especially for authentication, user sessions, and role-based access control.

Framer Motion — used to enhance user experience with smooth animations and interactive UI transitions.

5.4.2 Methodology

Project Setup

The frontend development started with a Next.js project structure to ensure scalability and modular architecture. The system supports three interfaces: Parent Portal, Therapist Dashboard, and Admin Panel.

API Integration

The API layer was implemented using Axios and Fetch API for secure communication with backend services, including AI-based autism screening and speech analysis models.

Component-Based Design

A component-based architecture was used for reusability and maintainability. Modules included questionnaires, reports, dashboards, result cards, and navigation components.

State Management

Global state management was implemented using React Context API to manage authentication, user roles, sessions, and results.

Routing System

Routing was handled using the Next.js file-based routing system for smooth navigation between login, screening, results, therapy sessions, and dashboards.

Key Features of the System

- Autism screening questionnaire interface
- AI-based speech analysis and post-assessment reports
- Therapist case review and therapy session tracking dashboard
- Admin management panel

5.5 Mobile App Development

5.5.1 Tools Used

Flutter / Dart — selected for building a cross-platform mobile application from a single codebase with a consistent user interface and native-like performance. It supports the development of a scalable mobile system for parents and experts.
GetX — used for state management, routing, and dependency handling, enabling a clean and modular application structure with role-based navigation and responsive UI updates.
Dio — used for HTTP communication with backend services, supporting authentication, assessment submission, appointment handling, and chat-related API interactions.
SharedPreferences — used for local storage of session-related data such as login state and user preferences, helping maintain continuity across app sessions.
Google Fonts / Material Design — used to maintain a consistent, modern, and accessible visual style across the mobile interface.
Packages for Media and File Handling — `record`, `image_picker`, `file_picker`, and `permission_handle` are used to support voice recording, media selection, file upload, and runtime permission handling within the mobile workflow.
Location Services — `flutter_map`, `geocoding`, and `latlong2` are used for location-based features such as expert location display and map interaction.
Web and External Handling — `webview_flutter` and `url_launcher` are used for in-app web content and external link handling, including payment and related web-based flows.
Document Handling — `pdf`, `open_file`, and `path_provider` are used to generate, store, and view document-based outputs where required by the application.

5.5.2 Methodology

Modular Development

The application was developed using a modular, feature-based methodology. Functional modules included authentication, parent workflow, expert workflow, speech assessment, child management, appointments, chat, payment, and settings. This improved organization and scalability.

Controller Architecture

A controller-based architecture using GetX was implemented. Each module had its own controller for managing state and business logic separately from the UI.

Navigation System

A centralized route system was used for smooth navigation between screens such as onboarding, login, home, appointments, and chat.

Reusable Components

Reusable widgets and shared constants were used for consistent design, reduced duplication, and easier maintenance.

Mobile-First Design

The application followed a mobile-first design with portrait orientation, touch-friendly controls, and layouts optimized for smaller screens.

Key Features of the System

- Parent and expert role-based interfaces
- Speech assessment and questionnaire workflow
- Child profile and progress management
- Appointment booking and consultation support
- In-app chat and payment integration
- Location-based expert discovery and map support

5.6 Machine Learning Model Training

5.6.1 Questionnaire Based Screening

Tools Used

Python — used for data preprocessing, model training, and prediction pipeline development.
Scikit-learn — used to implement the Random Forest classifier and evaluate model performance.
Pandas — used for loading, cleaning, and organizing questionnaire dataset values.
NumPy — used for numerical operations and feature array handling during model processing.
Joblib / Pickle — used to save and load the trained Random Forest model for deployment.
Matplotlib / Seaborn — used for data visualization, analysis, and performance comparison.

Methodology

Data Collection Questionnaire responses were collected from the dataset and prepared as structured input for the model. Each record represents a child's behavioural and developmental response pattern.
Data Preprocessing The dataset was cleaned by handling missing values, encoding categorical responses, and converting all features into numerical form suitable for machine learning.
Model Training A Random Forest classifier was trained on the processed questionnaire data. Multiple decision trees were built using different data subsets to learn patterns associated with autism risk.
Prediction and Evaluation After training, the model was tested on unseen data to generate screening results. Performance was evaluated using metrics such as accuracy, precision, recall, and confusion matrix.

5.6.2 Speech Analysis

Tools Used

Python — the primary programming language used to develop the automated feature extraction and inference pipelines.
TensorFlow / Keras — the deep learning framework utilized to design, load, and execute the Convolutional Neural Network (CNN) architecture.
Librosa — an audio analysis library used to process raw `.wav` files and extract spectral and energy-based features.
Parselmouth (Praat) — a specialized phonetic acoustic library used for precise extraction of pitch, formants, and vocal jitter.
Scikit-Learn — used for preprocessing tasks such as feature scaling using `StandardScaler` to normalize the 49 acoustic features.
Pandas & NumPy — used for structuring vocal data into 100×49 matrices and performing ensemble-based mathematical averaging.

Methodology

Audio Processing Raw audio recordings were processed to remove background noise and isolate speech using dynamic decibel thresholds.
Feature Extraction The system extracted 49 acoustic biomarkers such as pitch, resonance, and vocal energy from speech segments. These features included:

Pitch & Prosody Features (1–10)

Purpose: Pitch (Fundamental Frequency or F0) measures the perceived highness or lowness of the voice. Individuals with ASD often exhibit atypical prosody, such as overly monotonic

speech (low variance) or exaggerated, sing-song intonation (high variance).

#	Feature Name	Description & Clinical Purpose
1	mean_Pitch	Average fundamental frequency. Establishes baseline pitch level of the speaker.
2	Pitch_var	Variance of pitch across audio. Low variance correlates with monotonic speech.
3	NormedPitch_var	Pitch variance normalized by mean pitch for speaker-independent comparison.
4	mean_MinPitch	Average of minimum pitch values in vocal segments.
5	minPitch_var	Variance of minimum pitch values.
6	mean_MaxPitch	Average of maximum pitch values in vocal segments.
7	MaxPitch_var	Variance of maximum pitch values.
8	mean_VoicedPitch	Average pitch during voiced (tonal) segments only.
9	VoicedPitch_var	Variance of pitch during voiced segments.
10	MeanVoicedPitch_var	Secondary variance metric for voiced pitch capturing extended variations.

Voicing Stability Features (11–12)

Purpose: Voicing strength measures the periodicity of vocal fold vibration. Atypical voicing may indicate hoarseness, breathiness, or tension, which can be associated with neurodivergent speech patterns.

#	Feature Name	Description & Clinical Purpose
11	mean_Voicing	Average strength of voiced segments. Measures overall vocal clarity.
12	Voicing_var	Variance in voicing strength. High variance may indicate vocal instability or interrupted airflow.

Formants & Resonance Features (13–22)

Purpose: Formants are concentrations of acoustic energy around specific frequencies and correspond to the shape of the vocal tract (tongue, jaw, lips). Atypical articulation or vowel space centralization in ASD can be detected using formant dynamics.

#	Feature Name	Description & Clinical Purpose

13	mean_F1	Average first formant frequency. Inversely related to vowel height (tongue position).
14	F1_var	Variance of first formant (F1), indicating vertical tongue movement.
15	mean_F2	Average second formant frequency related to tongue front/back position.
16	F2_var	Variance of second formant (F2), indicating horizontal tongue movement.
17	mean_|F1-F2|	Average absolute difference between F1 and F2, representing vowel space separation.
18	|F1-F2|_var	Variance of F1-F2 difference reflecting articulation dynamics.
19	mean_Bandwidth1	Average bandwidth of the first formant, related to energy dissipation in vocal tract.
20	Bandwidth1_var	Variance of first formant bandwidth.
21	mean_Bandwidth2	Average bandwidth of second formant.
22	Bandwidth2_var	Variance of second formant bandwidth.

Spectral Slope Features (23–30)

Purpose: Spectral slope measures how quickly acoustic energy decreases at higher frequencies. It is a strong indicator of voice quality (e.g., breathy, pressed, or creaky voice). Children with ASD may exhibit distinctive voice qualities captured through spectral tilt variations.

#	Feature Name	Description & Clinical Purpose
23	mean_SpecSlope20_500	Average energy drop-off in the low-frequency range (20–500 Hz).
24	SpecSlope20_500_var	Variance of low-frequency spectral slope.
25	mean_SpecSlope500_1500	Average energy drop-off in the mid-frequency range (500–1500 Hz).
26	SpecSlope500_1500_var	Variance of mid-frequency spectral slope.
27	mean_SpecSlopeVoiced20_500	Average low-frequency slope during voiced frames.
28	mean_SpecSlopeUnvoiced20_500	Average low-frequency slope during unvoiced frames (fricatives, breath).
29	mean_SpecSlopeVoiced500_1500	Average mid-frequency slope during voiced frames.
30	mean_SpecSlopeUnvoiced500_1500	Average mid-frequency slope during unvoiced frames.

Vocal Perturbation / Roughness Features (31–32)

Purpose: Jitter measures cycle-to-cycle variations in fundamental frequency. High jitter is associated with a rough, harsh, or unstable voice quality and reflects micro-instabilities in vocal fold vibration.

#	Feature Name	Description & Clinical Purpose
31	mean_Jitter	Average local jitter, quantifying microscopic instability in vocal fold vibrations.
32	Jitter_var	Variance of jitter, capturing fluctuations in vocal roughness over time.

Energy & Amplitude Dynamics Features (33–40)

Purpose: Energy features capture the loudness of speech. Delta features represent the rate of change in loudness. Atypical volume control, such as speaking too loudly or too softly, is commonly observed in ASD and can be quantified using these measures.

#	Feature Name	Description & Clinical Purpose
33	mean_Energy	Average RMS energy representing baseline loudness of speech.
34	mean_DeltaEnergy	Mean of first derivative of energy capturing rate of loudness change.
35	mean_DeltaDeltaEnergy	Mean of second derivative of energy capturing acceleration of loudness changes.
36	mean_|DeltaEnergy|	Mean absolute change in energy representing magnitude of volume shifts.
37	Energy_var	Variance in energy reflecting consistency of speech loudness.
38	DeltaEnergy_var	Variance of energy change rate.
39	DeltaDeltaEnergy_var	Variance of energy acceleration.
40	|DeltaEnergy|_var	Variance of absolute energy changes.

Zero-Crossing Rate Features (41–46)

Purpose: Zero-Crossing Rate (ZCR) measures how often the audio signal changes sign. It is a key feature for distinguishing between voiced speech (low ZCR) and unvoiced speech (high ZCR), such as fricatives like *"s"* or *"sh"*.

#	Feature Name	Description & Clinical Purpose

41	mean_ZCR	Average zero-crossing rate across the entire audio signal.
42	ZCR_var	Variance of zero-crossing rate, indicating variation in speech articulation.
43	mean_VoicedZCR	Average ZCR during voiced speech segments only.
44	VoicedZCR_var	Variance of ZCR during voiced speech segments.
45	mean_UnvoicedZCR	Average ZCR during unvoiced speech segments.
46	UnvoicedZCR_var	Variance of ZCR during unvoiced speech segments.

Conversational & Macro Metrics (47–49)

Purpose: These macro-level features describe the structural properties of speech recordings, such as the amount and distribution of vocalization. Reduced vocal output or irregular utterance patterns may indicate developmental or social communication delays associated with ASD.

#	Feature Name	Description & Clinical Purpose
47	mean_LengthInRange	Average duration of valid vocalization segments within the recording.
48	LengthInRange_var	Variance in duration of vocalization segments.
49	Num_voc_segs	Total number of detected vocalization segments, representing speech output frequency.

Data Preparation An Adaptive Temporal Drift algorithm was applied to short 5–10 second clips, creating a fixed 100×49 data matrix for analysis.

CNN Model Training A Convolutional Neural Network was trained on 136 recordings using 5-fold cross-validation to learn reliable speech patterns.

Prediction and Evaluation During deployment, the trained model analyzed processed data to generate predictions and calculate a Social Affect (SA) severity score for communication assessment.

5.7 Summary

The SpeechSpectrum system employs a layered technical stack, including a secure Supabase/PostgreSQL database utilizing Row-Level Security and a Node.js/Express.js backend for payment, notification, and role-based access control. Cross-platform applications built with Next.js (web) and

Flutter (mobile) provide dedicated interfaces for parents, experts, and administrators. Core clinical functions are powered by machine learning: a Random Forest model handles questionnaire-based screening, and a Convolutional Neural Network (CNN) analyzes 49 extracted acoustic biomarkers from speech recordings to calculate a reliable Social Affect severity score.

Chapter 6
Deployment

6.1 Introduction

This chapter outlines the deployment of the SpeechSpectrum system, covering the PostgreSQL database provisioned via Supabase, the Vercel-hosted Node.js backend and Next.js web application, and the build process for the Flutter mobile app.

6.2 Database Deployment

6.2.1 Introduction

SpeechSpectrum's database is the backbone of the system, responsible for storing user profiles, children's health records, ASD screening submissions and results, appointments, payments, and push notification data. PostgreSQL was selected as the database engine for its robustness, support for advanced data types such as JSONB and generated columns, and strong enforcement of relational constraints.

6.2.2 Provisioning the Database

The PostgreSQL database was provisioned through Supabase, a managed Backend-as-a-Service platform built on top of PostgreSQL. No separate database server setup was required; the Supabase project was created through the Supabase dashboard at `supabase.com`, which automatically provisioned a dedicated PostgreSQL instance in the selected region.

6.2.3 Connection Configuration

The backend was connected to the Supabase database by configuring the following environment variables in the Vercel project dashboard:

- `SUPABASE_URL` — the unique project URL for the Supabase instance
- `SUPABASE_API_KEY` — the anonymous public key for authenticated client operations
- `SUPABASE_SERVICE_KEY` — the service role key for privileged operations such as webhook processing

6.3 Backend Deployment

6.3.1 Introduction

The backend of SpeechSpectrum, developed with Node.js and Express.js using ES Modules, provides the REST API layer connecting the Flutter mobile app and Next.js web application to the database and all external services.

6.3.2 Preparing the Environment

The backend was deployed to Vercel, a serverless cloud platform. Vercel was selected for its seamless GitHub integration, automatic deployments on every push, serverless Node.js runtime support, built-in HTTPS, and a global edge network. No server provisioning or containerisation was required.

6.3.3 Deployment Process

- The backend code was pushed to a GitHub repository.
- The repository was connected to a Vercel project through the Vercel dashboard, enabling continuous deployment on every push to the main branch.
- A `vercel.json` configuration file was added to the repository root, routing all incoming requests to `src/app.js` and specifying `npm install --legacy-peer-deps` as the install command to resolve peer dependency conflicts.
- All environment variables — including Supabase credentials, Cloudinary credentials, Safepay API keys, Zoom credentials, and Firebase service account details — were configured through Vercel's Environment Variables panel under Project Settings, with separate values set for Production and Preview environments.
- The deployed backend was verified by calling the `/api/health` endpoint, which returns a 200 response confirming the server is running, and by testing all critical endpoints through Postman using the production URL.

6.4 Mobile Application Build

6.4.1 Introduction

Release builds bundle optimized Dart code, assets and fonts, apply production API endpoints, and include platform permissions and signing so binaries are ready for distribution on Google Play and App Store.

6.4.2 Preparing the Build

Prepare by bumping the version in `pubspec.yaml`, setting production endpoints via `--dart-define` or environment files, verifying assets and fonts in `pubspec.yaml`, configuring the Android keystore and iOS provisioning, ensuring required permissions in `AndroidManifest.xml` and `Info.plist`, checking native plugin settings, and running `flutter analyze` plus smoke tests on devices.

6.4.3 Building the App

Build signed artifacts locally or in CI:

- **Android (AAB recommended):** `flutter build appbundle --release --dart-define=API_BAS`
- **iOS:** `flutter build ipa --release --dart-define=API_BASE_URL=<prod>` (or archive via Xcode).

6.5 Web Application Deployment

6.5.1 Introduction

Although the primary focus of the system is a web-based platform, a fully responsive multi-role web application was deployed to support different stakeholders, including parents, therapists, and administrators. The web interface enables users to access autism screening results, manage profiles, and monitor developmental progress in a centralised system.

6.5.2 Preparing the Web Frontend

The frontend was developed using Next.js (a React-based framework) to ensure performance, scalability, and server-side rendering capabilities. The application was designed to be:

- Fully responsive across desktop, tablet, and mobile devices
- Modular, with reusable UI components for dashboards, questionnaires, and result views
- Integrated with a centralised backend API for authentication, screening, and analytics

The frontend was connected to the deployed backend services, ensuring consistent data flow between the ML-based prediction system and user dashboards.

6.5.3 Hosting the Web Application

The web application was deployed using Vercel, which provides:

- Global Content Delivery Network (CDN) for fast performance
- Seamless Git-based deployment workflow
- Automatic builds and continuous deployment support
- SSL encryption for secure communication

6.5.4 Final Checks

After deployment, comprehensive end-to-end testing was conducted on the live system to ensure reliability and performance. The testing process focused on:

- User authentication and role-based access (Parent, Therapist, Admin)
- Questionnaire submission and AI-based prediction results
- Therapist dashboard functionality and case review flow
- Admin panel data management and system monitoring
- Cross-device responsiveness and UI consistency

Any identified issues were resolved to ensure a stable and production-ready system.

6.6 Machine Learning Deployment

6.6.1 Questionnaire Based Analysis Model

Cloud Deployment on Vercel

To integrate the questionnaire-based autism screening model with the user-facing application, the trained Random Forest pipeline was deployed on Vercel. This deployment allows the web application to send questionnaire responses to the backend API, where the input is processed and a prediction is returned in real time.

- **Cloud Hosting:** The questionnaire model was hosted on Vercel to provide fast, scalable, and reliable access to prediction services.
- **API Integration:** The model was exposed through an API endpoint, enabling the frontend to send user responses and receive autism screening predictions in JSON format.

- **Real-Time Prediction:** Once the questionnaire is submitted, the backend processes the input immediately and returns the classification result to the user.
- **Reliability and Scalability:** Using Vercel ensures automatic scaling, simple deployment, and stable performance for lightweight machine learning inference.

6.6.2 Speech Analysis Model

Cloud Deployment on DigitalOcean

To seamlessly integrate the Social Affect (SA) speech analysis model with the user-facing application, the entire machine learning pipeline was deployed to the cloud utilising DigitalOcean. This architecture ensures that computationally intensive tasks — such as audio feature extraction and deep learning inference — are offloaded from the client's device to a dedicated, high-performance server.

- **Cloud Hosting:** The Social Affect (SA) speech analysis model was deployed on DigitalOcean to handle heavy processing tasks on a dedicated server instead of the client device.
- **Containerization:** The pipeline was containerised using Docker, packaging the Python environment, trained CNN model, and required libraries such as TensorFlow, Librosa, and Parselmouth.
- **RESTful API:** The model was exposed through a RESTful API developed with FastAPI, allowing secure communication between the application and server.
- **Real-Time Inference:** Uploaded speech clips were processed instantly for feature extraction, CNN prediction, and generation of the Social Affect severity score returned in JSON format.
- **Security & Reliability:** The API was hosted with Gunicorn or Uvicorn and protected using SSL/HTTPS encryption to ensure privacy and reliable performance.

6.7 Summary

The deployment finalised the Supabase/PostgreSQL database, Vercel-hosted backend and web application, and mobile builds. Machine learning models were integrated by deploying the questionnaire model on Vercel and containerising the speech analysis model on DigitalOcean for robust, real-time inference.

Chapter 7
Results

7.1 Introduction

In this chapter, we present the results of our project, with the results illustrated through screenshots and short descriptions of our mobile application.

7.2 Application Results

7.2.1 Login and Signup

Login

The Login screen of the SpeechSpectrum application presents users with a clean, teal-branded interface. Users are prompted to enter their Email Address and Password, with a password visibility toggle and a Forgot Password link for account recovery. A prominent Log In button completes the sign-in process, while a Sign Up link is available for new users who do not yet have an account.

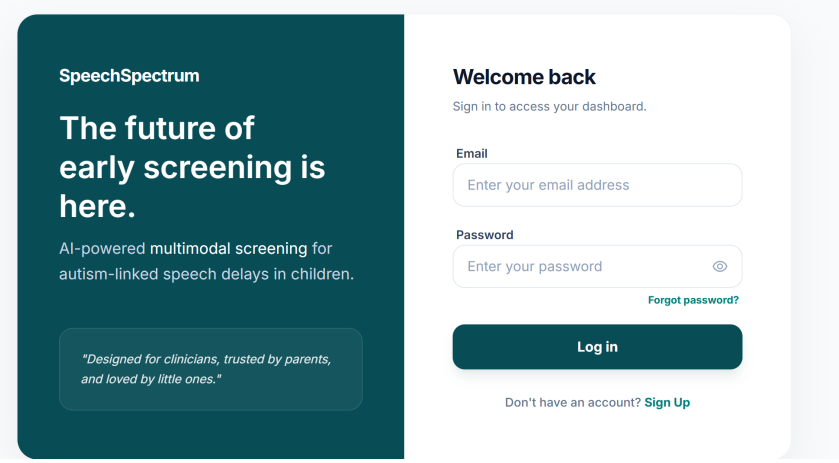


Figure 7.1: Web Login

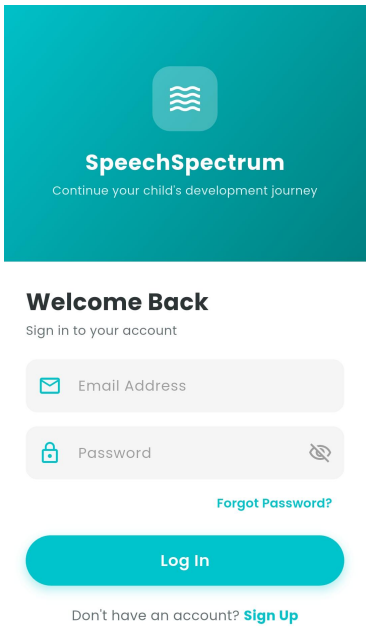


Figure 7.2: Mobile Login

Signup as parents

The Parent signup flow begins with a role-selection screen (Step 1 of 5), where the user chooses between Parent and Therapist to continue. This is followed by Step 2, which collects the user's first and last name. Step 3 In step 3 the application ask for your Phone Number Step 4 (Add Your Children) allows the parent to enter each child's full name, date of birth, gender, and optional health details including blood group, weight, height, known allergies, genetic disorders, chronic conditions, and family history of ASD, speech disorders, or hearing loss. Medical documents in PDF format can also be attached. The final step (Step 5) prompts the parent to set up login credentials by entering and confirming an email address and password, after which the account is created by tapping Register.

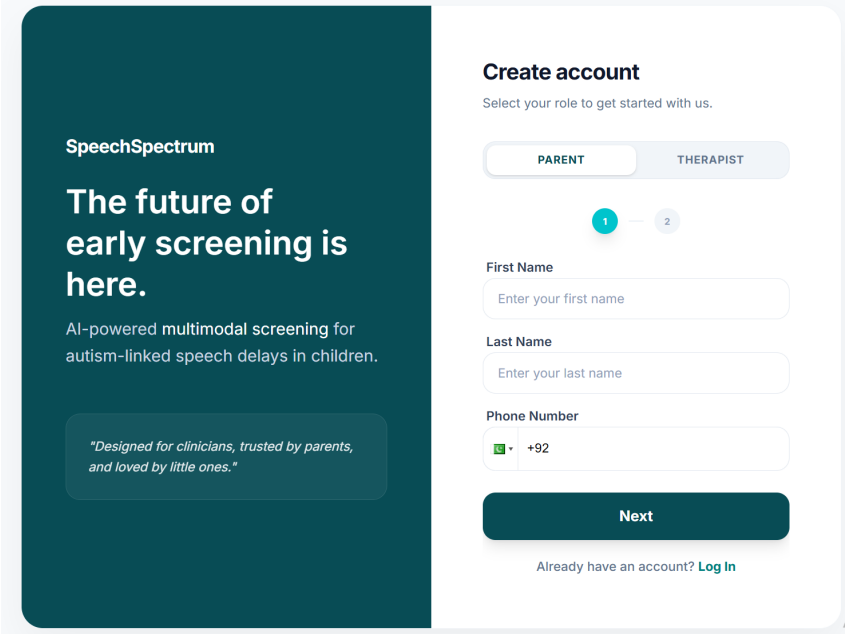


Figure 7.3: Web Signup

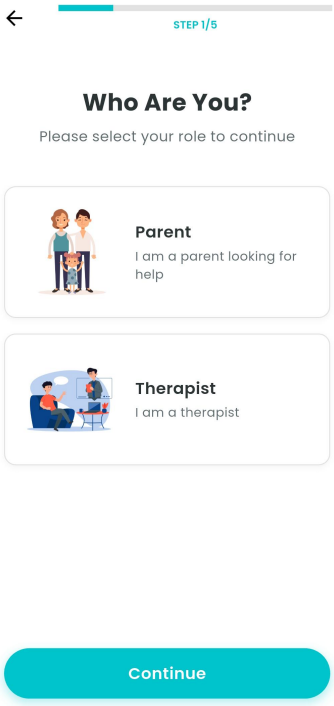


Figure 7.4: Sign Up As Parent Step 1

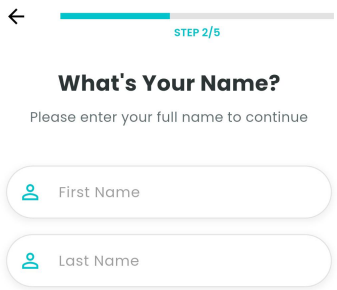


Figure 7.5: Sign Up As Parent Step 2

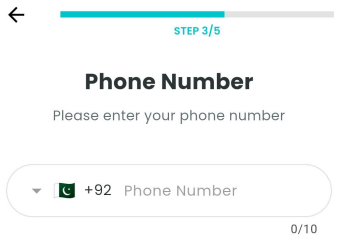


Figure 7.6: Sign Up As Parent Step 3

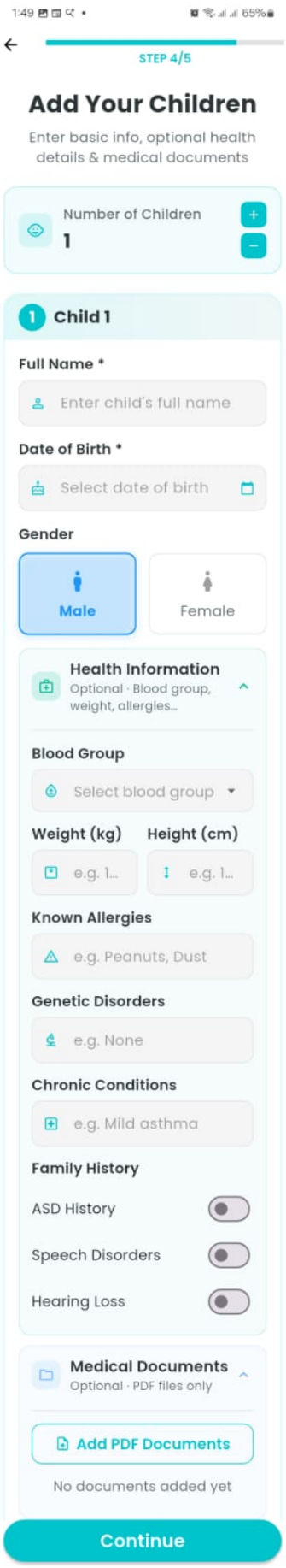


Figure 7.7: Sign Up As Parent Step 4

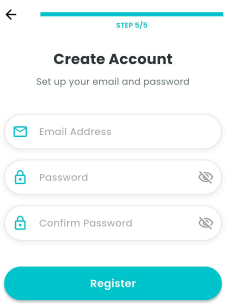


Figure 7.8: Sign Up As Parent Step 5

Signup as Expert

The Expert signup flow begins at Step 1 of 6 with the same role-selection screen, this time with the Therapist option selected. Step 2 collects the expert's full name. Step 3 captures their phone number using a country-code selector. Step 4 gathers professional details including specialisation (e.g., Speech Therapy), organisation name, contact email, and PMDC licence number. Step 5 requires the expert to upload a profile image, a degree certificate (PDF only), and a professional certificate (PDF only). The final Step 6 sets up the account credentials with email, password, and confirmation fields, completed by tapping Register.

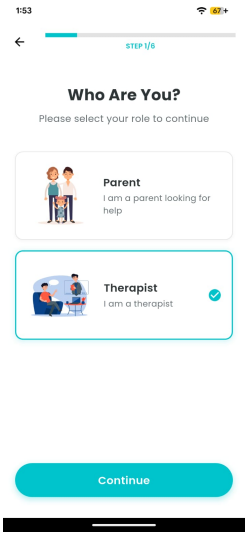


Figure 7.9: Signup as Expert step 1

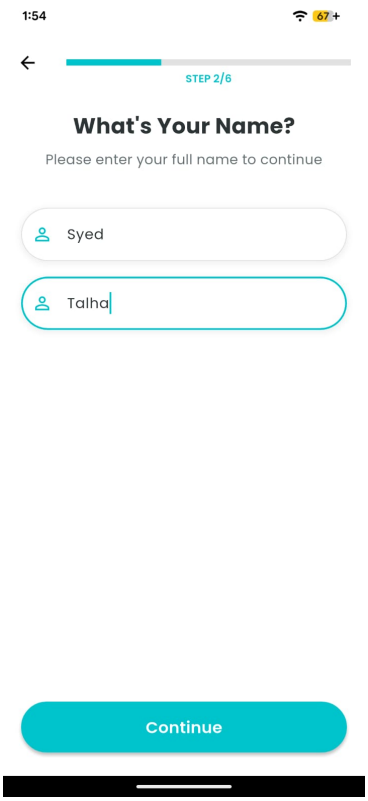


Figure 7.10: Signup as Expert step 2

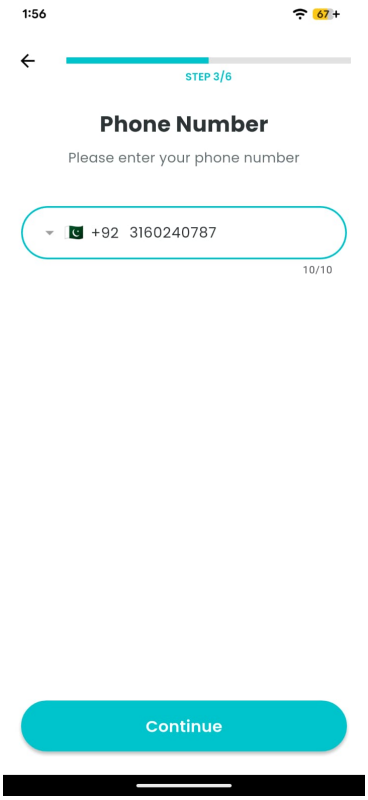


Figure 7.11: Signup as Expert step 3

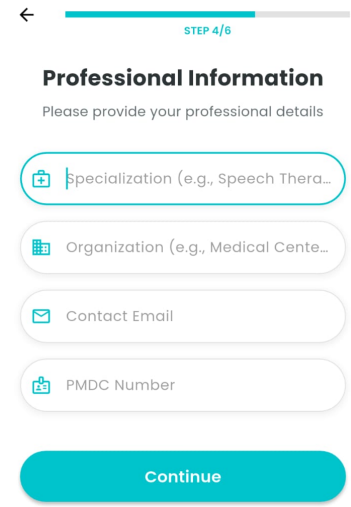


Figure 7.12: Signup as Expert step 4

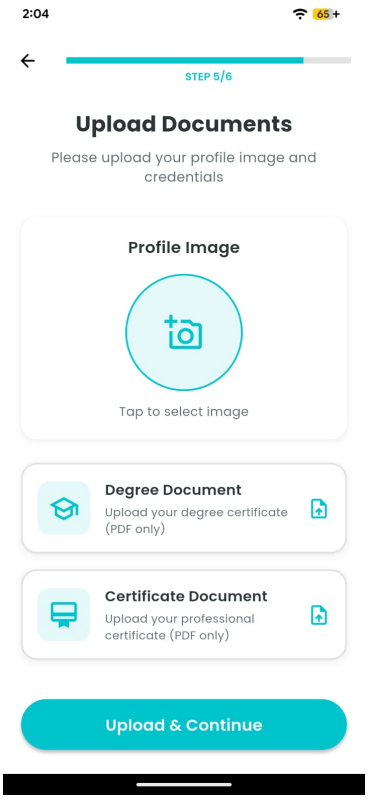


Figure 7.13: Signup as Expert step 5

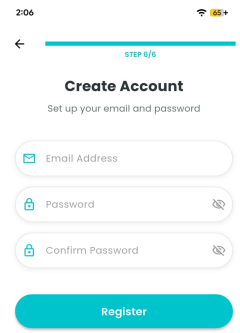


Figure 7.14: Signup as Expert step 6

7.2.2 Parent View

Parents Dashboard

The Parents Dashboard is the main home screen for the parent role, featuring the SpeechSpectrum branding and a teal hero banner that highlights the app's purpose of early autism screening. The dashboard presents two screening methods — Structured Questionnaire and Voice Sample Analysis — as numbered options. A Quick Actions section provides shortcuts to Questionnaire History and Speech Submissions. An Important Information notice at the bottom reminds parents that results are for screening purposes only and that a qualified healthcare professional should be consulted for a formal diagnosis.

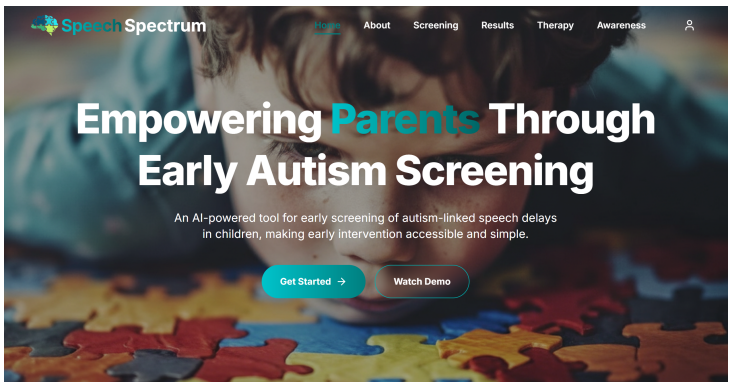


Figure 7.15: Web Dashboard

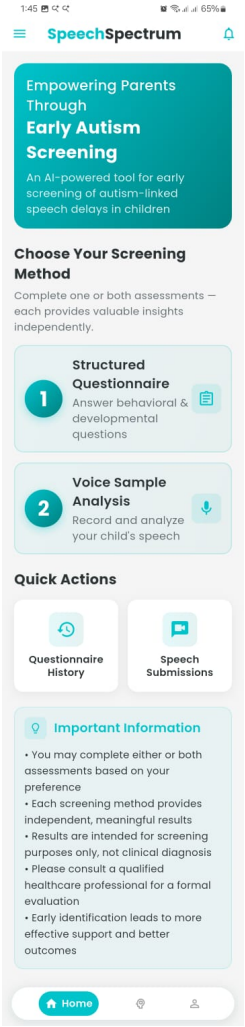


Figure 7.16: Mobile dashboard

7.2.2.0.1 Parent's Sidebar The Parent sidebar slides in from the left and displays the currently logged-in parent's name and email at the top. Navigation is divided into two sections: the Main section includes Dashboard, My Children, and Progress and Results links; the Therapists Services section contains Find Therapists and My Appointments. A Settings menu with expandable options, along with About and Help links, appears under a separate Settings group. A Logout Account button is shown in red at the bottom.

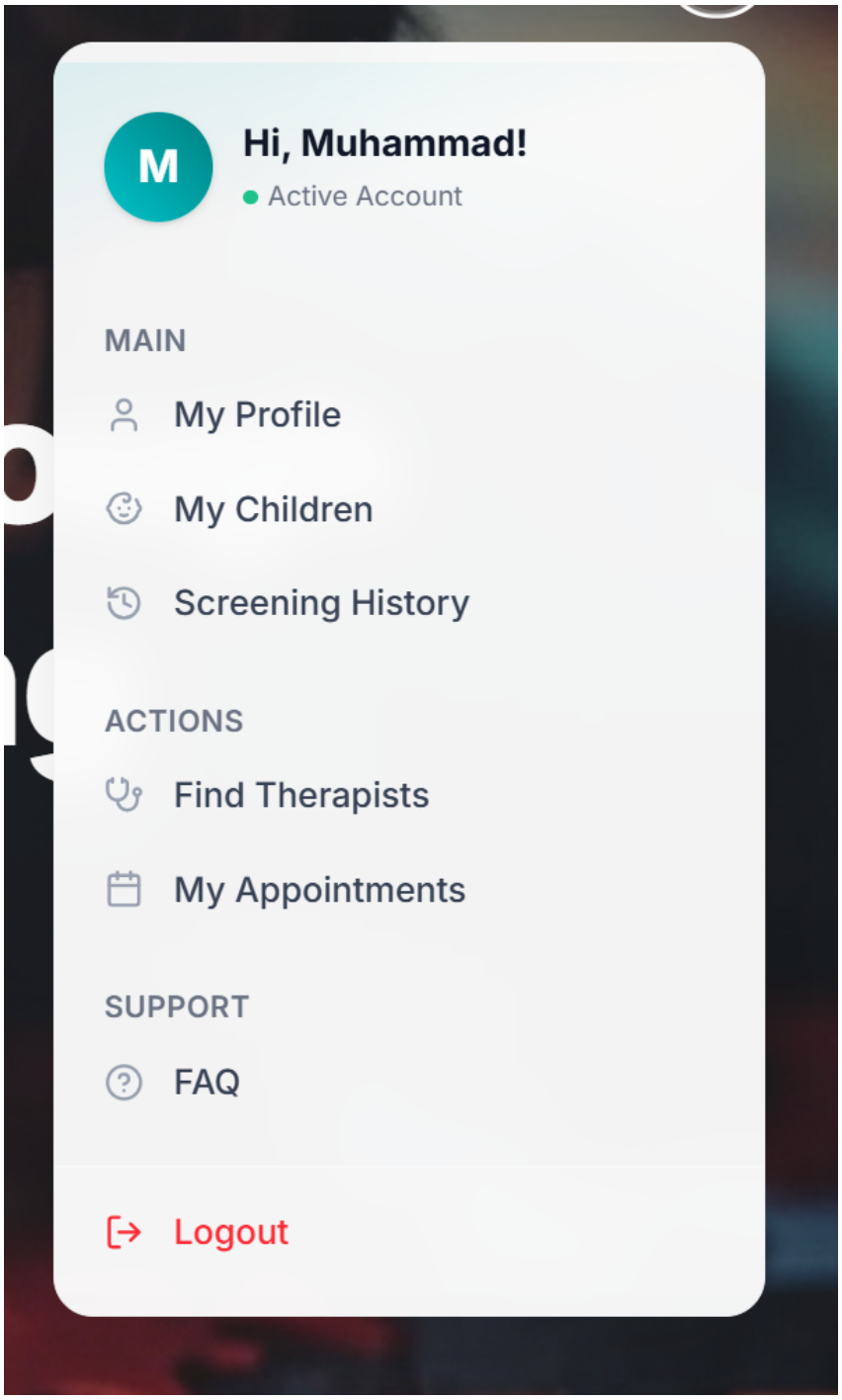


Figure 7.17: Web Parent's Sidebar

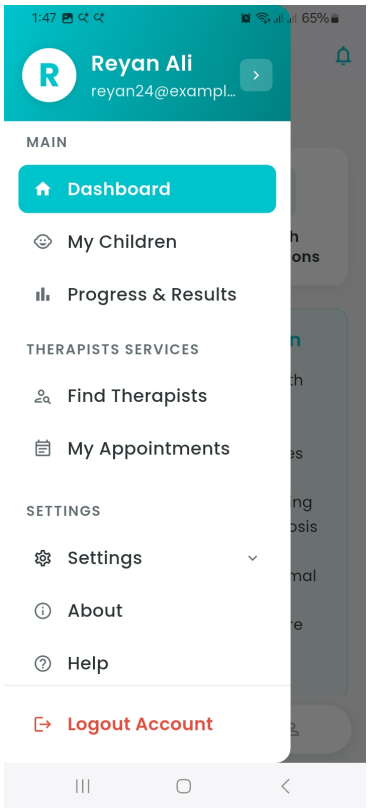


Figure 7.18: Mobile Parent's Sidebar

My Children Tab

The My Children tab displays a searchable list of all children registered under the parent's account. Each child entry shows the child's name, age, and date of birth, along with edit and delete action icons. An Add Child floating button at the bottom right allows the parent to register a new child at any time.

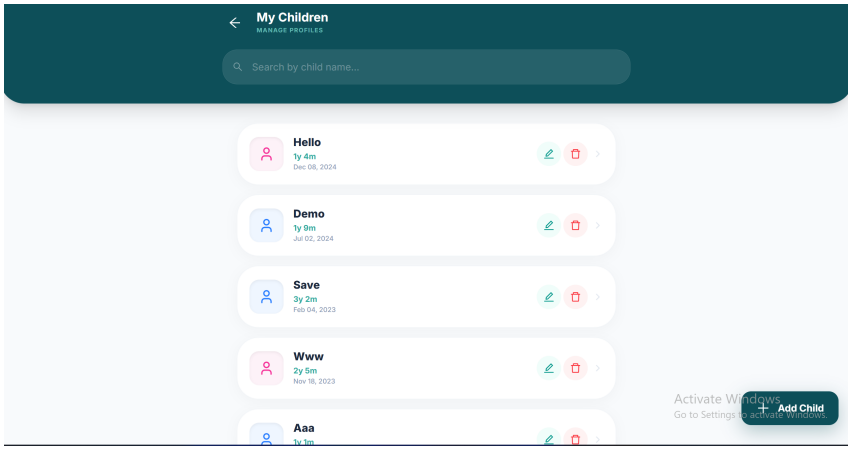


Figure 7.19: My children page

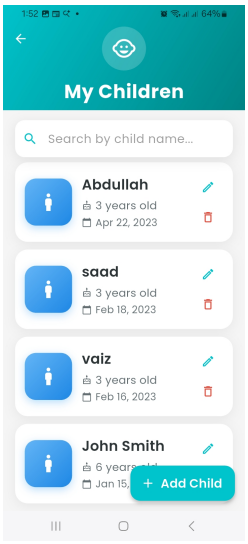


Figure 7.20: My children Tab

Progress and Result Tab

The Progress and Result tab gives parents an overview of their child's assessment history. It presents two category cards: Screening History, which covers questionnaire-based developmental assessments, and Speech Assessment History, which stores voice recordings and their analysis results. Each card includes a summary count and a View All link. An About Progress Tracking note at the bottom explains that all data is securely stored and automatically tracked over time to support personalised therapist guidance.

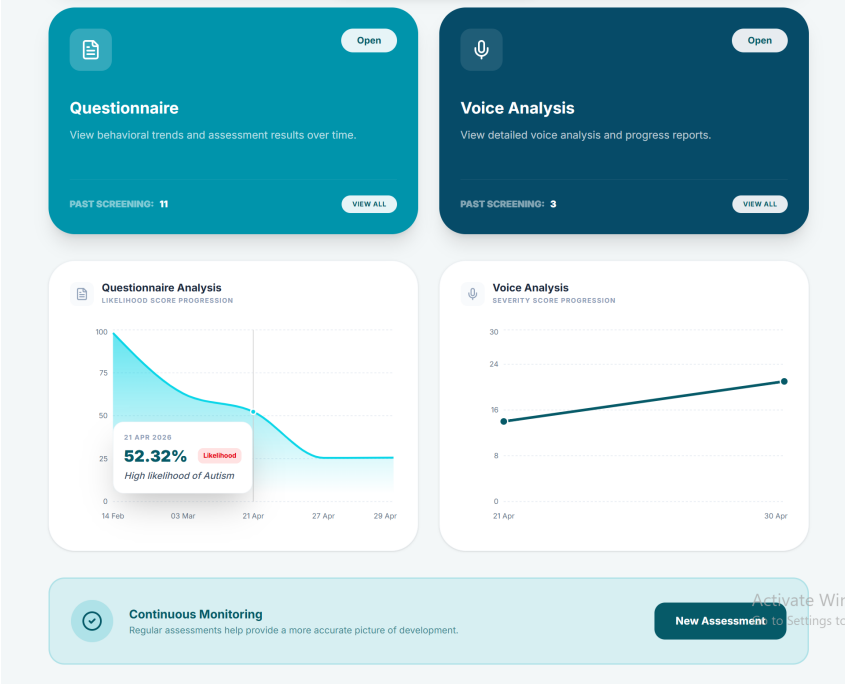


Figure 7.21: Web progress and Results view

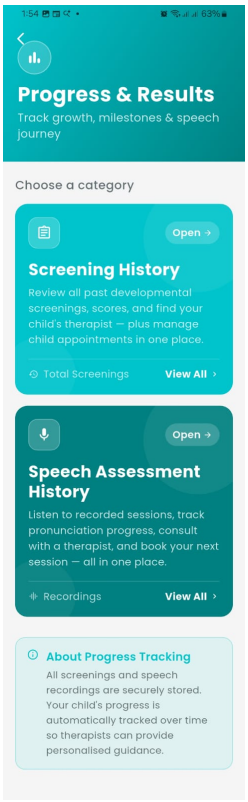


Figure 7.22: Mobile progress and Results view

Find Therapist Tab

The Find Therapist tab presents a searchable directory of available therapists. Each therapist entry displays the therapist's name, specialisation (e.g., ASD, Speech Therapy), and affiliated organisation. Users can tap any listing to view the full therapist profile and book an appointment.

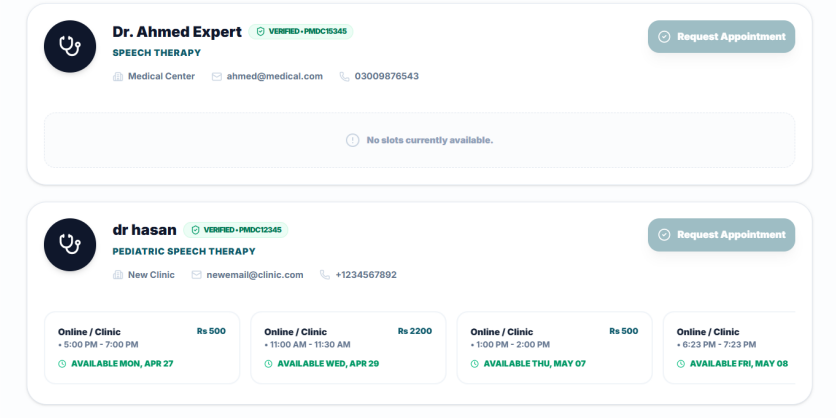


Figure 7.23: Web Find Therapist view

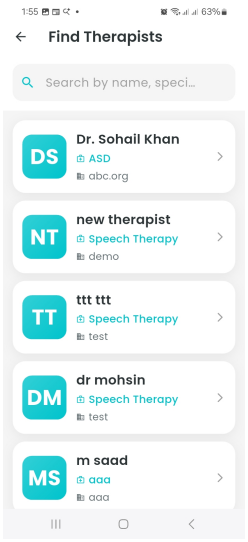


Figure 7.24: Mobile Find Therapist View

My appointments tab

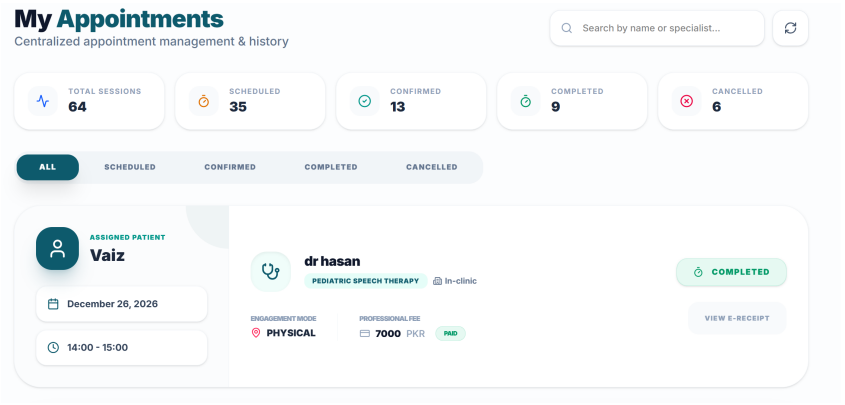


Figure 7.25: Web My Appointments

7.2.2.0.2 Requested The My Appointments screen, opened to the Requested tab, lists all appointments that have been submitted by the parent but not yet confirmed by the therapist. Each card shows the child's name, the assigned therapist, the appointment date and time, the consultation mode (Online or Physical), and the session fee in PKR. The tabs at the top allow filtering between All, Requested, Confirmed, Completed, and other statuses.

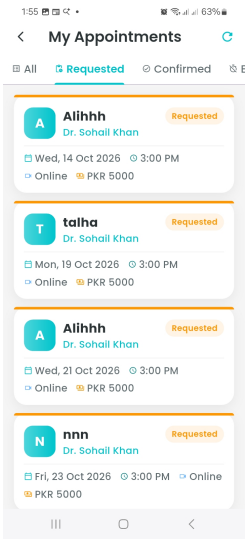


Figure 7.26: Requested

7.2.2.0.3 Completed The Completed tab shows appointments that have been fully attended and concluded. Each entry displays the child's name, the treating therapist, the session date, consultation mode, and the fee paid. Only sessions that have reached their scheduled end and been marked as completed by the therapist appear in this view.

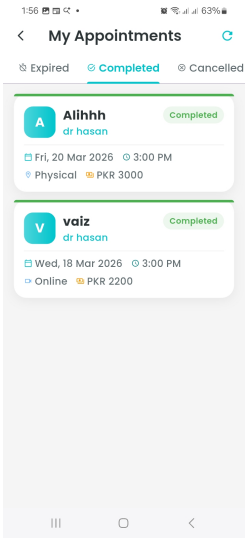


Figure 7.27: Completed

7.2.2.0.4 Confirmed The Confirmed tab of the My Appointments screen lists all appointments that have been accepted and paid. Each confirmed card shows the child's name, therapist, session date and time, consultation mode, and fee. A Payment Complete badge appears for settled sessions. For online appointments, a Join Meeting button with a copy/open link is also displayed so the parent can access the video call directly.

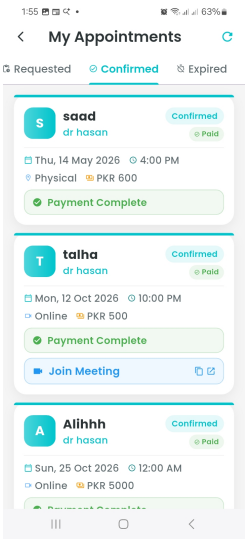


Figure 7.28: confirmed

Setting Tab

The Settings tab provides account and preference management in three grouped sections. Under Account, users can Edit Profile and Change Password. Under Preferences, options for Notifications and Language are available. The Legal section contains links to the Terms and Conditions and Privacy Policy documents.

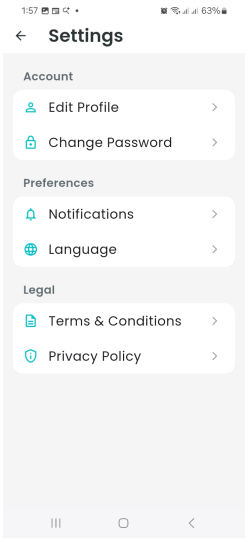


Figure 7.29: Setting

Questionnaire Screening

The Questionnaire Screening flow begins with a Patient Information form where the parent selects the child from a dropdown, confirms the child's age in months and gender, and answers two background health questions regarding jaundice history and family history of ASD. This is followed by the ASD Screening questionnaire, which presents ten behavioural questions —

such as whether the child responds to their name or uses simple gestures — each with multiple-choice frequency options (Always, Usually, Sometimes, Rarely, Never). A progress bar at the top tracks completion across all ten questions.

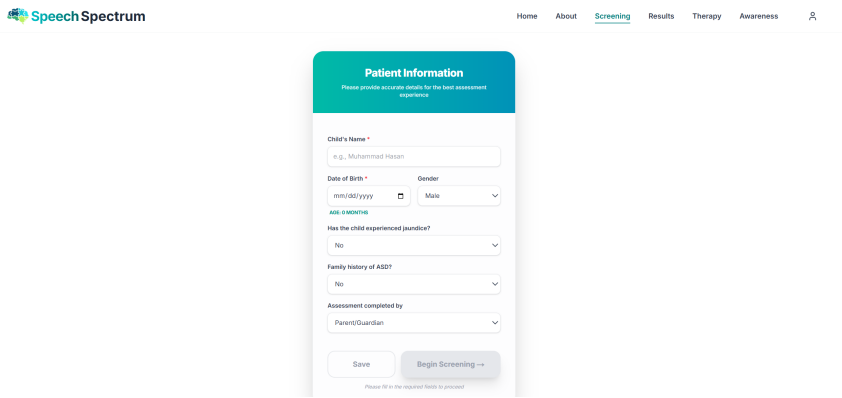


Figure 7.30: Web Questionnaire Screening

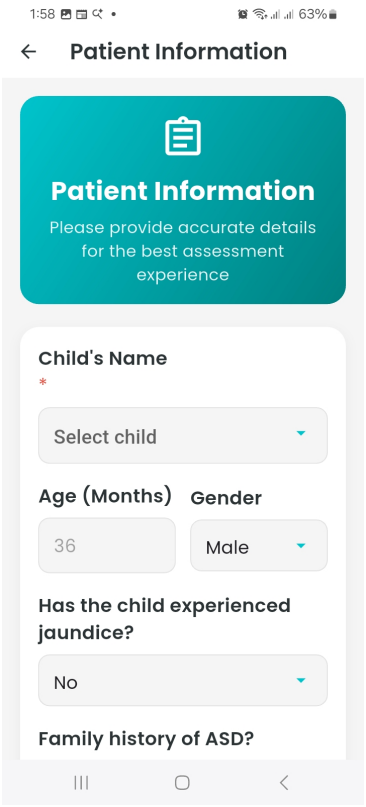


Figure 7.31: Mobile Questionnaire Screening step 1

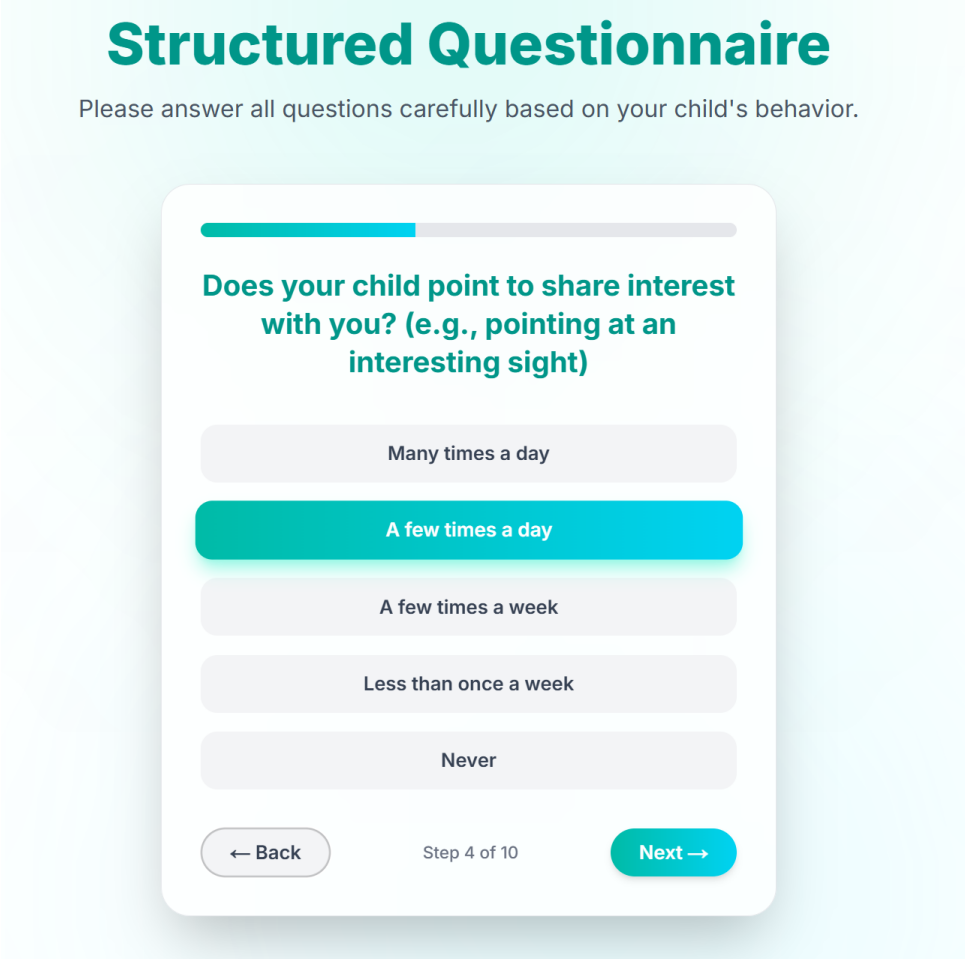


Figure 7.32: Web Questionnaire Screening step 2

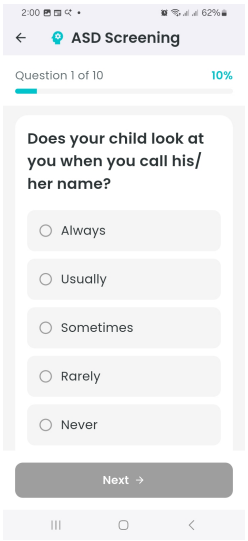


Figure 7.33: Mobile Questionnaire Screening step 2

Questionnaire Screening Result

After completing the questionnaire, the Screening Results screen displays the AI-generated probability score as a circular progress indicator. The result shown indicates a 63.23 percent probability with a classification of High likelihood of Autism. A recommendation card below advises the parent to consult a qualified expert, and a View All Experts button links directly to the therapist directory. The lower portion of the screen lists each questionnaire response, colour-coded as Positive (green) or Negative (red), giving a transparent breakdown of how the score was calculated.

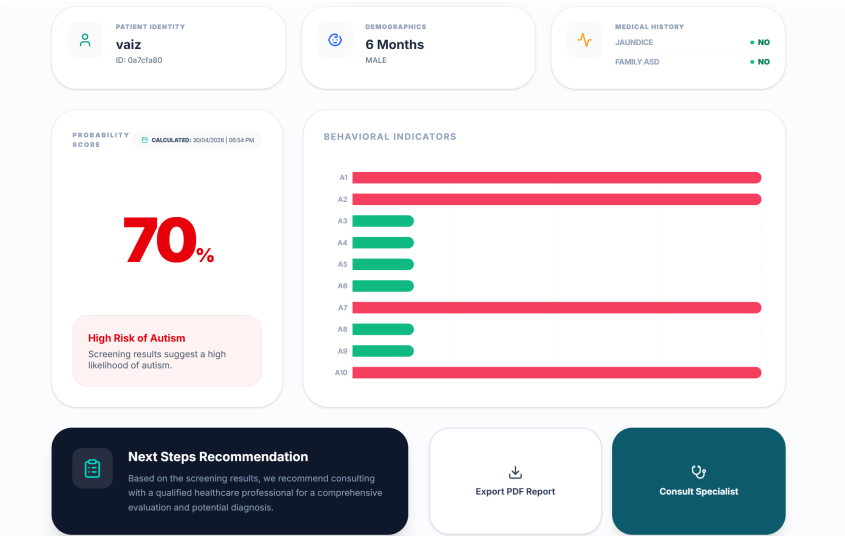


Figure 7.34: Web Questionnaire screening Result

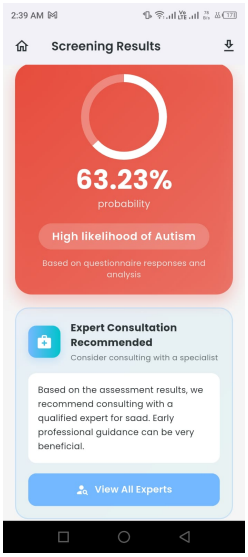


Figure 7.35: Mobile Questionnaire screening Result

Questionnaire Screening Report

Tapping Download PDF Report on the results screen opens a formatted ASD Screening Report in the device's PDF reader. The report is branded with the SpeechSpectrum logo and labelled as a Confidential Assessment. It includes the child's name, prediction label, probability percentage, and assessment date, followed by a Submission Information section detailing age, gender, jaundice history, and family ASD history. All ten questionnaire responses are listed with their Positive or Negative classification, and a disclaimer at the bottom clarifies that this is a screening tool and not a clinical diagnosis.

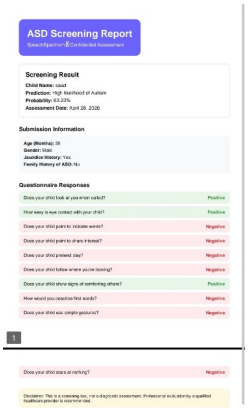


Figure 7.36: Questionnaire Screening Report

Speech Analysis

The Speech Analysis screen allows the parent to select a child from a dropdown and then choose between recording a voice sample live or uploading an existing audio file. When recording, a microphone waveform visualiser and countdown timer are displayed, with a minimum recording duration of 5 seconds enforced. Step-by-step instructions guide the parent through finding a quiet environment, tapping the microphone, speaking clearly, and stopping when finished.

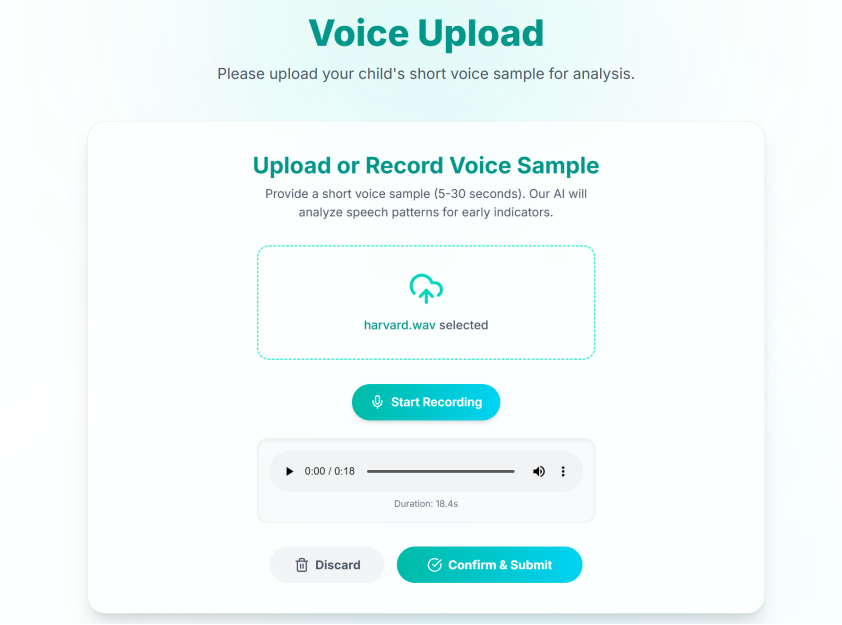


Figure 7.37: Web Speech Analysis view

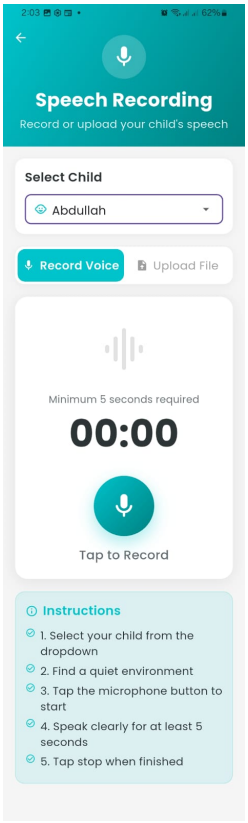


Figure 7.38: Mobile Speech Analysis view

Speech Analysis Result

The Speech Analysis Result screen presents the outcome of the AI speech analysis. Recording information including the child's name, duration, date, time, and audio format is displayed at

the top. The Analysis Results section shows a circular score gauge Action buttons allow the parent to Find a Therapist, Download the PDF Report, or Delete the submission.

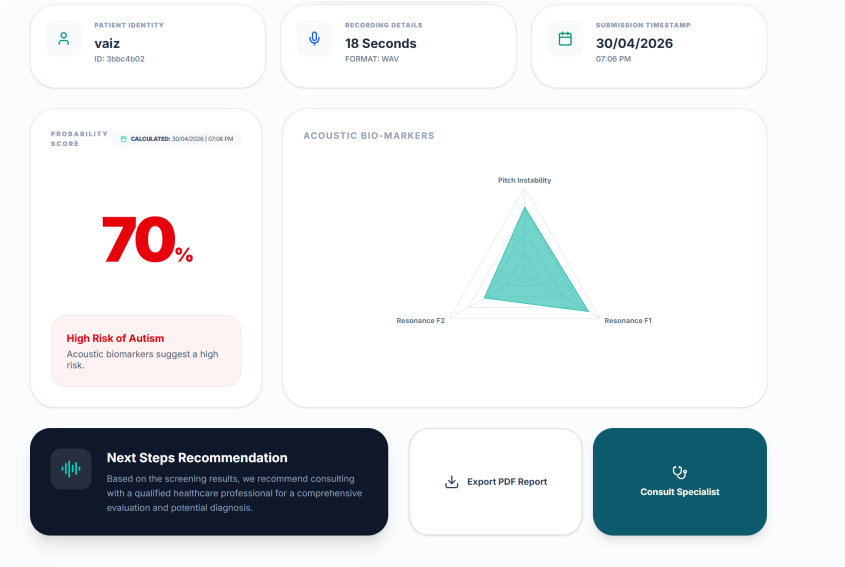


Figure 7.39: Web Speech Analysis Result view

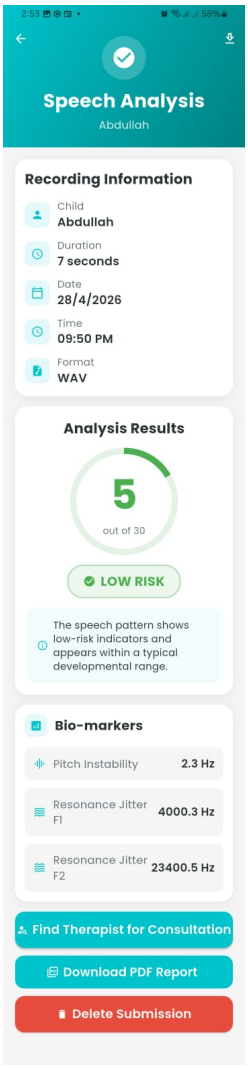


Figure 7.40: Mobile Speech Analysis Result view

Speech Analysis Report

The Speech Analysis Report is generated as a downloadable PDF document. It carries the SpeechSpectrum branding and Confidential Assessment label. The report captures the child's name, risk level, severity score, assessment date, recording duration, and file format. An Interpretation paragraph summarises the findings in plain language, and a disclaimer reminds the reader that professional evaluation is recommended.



Figure 7.41: Speech Analysis Report

Questionnaire Assessment History

The Questionnaire Assessment History screen provides a complete record of all past ASD questionnaire submissions, grouped by child. Each child's card shows the total number of submissions, the latest risk classification (e.g., High likelihood of Autism), the probability percentage, and the date of the most recent assessment. Parents can tap a child's entry to drill down into individual submission details.

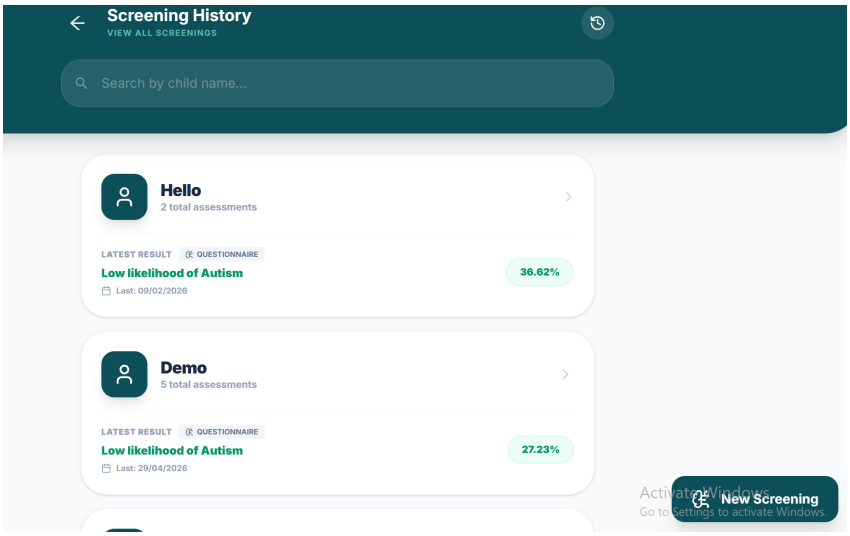


Figure 7.42: Web Questionnaire Assessment History view

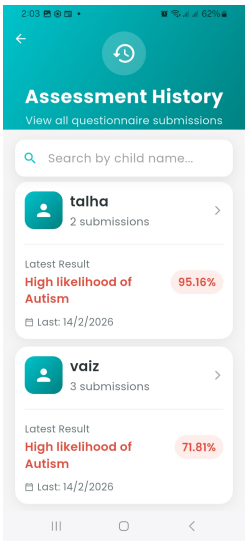


Figure 7.43: Mobile Questionnaire Assessment History view

Speech Assessment History

The Speech Assessment History screen lists all voice recording submissions across all children. Each entry displays the child's name, recording duration, submission date, and risk level classification (LOW RISK, MODERATE RISK, etc.) alongside a colour-coded severity score out of 30. A New Recording floating button is provided for quick access to the recording flow. Parents can tap the three-dot menu on any entry to view or delete it.

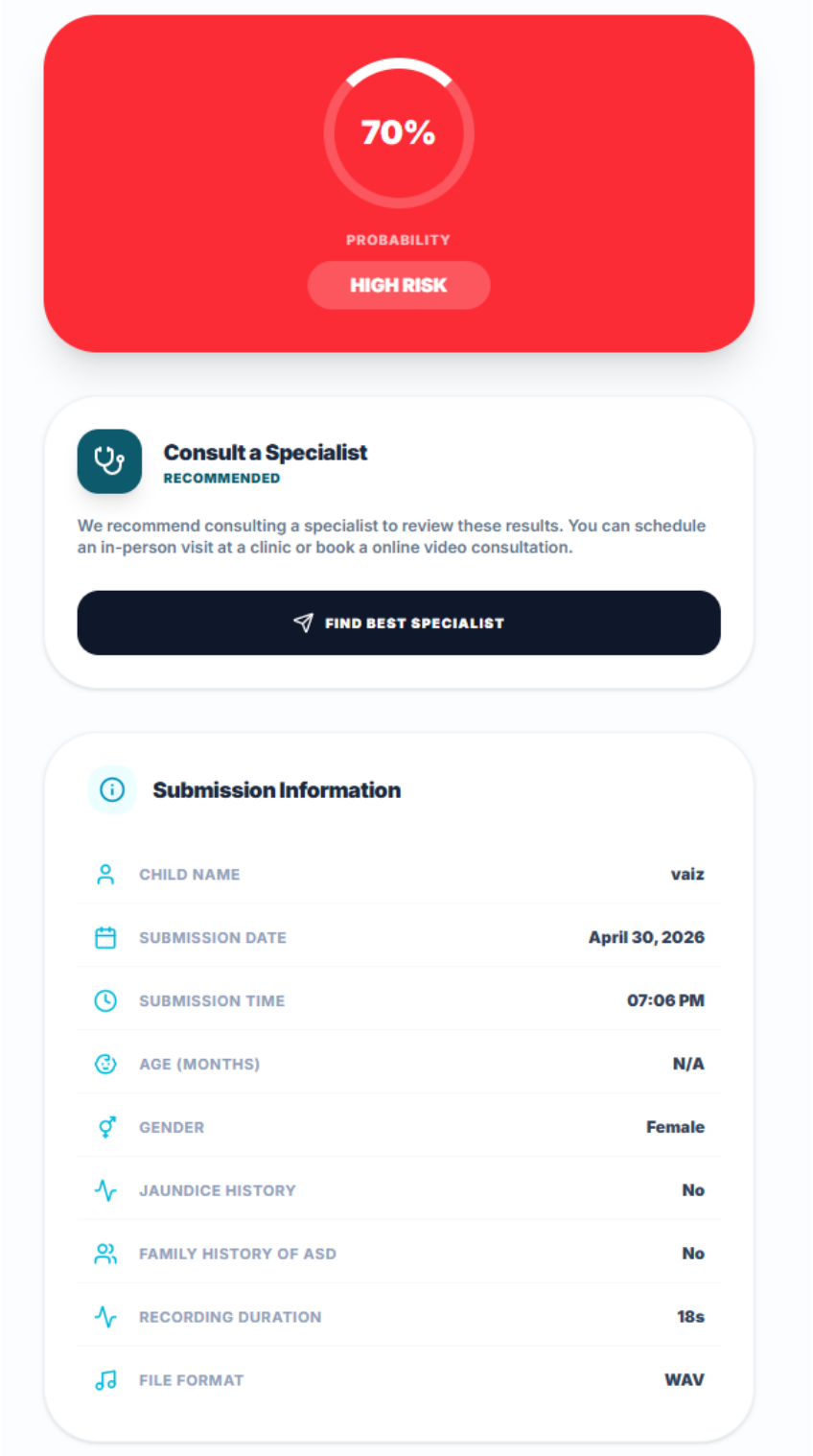


Figure 7.44: Web Speech Assessment History view

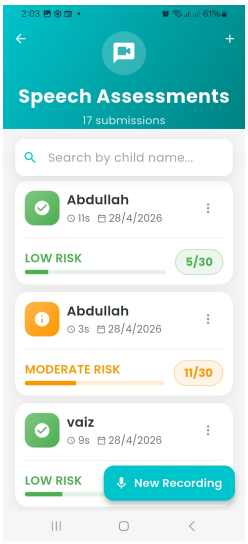


Figure 7.45: Mobile Speech Assessment History view

Parent Profile Management

The Parent Profile Management screen displays the logged-in parent's profile picture initial, full name, and role badge (Parent). Contact information including the phone number is listed. Quick-access links are provided for My Children, Speech Assessment, and Screening History. Notification and Help and Support options are also accessible from this screen. An Edit Profile button in the header allows the parent to update their details, and a Logout option is available at the bottom.

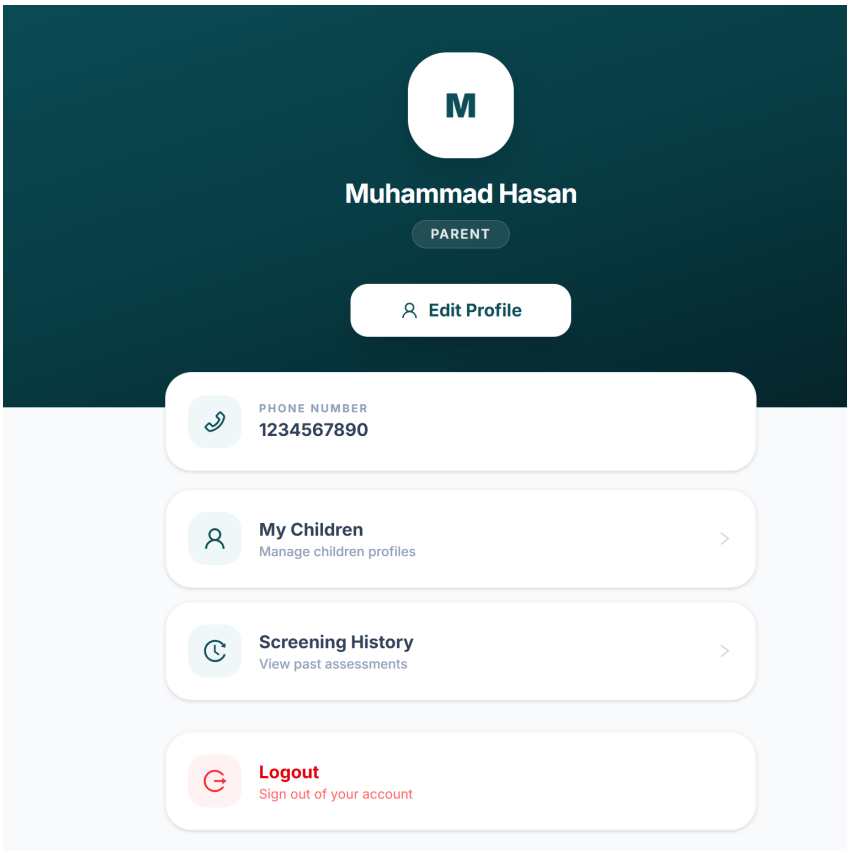


Figure 7.46: Web Parent Profile Management view

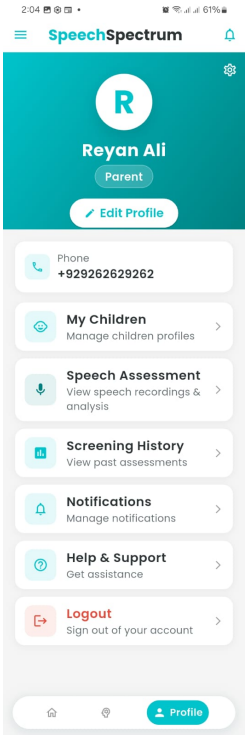


Figure 7.47: Mobile Parent Profile Management view

Expert Dashboard

The Expert Dashboard is the home screen for the therapist role. It greets the logged-in expert by name and displays an Approved Expert verification badge. A Quick Overview section provides three shortcut tiles for Slots, Appointments, and Locations. A Management section below contains full-page links to My Slots, My Appointments, and My Locations, each with a brief description. The bottom navigation bar gives persistent access to Home, Appointments, and Profile tabs.



Figure 7.48: Web Expert Dashboard view

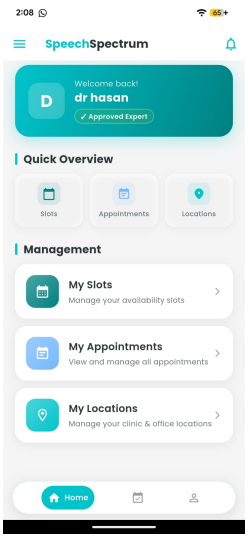


Figure 7.49: Mobile Expert Dashboard

Expert Sidebar

The Expert sidebar is accessed via the hamburger menu icon. It shows the expert's name, specialisation, and Approved status badge at the top. The Main section contains the Dashboard link. The Management section lists My Locations, My Slots, and My Appointments. The Account section provides access to My Profile, Settings, About, and Help. A Logout Account button in red appears at the bottom of the drawer.



Figure 7.50: Web Expert Sidebar view

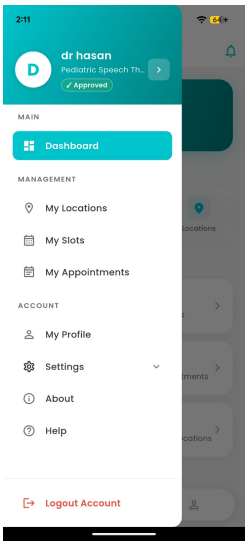


Figure 7.51: Mobile Expert Sidebar

Locations Tab

The Locations Page (My Locations) displays all clinic and office locations saved by the expert, showing a count summary banner at the top. Each location card lists the label name, street address, and city, along with action buttons to View Map, Copy Link, Edit, or Delete the entry. An Add Location floating button at the bottom allows the expert to register a new location. When adding a location, a bottom sheet presents options to select an existing saved location, pick a point directly on an interactive map, or fill in the address manually. The map picker displays a full-screen map with a draggable pin; once a point is selected, the address and coordinates are auto-filled and the expert can confirm by tapping Use This Location.

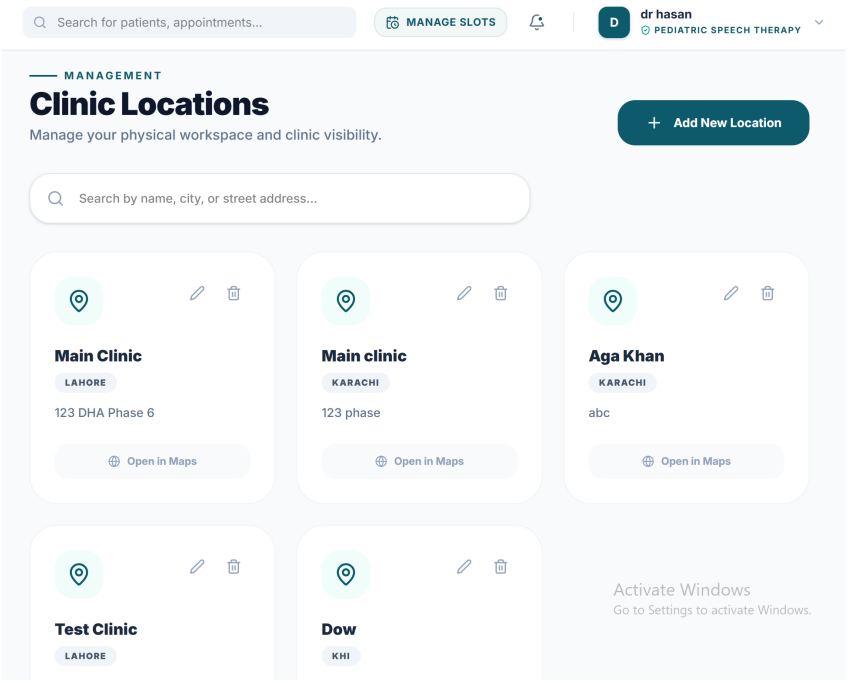


Figure 7.52: Web locations view

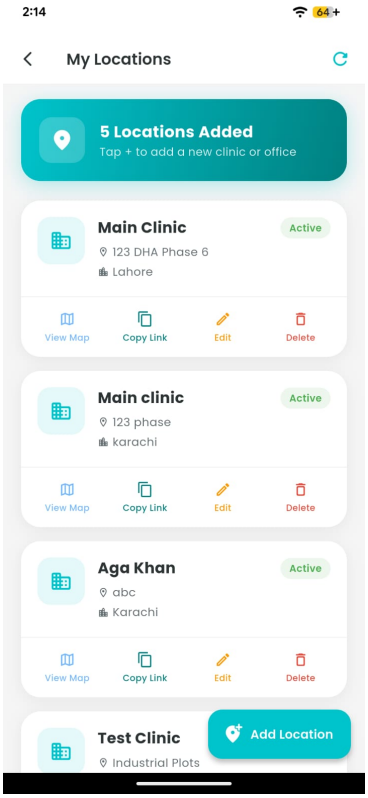


Figure 7.53: Mobile locations Tab 1 view

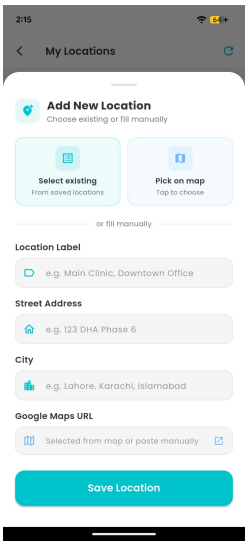


Figure 7.54: Mobile locations Tab 2 view

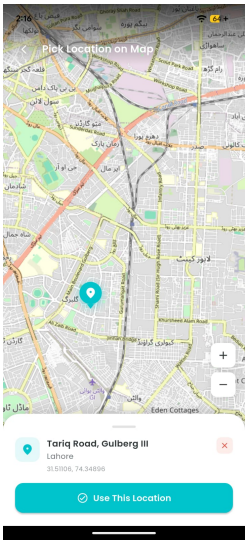


Figure 7.55: Mobile locations Tab 3 view

Slots Page

The Slots Page (My Slots) summarises the expert's availability schedule. A header banner shows the total number of slots created and how many days of the week are covered. A Weekly Coverage row shows toggle buttons for each day (Mon–Sun), with covered days highlighted in teal. Below this, individual slot cards list the date, time range, booking status (Booked or Available), consultation mode (Online, Physical, or Both), fees for each mode, recurrence pattern (e.g., Weekly Every Monday), and Edit/Delete controls. Tapping Add Slot opens a Create Slot form where the expert sets the date, start and end times, consultation mode, online fee, physical fee, currency, optional location, and availability status.

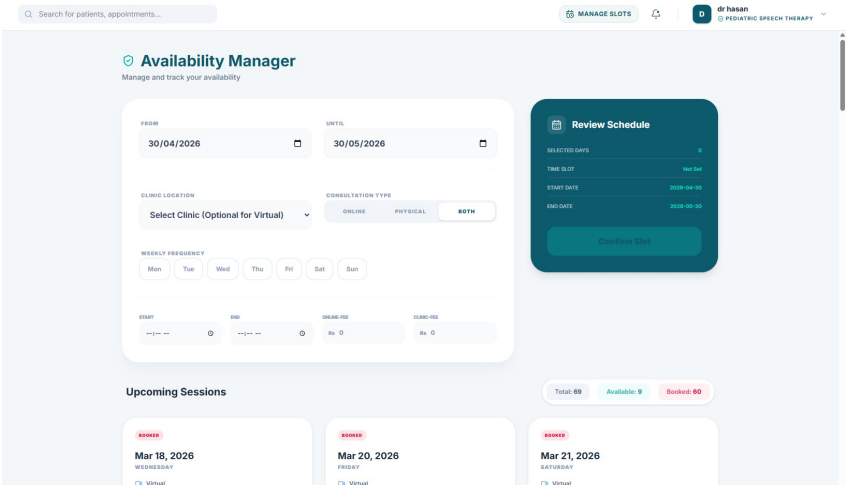


Figure 7.56: Web Slots Page

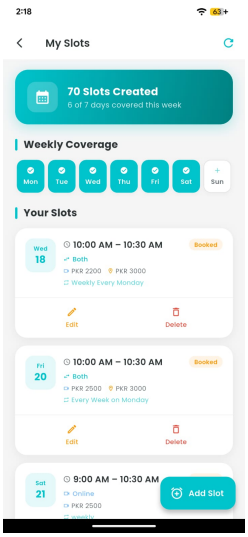


Figure 7.57: Slots Page 1

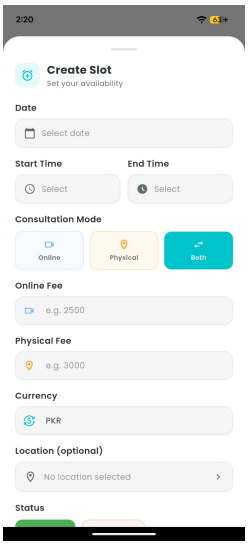


Figure 7.58: Slots Page 2

Appointments Page

The Appointments Page for the expert lists all incoming and past appointments. The Requested tab shows appointments awaiting the expert's action, each displaying the child's name, the parent's therapist, specialisation, scheduled date and time, consultation mode, fee, and links to view the Child Profile or full appointment details. The Confirmed tab lists accepted appointments, distinguishing between Unpaid (payment pending) and Paid entries; online confirmed appointments additionally show a Tap to Join Meeting button. The Completed tab lists all finished sessions with options to view or add session notes.

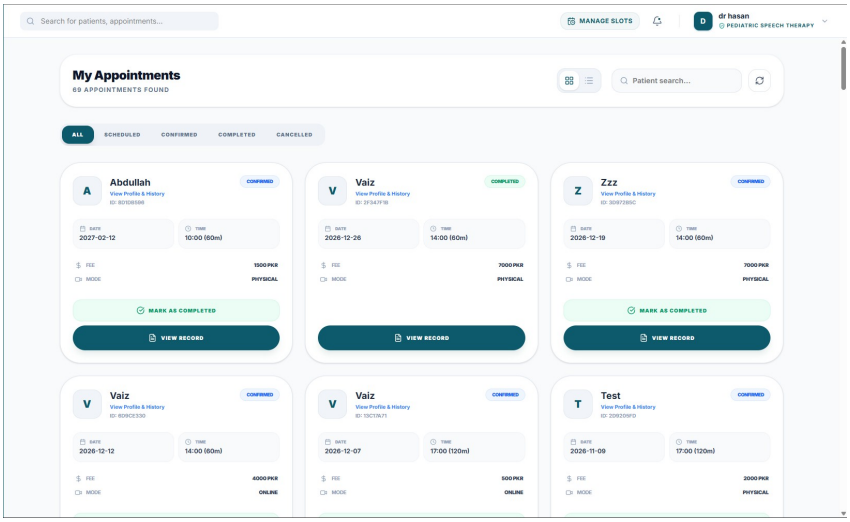


Figure 7.59: Web Appointments Page

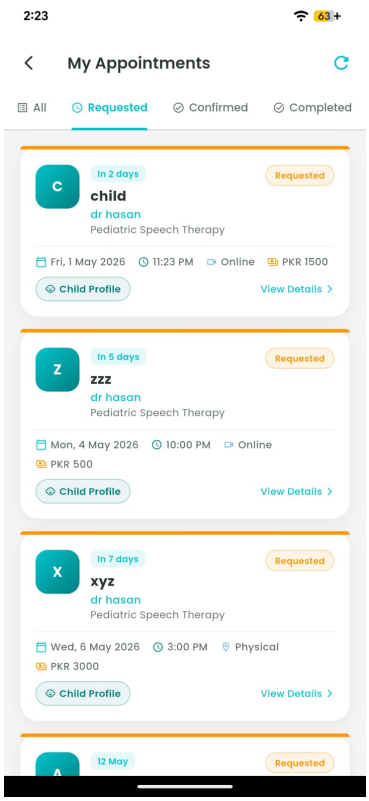


Figure 7.60: Mobile Appointments Page 1

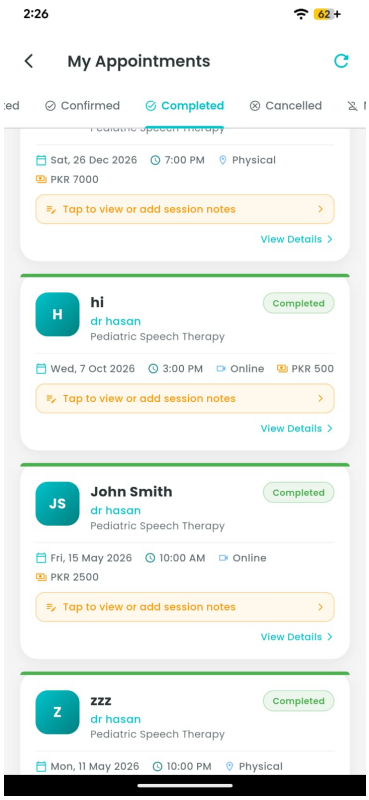


Figure 7.61: Mobile Appointments Page 2

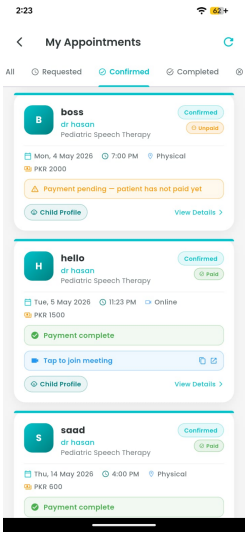


Figure 7.62: Mobile Appointments Page 3

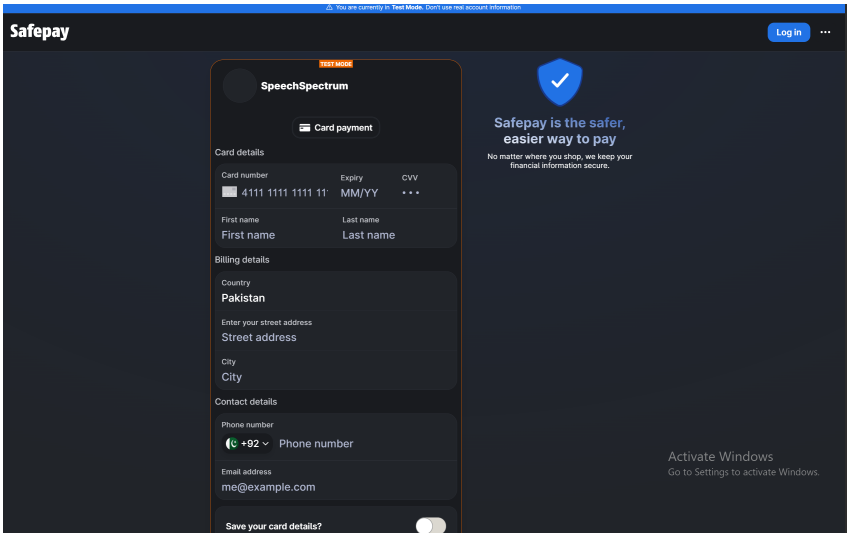


Figure 7.63: Payment Gateway

Appointment Detail Page

The Appointment Detail Page gives the expert a full breakdown of a single appointment. The header shows the child's name, the expert's name, specialisation, and current status badges (Confirmed / Unpaid). Below, action buttons allow the expert to mark the session as Complete, Cancel it, or flag a No Show. The Payment Status section details the session fee and payment state, with a note indicating whether the patient has paid. The Appointment Details card shows the date, time, duration, consultation mode, and fee. A Slot Information card confirms the linked slot's date, time range, mode, and booking status. An Expert Information section at the bottom lists the therapist's contact details.

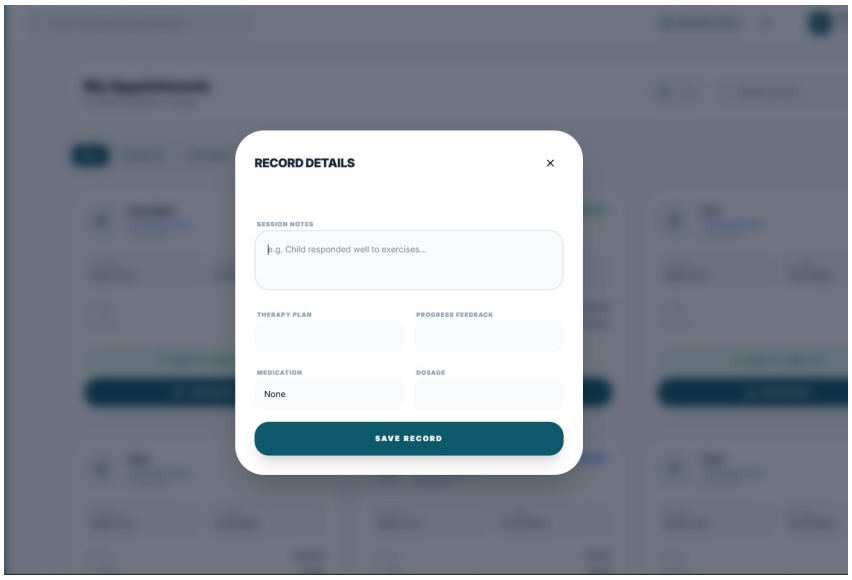


Figure 7.64: Web Appointment Detail Page

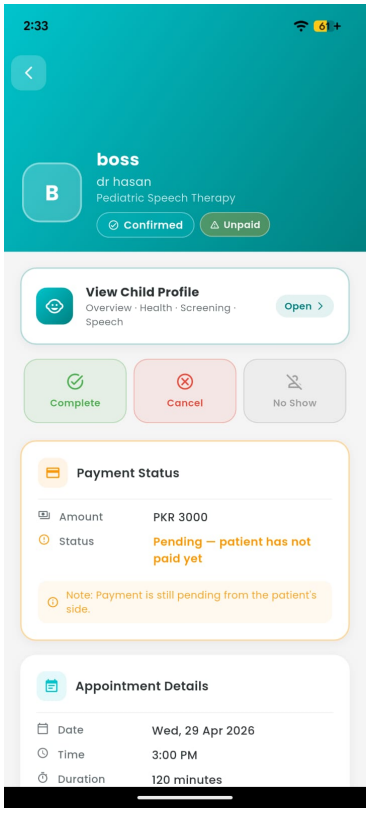


Figure 7.65: Appointment Detail Page 1

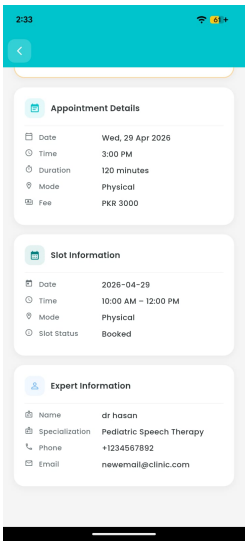


Figure 7.66: Appointment Detail Page 2

Expert Profile Management

The Expert Profile Management screen displays the expert's profile avatar, full name, specialisation (Pediatric Speech Therapy), and an Approved Expert badge. Contact details including phone number, email, organisation, and specialisation are listed in a card. A verified PMDC Number is displayed with a blue verification tick. An Edit Profile button in the header opens the Edit Profile form, which allows the expert to update their profile photo, name, phone number, email, specialisation, and organisation. A Save Changes button commits the updates, and a Delete Account option is available at the bottom of the edit form.

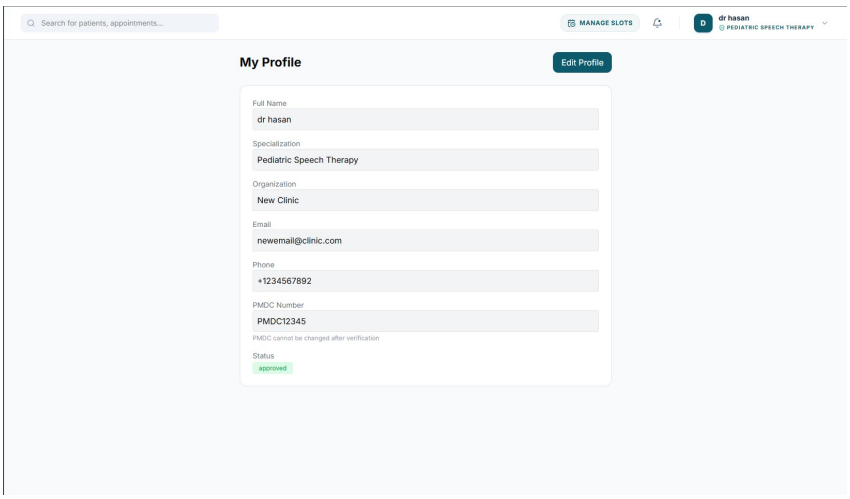


Figure 7.67: Web Expert Profile Management

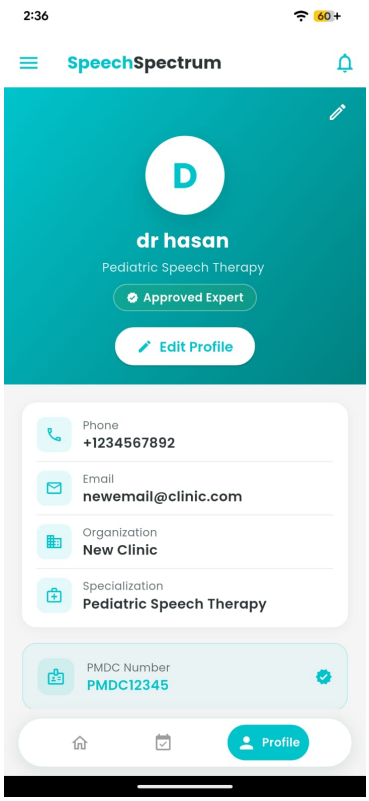


Figure 7.68: Expert Profile Management

Future Enhancements

Following the implementation and evaluation of the current *SpeechSpectrum* system, several key areas have been identified for future development to enhance diagnostic precision and clinical utility:

1. **Multilingual and Cross-Cultural Speech Model**: The current CNN model was trained on a limited dataset and does not yet account for extensive language or dialect variation. Given that the system is designed for the Pakistani context—utilizing Urdu-validated questionnaires and PMDC registration fields for therapists—future iterations should include training on Urdu, Sindhi, and Punjabi speech data. This would significantly improve diagnostic accuracy and reduce algorithmic bias for the primary user demographic.

2. **Longitudinal Progress Tracking with Predictive Reassessment**: While the system currently supports manual re-assessment, it does not yet model developmental trajectories over time. Integrating a time-series analysis layer to track a child's questionnaire responses and Social Affect (SA) scores across multiple sessions would allow the system to detect if a child is improving, plateauing, or regressing. This provides therapists and parents with data-driven insights for intervention adjustments rather than isolated snapshots.

3. **AI-Generated Session Summaries for Appointments**: To streamline the workflow for Expert Users, the system can be enhanced with an NLP-based summarization engine. This feature would automatically transcribe and analyze consultation sessions between therapists and parents, generating concise summaries of clinical observations, recommended intervention activities, and upcoming milestones. This reduces administrative overhead for specialists while ensuring parents have a clear, actionable record of every professional consultation.

References

[1] L. D. Shriberg, R. Paul, J. L. McSweeny, A. Klin, D. J. Cohen, and F. R. Volkmar, "Speech and prosody characteristics of adolescents and adults with high-functioning autism and Asperger syndrome," *Journal of Speech, Language, and Hearing Research*, vol. 44, no. 5, pp. 1097–1115, Oct. 2001. [Online]. Available: https://digitalcommons.sacredheart.edu/cgi/viewcontent

[2] "Distinctive prosodic features of people with autism spectrum disorder: A systematic review and meta-analysis," *Scientific Reports*, 2021. [Online]. Available: https://www.nature.com/articles/s41598-021-02487-6

[3] "Language differences at 12 months in infants who develop autism spectrum disorder," *Journal of Autism and Developmental Disorders*, Mar. 2016. [Online]. Available: https://eric.ed.gov/?id=EJ1090376

[4] "Acquisition of voice onset time in toddlers at high and low risk for autism spectrum disorder," *PubMed*, 2017. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/28339140/

[5] "Early detection of autism using digital behavioral phenotyping," *Nature Medicine*, 2023. [Online]. Available: https://www.nature.com/articles/s41591-023-02574-3

[6] E. A. Fuller and A. P. Kaiser, "The effects of early intervention on social communication outcomes for children with autism spectrum disorder: A meta-analysis," *J. Autism Dev. Disord.*, vol. 50, no. 5, pp. 1683–1700, May 2020. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7350882/

[7] M. Eni, I. Dinstein, M. Ilan, I. Menashe, G. Meiri, and Y. Zigel, "Estimating autism severity in young children from speech signals using a deep neural network," *IEEE Access*, vol. 8, pp. 139489–139500, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9151117

[8] J. J. Simacek *et al.*, "Examining behavioral interventions for infancy and early toddlerhood," *J. Autism Dev. Disord.*, 2025. [Online]. Available: https://doi.org/10.1007/s10803-025-06804-0

[9] F. A. Bivarchi, V. Kehyayan, and S. M. Al-Kohji, "Barriers to the early detection and intervention of children with autism spectrum disorders," *J. Nursing Educ. Pract.*, vol. 11, no. 11, pp. 72–80, Jul. 2021. [Online]. Available: https://doi.org/10.5430/jnep.v11n11p72

[10] T. A. Farooq and S. Ahmed, "Sociocultural barriers to early diagnosis of autism spectrum disorder," *Life and Science*, 2018. [Online]. Available: https://doi.org/10.37185/LnS.1.1.106

[11] N. E. Mandell, K. M. Ittenbach, M. Levy, and J. M. Pinto-Martin, "Delay in diagnosis among Medicaid-autistic children correlates with access to specialty services," *Molecular Autism*, 2018. [Online]. Available: https://molecularautism.biomedcentral.com/articles/10.1186/s13229-018-0201-0

[12] I. Magiati *et al.*, "The psychometric properties of the Q-CHAT," *Molecular Autism*, vol. 6, p. 40, 2015. [Online]. Available: https://doi.org/10.1186/s13229-015-0032-1

[13] A. Allison, B. Auyeung, and S. Baron-Cohen, "Toward brief red flags for autism screening," *J. Am. Acad. Child Adolesc. Psychiatry*, 2012. [Online]. Available: https://www.autismalert.org/uploads/PDF/SCREENING–AUTISM–QCHAT-10

[14] "Translation and validation of autism spectrum screening questionnaire in Urdu," *Pakistan Journal of Medical & Health Sciences*. [Online]. Available: https://pjmhsonline.com/index.php/pjmhs/article/view/1362

[15] "Translation and validation of M-CHAT-R/F in Urdu," *TPMA Journal*. [Online]. Available: https://tpmap.org/submission/index.php/tpm/article/download/447/322/939

[16] "Screening autism spectrum disorder using machine learning on speech transcripts," *Scientific Reports*, 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-01500-6

[17] G. Tartarisco *et al.*, "Machine learning for Q-CHAT-based autism screening," *Diagnostics*, vol. 11, no. 3, 2021. [Online]. Available: https://doi.org/10.3390/diagnostics11030574

[18] L. E. K. Achenie *et al.*, "A machine learning strategy for autism screening in toddlers," *J. Dev. Behav. Pediatr.*, vol. 40, no. 5, pp. 369–376, 2019. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6579619/

[19] C. Feltner *et al.*, "Screening for speech and language delay," *JAMA*, vol. 331, no. 4, pp. 335–351, Jan. 2024. [Online]. Available: https://jamanetwork.com/journals/jama/fullarticle/2814125

[20] A. P. Kaiser *et al.*, "A case for early language and behavior screening," *Policy Insights Behav. Brain Sci.*, 2022. [Online]. Available: https://files.eric.ed.gov/fulltext/ED620207.pdf

[21] American Speech-Language-Hearing Association, "Assessment tools, techniques, and data sources," 2023. [Online]. Available: https://www.asha.org/practice-portal/resources/assessment-tools-techniques-and-data-sources

[22] S. Jullien, "Screening for language and speech delay," *BMC Pediatrics*, 2021. [Online]. Available: https://link.springer.com/content/pdf/10.1186/s12887-021-02817-7.pdf

[23] P. James and L. Shiels, "Audiological management of autistic children," *ENT & Audiology News*, 2022. [Online]. Available: https://www.entandaudiologynews.com/features/audiology-features/post/audiological-management-of-autistic-children

[24] Y. Cho *et al.*, "Natural speech-based features for autism detection," in *Proc. Interspeech*, 2019. [Online]. Available: https://www.isca-archive.org/interspeech2019/cho19interspeech.pdf

[25] M. Briend *et al.*, "Acoustic analysis of nonword repetition," *PubMed*, 2023. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/37422467

[26] H. Chen *et al.*, "Infant cry analysis using convolutional neural networks," *J. Autism Dev. Disord.*, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s10803-025-06811-1

[27] M. Fusar-Poli *et al.*, "Prosodic patterns in autism spectrum disorder," *Behavioral Sciences*, 2024. [Online]. Available: https://www.mdpi.com/2076-328X/14/2/90

[28] A. Perochon *et al.*, "Multimodal audio-visual fusion for developmental disorder detection," *Diagnostics*, 2025. [Online]. Available: https://www.mdpi.com/2075-4418/13/12/2107

[29] M. Akhter *et al.*, "Integration of therapies in autistic children in Pakistan," *JPMA*, 2018.

[30] M. Gitimoghaddam *et al.*, "Applied behavior analysis: A scoping review," *Perspectives on Behavior Science*, 2022.

[31] Y. Hao *et al.*, "Telehealth social communication interventions," *Rev. J. Autism Dev. Disord.*, 2023.

[32] N. P. Rao *et al.*, "Telepractice experiences in autism therapy," *Int. J. Lang. Commun. Disord.*, 2025.

[33] A. S. Sandbank *et al.*, "Meta-analyses on developmental behavioral interventions," *Rev. J. Autism Dev. Disord.*, 2024.

[34] T. Carneiro *et al.*, "Serious games for children with autism," *Healthcare*, 2024.

[35] H. Atturu *et al.*, "AI-based platform for autism therapies," *JMIR Neurorehabilitation*, 2024.